Singularity: Certified Experiments and Observable States

NX36 implementation companion: proofs, contracts, pilots and limits

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Chapter 1

NX36: states, experiments, evidence and limits

1.1 What this companion establishes

The NX36 programme develops a single question across graph interfaces, dynamical systems, arithmetic populations and measured application pilots: which differences can a declared experiment language reveal, at what cost, and with what uncertainty? This implementation companion records the contracts that make that question decidable in bounded cases. It accompanies the graph monograph and the independent frontier companion. It does not replace their definitions or turn their remaining reconstruction questions into solved problems.

An experiment has a population, an intervention, an observation and a cost. Each part is needed. A rank calculation without an uncertainty set does not certify reliable identification. A panel optimum without its candidate catalogue does not certify optimality over every legal physical experiment. A finite coefficient prefix without a generation theorem does not certify equality of infinite series. The modules below return explicit finite answers, witnesses, or resource-bound refusals under recorded assumptions.

Let the original candidate population be Ω, the target be $\tau : \Omega \to \mathcal{T}$, and the possible outputs of experiment e on candidate ω be $Y_e(\omega)$. For independently reset scalar experiments a panel P identifies the target exactly when

$$\forall \omega, \omega' \in \Omega, \quad \tau(\omega) \neq \tau(\omega') \Longrightarrow \exists e \in P : \; Y_e(\omega) \cap Y_e(\omega') = \emptyset.$$

This statement concerns closed output sets: touching endpoints are ambiguous. When the same nuisance variable persists across observations, the correct object is the joint attainable set $Y_P(\omega)$; scalar interval projections can discard information. When interventions share physical history, the state and nuisance variables must persist through the whole word. The implementation keeps these models distinct.

1.2 The common machinery

Object	Mathematical representation	What the representation preserves	Necessary qualification
Original population	Labeled finite objects, Boolean masks, integer fibres, grammar words	Original identities, multiplicities, rank/unrank witnesses	Quotient classes are reported with their fibres

Object	Mathematical representation	What the representation preserves	Necessary qualification
Linear state	$v \in V$, with intervention M_e and reading ℓ	Chronology through $\ell M_{e_m} \cdots M_{e_1} v$	Types, reset rules and boundary legality remain explicit
Persistent uncertainty	$y = c + Gz$, $z \in [-1,1]^k$	Correlations across readings	Different hypothetical worlds may have independent feasible nuisance values
Independent uncertainty	$y_e \in [c_e - r_e, c_e + r_e]$	Candidate-specific closed intervals	No averaging benefit is assumed for adversarial errors
Target recovery	$t = H\alpha$, observations $y = A\alpha + \eta$	Recovery of selected functionals even when α is hidden	Conditioning and admissible-state restrictions matter
Exact experiment search	Cover states, typed row closure, finite dynamic programmes	Real experiment words and costs	The declared language and resource ceiling bind the guarantee
Physical transport	$x_{k+1} = Ax_k + Bu_k$	Exact source-derived discrete evolution	Physical calibration is a separate premise
Parameter strata	Polynomial identities, determinantal divisors, guarded rational charts	Exceptional points and regular cells	Refused symbolic jobs imply no interval theorem
Infinite-sequence equality	Proved recurrence, periodic numerator, finite invariant state	All later coefficients from an explicit cutoff	A long successful sample alone supplies no such rule
Measured pilot	Frozen source, target, split and matched baseline	Auditable held-out observations	A small negative or positive pilot does not establish general predictive skill

For a finite state language, observable rows are generated by

$$W_0 = \operatorname{rowspan}(L), \qquad W_{k+1} = W_k + \sum_e W_k M_e.$$

At stabilization $W_{k+1} = W_k$, every future legal reading lies in W_k. Its annihilator $N = \bigcap_{w \in W_k} \ker w$ is invariant under the generators, so V/N is an exact observational quotient. Retained basis rows carry actual words; linear combinations certify completeness but are not silently treated as executable experiments. Positive cost ordering permits span pruning for cheapest single witnesses. Robust maximum-margin selection and shared-panel optimization require additional searches because a longer dependent experiment may amplify a signal or share costs differently.

For persistent zonotopic uncertainty, a rational contrast a certifies disjointness if

$$|a^{\mathsf{T}}(c_\omega - c_{\omega'})| > \|G_\omega^{\mathsf{T}} a\|_1 + \|G_{\omega'}^{\mathsf{T}} a\|_1.$$

The two generator matrices represent feasible nuisance variables in two alternative worlds. Within either world their columns remain shared across the panel. Feasibility certificates and contrast certificates give complementary primal and dual evidence. An optimizer timeout supplies neither a disjointness theorem nor an overlap theorem.

For target recovery, an exact decoder D satisfies $DA = H$ on the admissible state subspace. This identity proves noiseless recovery. A separate norm bound controls $D\eta$. Two panels can have the same rank and dramatically different amplification. Coordinate rescaling changes coefficient norms, so source units and the declared observation metric are part of a conditioning certificate.

1.3 Implementation map

Tracks	Implemented question	Main mathematical device	Evidence boundary
NX01–NX02	Which panels separate uncertain candidate outputs?	Closed-box strict gaps; exact polytope feasibility and rational contrasts	Finite catalogues and declared nuisance geometry
NX03–NX04	Which discrete and continuous faults remain possible?	Original-mask completion counts; exact parameter-domain covers	Bounded masks and explicit parameter cells
NX05–NX06	Can branching or shared history reduce experimental cost?	Observation-cell Bellman recursion; persistent affine beliefs	Finite histories, complete cost/reset accounting
NX07–NX09	Which capabilities, sensors, evaluator ranks and arithmetic should be purchased?	Joint installation/runtime cover; target recovery; source-bound error menus	Finite capability and evaluator menus
NX10–NX12	When should experiments stop, and which protocols were considered?	Certified tails; rational remote queries; grammar completion dynamic programming	Fixed blocks, checked tail hypotheses and complete declared grammars
NX13–NX14	What do typed and parameter-dependent languages reveal?	Rectangular typed closure; module identities and exceptional strata	Legal typed paths; guarded specialization domains
NX15–NX18	Which formal graph ambiguities are realizable and affordable?	Structural exclusions; local coefficient updates; rank-one fault transport; recovery norms	Explicit graph families and exact discrete source models
NX19–NX23	How do integrality, arithmetic, symmetry and multiplicity change observation?	Smith normal form; residue automata; rational channels; phase transforms; posterior fibres	Integer constraints, field conventions, degree bounds and priors
NX24	When does a finite prefix prove full equality?	Periodic numerators, finite-state closure and proved comparison recurrences	Only the stated generation mechanisms
NX25–NX30	Can these contracts improve application design?	Forecast targets; nonlinear fibres; scale closure; reactive and movement observability; ordered signatures	Measured pilot provenance and held-out targets stated separately from structural theorems
NX31–NX33	How should period targets move across parameter space?	Pinned differential closure, validated transport, monodromy and moving gauges	Source connection, normalization, chart exclusions and actual loop evidence
NX34–NX36	How can analysis, refinement and decisions be certified?	Rational analytic majorants; optimistic/conservative covers; independent certificate checking and Lean soundness	Local analytic discs, finite menus and the precise formal theorem

1.4 What changed in practice

The thermal work expands from a six-candidate example into several distinct experimental models. The original 112-test pool remains available for fair comparison. A

separate 79-member fault population keeps all masks with at most two faults on twelve explicitly selected plate bonds. Some finite catalogues identify fewer response classes than original masks; the receipts retain those blind pairs. A maximal observable quotient is useful, but it does not identify faults that the experiment language cannot distinguish.

Persistent-error geometry can certify a joint separation that every independent scalar projection misses. Adaptive policies can sometimes save experiments, but the retained original thermal pool gives an honest case where adaptive and static costs coincide. Shared-history experiments use a different contract and can exploit information collected before a reset erases it. Their savings are compared only with a static language using the same controls, sensors, initial-state assumptions and noise model.

Low-rank fault transport separates a reusable nominal calculation from each fault correction. If a blocked bond has incidence vector b, then its discrete transition has the form

$$A_\gamma = A_0 + \gamma b b^{\mathsf{T}},$$

with the sign determined by removing conductance from the nominal conservative update. The implementation checks this sign against independent neighbour-flux evolution. Resolvent or coefficient Woodbury identities operate on small fault-index matrices, while the full state and sensor dimensions remain in the source binding. A rank-one algebraic identity does not automatically transport a nominal uncertainty certificate to a damaged model; each claimed bound must be valid for the actual fault/sensor/protocol combination.

The graph frontiers gain stronger tools without assuming their general conjectures. Exact integer quotients expose congruence constraints hidden by rational row spaces. A rationally realizable state need not have an integer preimage. Parameter receipts distinguish rank at $q = 1/2$ from rank on an interval. Structural graph searches report their constructor bounds and can establish exclusions for promised families; failure to find a pair in a finite search does not rule out a larger realization. The effective-prefix work supplies genuine infinite conclusions in restricted sectors by proving the rule that advances coefficients.

The period work follows the same discipline. A differential observable is generated by covariant derivatives of a row, and a reduced connection must satisfy an exact moving-basis identity. If B is a row reduction of the original system $y' = Ay$, then

$$B' + BA = DB, \qquad r = \rho B$$

proves that $z = By$ obeys $z' = Dz$ and preserves target $ry = \rho z$. On an overlap $\widetilde{B} = GB$, the reduced connections must satisfy

$$\widetilde{D} = (G' + GD)G^{-1}.$$

Local gauge identities alone do not prove loop holonomy. The monodromy receipts therefore bind source equations, paths, normalization and independent validated continuation. Numerical proximity to an integer matrix does not authorize exact integer recovery without a lattice premise.

1.5 How to read the evidence

The executable status source is `handoff/nx36/roadmap.json`. Each acceptance unit records its artifacts, successful checks, mathematical status and implementation com-

mit. `handoff/nx36/recovery/CLAUDE_AUDIT.md` preserves the recovered starting point. Track notes below supply definitions, proofs and explicit limitations. JSON receipts preserve exact fractions and source hashes; wall-clock timings are descriptive measurements and are separated from deterministic equality checks.

Use the complete Python test suite for broad regressions, the NX36 receipt runner for retained experiments, the roadmap verifier for dependency/evidence integrity, and the dedicated formal checker for the finite decision theorem. Optional source adapters pin archived upstream engines and retain their licence provenance. Measured application pilots have separate source manifests; restricted athlete rows are excluded from the public archive.

1.6 Consequences for the wider programme

The project now has a reusable path from a mathematical observation to a declared experiment, a costed design, an uncertainty-aware distinction and a portable certificate. This makes the earlier graph interfaces operational and lets the same questions be asked of thermal faults, arithmetic states and period targets. It also makes negative results actionable: a blind direction says which information the current language lacks, and a poorly conditioned decoder says which experiment needs a stronger signal.

Three broad graph research questions remain separate from completing this implementation programme: efficient reconstruction beyond the graph classes already proved, completeness or a full-series collision for fixed pinning, and realization of surviving symmetric-interface ambiguities. These require new mathematics. The modules and receipts in this companion improve the tools, constraints and experimental evidence available for those questions.

Chapter 2

Candidate-specific uncertainty boxes for finite experiment panels (NX01)

This note covers track NX01 of the NX36 expansion. The existing thermal pilot (THERMAL_DIAGNOSTICS.md) computes an uncertainty radius for every experiment and candidate. It then deliberately replaces those radii by one common envelope per experiment before calling the panel optimizer (PANELS.md). The question here is whether keeping the individual radii changes the cheapest guaranteed diagnosis. It adds no new physical calibration, fault population, correlated-error model or global robustness claim.

Short answer: on the unchanged 112-protocol, six-candidate 12 by 12 plate pool, the cheapest guaranteed panel still costs 20 units, and the per-candidate radii do not change a single coverage decision under the primary contract. On a declared 12-protocol sub-pool of the 4 by 4 plate, the common envelope cannot identify the candidates at any cost, but the per-candidate boxes can, at cost 19. Both statements are exact finite computations, checked by independent oracles.

Implementation: `candidate_boxes.py` (production), `candidate_box_oracles.py` (independent oracles), `test_candidate_boxes.py`, receipt `candidate_box_thermal.py` with retained result `candidate_box_thermal.json`.

2.1 1. Objects and contract

A pool has experiments $e = 1, \dots, E$ with exact nonnegative rational costs κ_e, and original candidates $c = 1, \dots, K$, $K \geq 2$, with distinct identifiers. Each candidate has a target label $T(c)$; by default every candidate is its own target. For each experiment and candidate there is an exact central reading r_{ec} and a nonnegative rational radius ρ_{ec}. The radius matrix is indexed experiment first, candidate second. Its transpose is a different contract, even when $E = K$ makes the shapes agree.

Uncertainty contract. Running experiment e when the truth is c yields one reading in the closed interval

$$I_{ec} = [\, r_{ec} - \rho_{ec},\ r_{ec} + \rho_{ec} \,].$$

Experiments are independently reset, so the readings of a panel P range over the product box $Y_c(P) = \prod_{e \in P} I_{ec}$. The contract asserts that every reading actually

produced lies in this box. For the necessity half of Theorem 3, every point of the box must also be a possible reading. That holds for the thermal pilot (Section 6) but not for every radius that is merely an upper bound.

Required pairs. $R = \{\{a, b\} : a < b,\ T(a) \neq T(b)\}$. Pairs with a shared target need not be separated. Their candidates are still kept as distinct original hypotheses, with distinct identifiers.

Plain-language summary. Every candidate has its own "error bar" in every experiment. A set of experiments is good when, whatever the true candidate and whatever the errors within their bars, the readings rule out every candidate with a different target.

2.2 2. Geometry

Lemma 1 (closed intervals). For $\rho_a, \rho_b \geq 0$, the intervals $[r_a - \rho_a, r_a + \rho_a]$ and $[r_b - \rho_b, r_b + \rho_b]$ are disjoint if and only if $|r_a - r_b| > \rho_a + \rho_b$.

Proof. Without loss of generality $r_a \leq r_b$. Two closed intervals on the line are disjoint exactly when the upper end of the left-hand one is strictly below the lower end of the right-hand one. Their order is fixed by their centers when they are disjoint: if $r_a \leq r_b$ and the intervals are disjoint, then I_a lies to the left. So disjointness means $r_a + \rho_a < r_b - \rho_b$, that is $r_b - r_a > \rho_a + \rho_b$. Conversely, if $r_b - r_a \leq \rho_a + \rho_b$, the point $\max(r_a - \rho_a, r_b - \rho_b)$ is at most $\min(r_a + \rho_a, r_b + \rho_b)$, so the intersection is nonempty. □

Write the raw margin $\Delta_{eab} = |r_{ea} - r_{eb}| - \rho_{ea} - \rho_{eb}$. Experiment e separates $\{a, b\}$ iff $\Delta_{eab} > 0$. The cases this rules on are exactly the ones tested:

- Touching endpoints ($\Delta = 0$). The shared endpoint is a possible reading under both candidates, so this is not separation.
- Zero radii. $\rho = 0$ is a degenerate closed interval. Two exact readings separate iff they differ. A zero radius against a positive one separates iff the exact point lies strictly outside the other interval. The common backend requires strictly positive noisy margins; this module does not.
- Identical centers ($r_{ea} = r_{eb}$). Then $\Delta \leq 0$ whatever the radii, so the pair is never separated by e. The witness reading is the common center.

Lemma 2 (boxes). $Y_a(P) \cap Y_b(P) = \emptyset$ iff some $e \in P$ has $\Delta_{eab} > 0$.

Proof. A product of sets is empty-intersecting with another product exactly when some coordinate pair is disjoint. If every coordinate pair meets, choosing one common point in each coordinate gives a common point of the boxes. Apply Lemma 1 coordinatewise. □

For the empty panel the box is the one-point product, so no pair is separated. Hence the empty panel is feasible exactly when $R = \emptyset$.

2.3 3. Decoder and identification

The decoder for panel P and observation vector $y \in \mathbb{Q}^P$ returns

$$D_P(y) = \{c :\ |y_e - r_{ec}| \leq \rho_{ec} \text{ for all } e \in P\} = \{c : y \in Y_c(P)\}.$$

Its target output is $T(D_P(y))$.

Theorem 3 (cover iff guaranteed target identification). Suppose every point of every box is a possible observation. Then the following are equivalent:

1. every required pair is separated by some member of P;
2. for every candidate c and every possible observation $y \in Y_c(P)$, $T(D_P(y)) = \{T(c)\}$.

Proof. (1)⇒(2). The truth c is always in $D_P(y)$, since $y \in Y_c(P)$. If $d \in D_P(y)$ with $T(d) \neq T(c)$, then $y \in Y_c(P) \cap Y_d(P)$. By Lemma 2 this contradicts (1).

(2)⇒(1). Suppose $\{a, b\} \in R$ is unseparated. In each coordinate, the intersection $I_{ea} \cap I_{eb} = [\max(\text{lower ends}), \min(\text{upper ends})]$ is nonempty by Lemma 1. Let y_e be its midpoint. Then $y \in Y_a(P)$, so it is possible under truth a, and $D_P(y) \supseteq \{a, b\}$ has two targets. □

`candidate_box_ambiguity` returns exactly this witness (a, b, y) for the first unseparated required pair. The direction (1)⇒(2) needs only the enclosure half of the contract. If radii are only upper bounds, a covering panel still certifies identification, but an ambiguity witness might not be physically attainable. Sharper radii without a proof change the problem; they are not a refinement.

An empty decoder output, which is possible when an observation lies outside every box, signals that the model contract failed. It does not identify an unmodelled fault.

2.4 4. Comparison with the common envelope

Let $\varepsilon_e = \max_c \rho_{ec}$ (`common_envelope`).

Proposition 4 (monotonicity). If $|r_{ea} - r_{eb}| > 2\varepsilon_e$ then $\Delta_{eab} > 0$. Hence every common-envelope covering panel is a heterogeneous covering panel, and the heterogeneous optimum costs at most the common-envelope optimum. Both comparisons hold for the same readings, costs, population, targets and contract.

Proof. $\rho_{ea} + \rho_{eb} \leq 2\varepsilon_e$. □

Corollary 5 (where gains can occur). If $\rho_{ea} = \rho_{eb} = \varepsilon_e$, then experiment e has the same decision on $\{a, b\}$ under both contracts. Gains can occur only on pairs containing a candidate whose radius is below the row maximum.

On equal common radii, the heterogeneous masks coincide with `panels.coverage_masks`, which is tested whenever the envelope is positive. The tests also compare minimum costs with `panels.minimum_cost_panel` on random common-radius pools.

2.5 5. Exact optimization

With $M_e = \{p \in R : e \text{ separates } p\}$, a panel covers iff $\bigcup_{e \in P} M_e = R$ (Lemma 2). Minimum-cost selection is therefore weighted set cover on required pairs. This problem is NP-hard in general.

Production solver (`minimum_candidate_box_panel`). This is a private branch-and-bound kernel over validated masks; the masks are always produced from readings and radii inside the module. It follows the design proved in PANELS.md:

- Dominance reduction. Experiment f dominates e when $M_f \supseteq M_e$ and $\kappa_f \leq \kappa_e$, with strict improvement or an index tie-break. Removing empty and dominated experiments keeps an optimum, because replacing a dominated member by its

dominator never increases cost or loses coverage. The tie-break makes the relation a strict partial order, so a kept dominator exists.
- Branching. Pick an uncovered pair p with fewest admissible coverers $e_1, \dots, e_m$, ordered by cost. Branch i includes e_i and excludes $e_1, \dots, e_{i-1}$. Every cover contains some coverer of p, and the branch indexed by its first coverer in this order is unique, so the branches partition the covers.
- Bounds. Pairs are taken greedily so that no admissible experiment covers two of them. Summing their cheapest admissible coverers gives the packing bound, a valid lower bound because those coverers are distinct experiments. Without packing, the bound is the maximum cheapest coverer.
- Termination with zero costs. Every branch removes at least one experiment from the admissible set, so the tree is finite whatever the costs. A pool of many zero-cost experiments separating nothing is pruned by the reduction and never searched as repeated words.
- Budget. `max_nodes=10**6` by default. Exceeding it returns `inconclusive` with the greedy-or-better incumbent as an upper bound and the root packing bound as a proved lower bound. An exhausted budget never produces `infeasible`. `infeasible` is returned only when some required pair has no separating experiment in the whole pool, and every such pair is listed.

Independent oracles (`candidate_box_oracles`). This module never imports the production module. It recomputes separation from interval endpoints, $\max(\text{lower}) \le \min(\text{upper})$, instead of the production margin formula:

- `exhaustive_box_panel` scans every subset (bounded by `max_subsets`, default 2^{16}) and tests product-box intersection directly.
- `cost_bounded_box_panel` scans every subset of total cost at most a declared bound. If none separates, no panel of cost at most the bound exists. With positive costs this is affordable on large pools when the bound is small.
- `cover_state_dijkstra` is Dijkstra's algorithm on reached sets of separated pairs, with one edge per experiment. Repeating an experiment never enlarges a covered set, so with nonnegative costs a cheapest path uses distinct experiments and equals the cheapest cover. For $K = 6$ there are at most 2^{15} states.

The tests compare all three oracles with the solver, under all four packing and reduction settings, on 300 seeded tiny rational pools. These pools have up to 7 experiments and up to 4 candidates, with mixed zero and positive radii, half-integer readings, zero costs and occasional shared targets. Dedicated tests cover dominated experiments, zero-cost combinations, budget exhaustion and the common-backend comparison. A greedy result is never reported as optimal: optimality is claimed only from completed search, and that search is checked by the oracles.

2.6 6. Thermal application

Unchanged pool. `protocol_pool` compiles the 12 by 12 source-pinned plate with its six candidates: nominal, conductance_half, heater_half, loss_double, and the two blocked bonds. The protocol pool is the original 112-member pool (horizons 1, 2, 3, 4, 6, 8, 12, 20; four sensors). The primary contract is initial radius $\rho_0 = 1/100$, sensor noise $\nu = 1/100$, process radius 0 and reduction error 0. `from_diagnostic_pool` reuses `DiagnosticPool.error_bounds` as the radius matrix without recomputing it. The receipt then checks the compilation independently:

- all 672 central readings against the separately written neighbor-flux stencil oracle of `scripts/thermal_diagnostics.py`, which reads no compiled matrix;

- all 672 radii against the closed form $\rho_{ec} = \nu + \rho_0(1-\lambda_c)^{T_e}$, where $\lambda_c = 1/50$ for loss_double and $1/100$ otherwise;
- the five retained rows of `results/thermal_diagnostics.json`, covering readings, costs and common radius ε_e, without regenerating that older receipt;
- the common-envelope optimum from the existing backend: cost 20, protocol index 78 (input 2 for eight steps, left-central sensor), as retained.

Why the closed form holds, and why it is attained. For every candidate, A_c has nonnegative entries and row sums $1-\lambda_c$; blocked bonds move conductance onto the diagonal. Thus $|A_c|\mathbf{1} = (1-\lambda_c)\mathbf{1}$, and the propagated bound is $b_T = \rho_0(1-\lambda_c)^T\mathbf{1}$. Every sensor row is nonnegative with sum one, so $|C_j|b_T = \rho_0(1-\lambda_c)^T$. The linear image of the initial box $[-\rho_0,\rho_0]^{144}$ under $C_jA_c^T$ is the whole interval $[-\rho_0(1-\lambda_c)^T, \rho_0(1-\lambda_c)^T]$, with endpoints at the same-sign corners. Adding independent sensor error in $[-\nu,\nu]$ fills I_{ec} exactly. So the thermal radii are attained, and the necessity half of Theorem 3 applies. For signed sparse models, $|A|$-propagation is safe but may be conservative. The test suite keeps the existing signed example, where the radius $1/20$ is attained.

Result (exact finite computation). The heterogeneous optimum is panel $\{78\}$ with cost 20, the same as the common envelope. The checks agree:

- the solver finishes with status `optimal`, lower bound 20, after 18 nodes;
- the cover-state Dijkstra oracle gives 20 after 23 settled states;
- the cost-bounded exhaustive scan, which enumerates all 3223 subsets of cost at most 20, finds no cheaper separating subset.

Under the primary contract, the heterogeneous and common-envelope masks have exactly the same 809 coverage bits. By Corollary 5, gains could only involve loss_double, the only candidate with a smaller radius, and none occurs here. Every corner of every candidate's box decodes to the true candidate under panel $\{78\}$. The zero-noise optimum $\{22\}$ of the old receipt is not robust here. Its witness pair is nominal versus loss_double, with common reading $57120893/200000000$, and both are retained.

The receipt also repeats the old receipt's grid: initial radius $0, 10^{-4}, 10^{-3}, 10^{-2}$ times sensor noise $0, 10^{-3}, 10^{-2}$. Each grid contract uses the exact additive form $\rho = |C|b + \nu$ on a pool compiled with zero sensor noise. The grid matches the direct compilation at the primary contract. In all 12 contracts, the heterogeneous optimum equals the common-envelope optimum, and each is confirmed by Dijkstra. Two contracts gain mask bits without changing the optimum: 915 versus 906 bits at $(10^{-2}, 0)$, and 898 versus 895 at $(10^{-2}, 10^{-3})$.

Small-plate contrast (exact finite example). Take the 4 by 4 plate, the 12 protocols `default_protocols((1,2,3,4,6,8))[1::5][:12]`, $\rho_0 = 1/100$ and $\nu = 1/1000$. The common envelope leaves nominal versus loss_double uncoverable: its best margin over the pool is negative, and both the backend and the exhaustive box oracle report infeasibility. With per-candidate radii, protocol 9 (input 2 for four steps, left-central sensor) separates that pair with margin $926083/10^{10}$, where the envelope margin is $-1448351/(5\cdot 10^9)$. The exact optimum is $\{4, 9\}$ at cost 19, confirmed by the 4096-subset exhaustive oracle and by Dijkstra. Readings and radii are replayed by the stencil oracle and the closed form. This shows the refinement can change feasibility. It is a property of this declared sub-pool, not of the 12 by 12 pool.

Interpretation. The gain comes from sharper but still proved uncertainty assumptions on the same model, not from a new measurement. In the source pool, the binding pairs are not ones involving the lower-radius candidate, so the cheapest guaranteed

diagnosis is unchanged. An improvement was not required for acceptance, and none is claimed.

2.7 7. Scope and limits

- **Valid scope.** The results hold for independent resets, scalar closed intervals and a product uncertainty set. A persistent correlated nuisance invalidates the necessity half: box separation remains sufficient for any smaller joint set, but a joint geometry can separate where no coordinate does. That is NX02, not this track.
- **Search space.** Optimality is over the supplied finite pool only, not over all thermal interventions.
- **Unknown faults.** These lie outside the six-candidate population.
- **Cost units.** Costs are declared model accounting units.
- **Wire format.** The receipt uses the NX interchange encoding `[numerator, denominator]` for every exact rational, as specified in `handoff/nx36/architecture/INTERCHANGE_CONTRACT.md`. The packet's earlier suggestion of canonical strings is superseded by that shared contract. The existing `singularity.thermal-diagnostics/1` receipt keeps its string encoding and is not migrated. The schema name is `singularity.candidate-box-panel/1`.
- **No refactor of the existing backend.** `panels.py` is unchanged. The new module has its own private kernel adapted from the PANELS.md design. Factoring a shared kernel out of `panels.minimum_cost_panel` remains an integration option. It would need the existing panel compatibility tests to pass unchanged.
- **Arithmetic growth.** Bit lengths of readings and radii grow roughly linearly with horizon and are recorded per horizon in the receipt. At horizon 20 they stay small enough for exact replay in about ten seconds.

2.8 Reproduction

```
PYTHONPATH=src python -m unittest tests.test_candidate_boxes -v
python scripts/candidate_box_thermal.py --check
```

The check recompiles everything and compares it with the retained JSON. It never rewrites the JSON; running without `--check` regenerates it as an explicit maintenance action.

Chapter 3

NX02: exact joint observation geometry

This module completes the three NX02 units with a bounded exact rational engine, a complete finite panel optimizer and a source-derived thermal pilot. Its uncertainty sets are deterministic sets, not covariance matrices or probability laws. The guarantee is relative to the stated affine nuisance model.

3.1 Contract and data orientation

`JointObservationFamily` records original candidate identifiers, ordered experiment identifiers, positive rational experiment costs, target labels, candidate-major central vectors and candidate-major generator matrices. For candidate c, its possible panel observations are

$$Y_c = \mu_c + G_c[-1,1]^{m_c}.$$

Generator columns, rather than rows, identify nuisance sources. A persistent offset has the same column in every reading coordinate. A separately reset initial condition has its own column block for that experiment. A sensor error has a separate diagonal column. `source_labels` identifies these distinctions; projection selects rows while retaining the complete generator-column identity. The `exact_set` flag distinguishes an exact affine image from a deliberately conservative enclosure. Every separation certificate remains safe for an outer enclosure; overlap of that enclosure need not be physically realizable.

Each competing candidate receives its own nuisance vector. Reusing a source label across hypotheses does not impose equality of its unknown value. The possible-world quantifier is

$$\exists \xi \in [-1,1]^{m_a},\ \eta \in [-1,1]^{m_b} : \mu_a + G_a\xi = \mu_b + G_b\eta.$$

Forcing xi=eta would change the model and could invent identification.

3.2 Support and completeness theorem

SNG-NX02-01. For a rational contrast w,

$$h_c(w) = \max_{z\in[-1,1]^{m_c}} w^T G_c z = \sum_j |(G_c^T w)_j|.$$

Each coordinate z_j may independently choose the sign of its coefficient, so the upper bound from the triangle inequality is attained. Consequently

$$|w^T(\mu_a - \mu_b)| > h_a(w) + h_b(w)$$

certifies disjoint sets. The strict inequality matters: equality can leave a common boundary point and is rejected by the checker.

`joint_pair_certificate` solves the exact overlap problem, rather than merely searching a list of sufficient contrasts. Set A=[G_a,-G_b] and b=mu_b-mu_a. The solver converts Az=b and -1<=z<=1 into rational inequalities. It repeatedly eliminates the last variable by pairing a positive-coefficient inequality with a negative-coefficient inequality. A zero-coefficient inequality is retained. If one side has no bound, elimination imposes no extra condition from that side. These are exactly the conditions that the lower bounds on the eliminated variable do not exceed its upper bounds. Induction proves equivalence of the projected feasible set and the eliminated system.

Every generated row carries nonnegative multipliers on the original rows. An impossible constant row therefore supplies an exact Farkas certificate. For a feasible final system, back substitution selects zero where possible and otherwise an available rational endpoint. The returned point is replayed against every original inequality before being reported.

To obtain a separating contrast from an infeasibility certificate, let lambda_i be the multiplier on the positive equality row minus that on its negative copy. The nonnegative box-bound multipliers have total at least ||A^T lambda||_1, because their signed difference cancels A^T lambda. The negative constant row implies

$$\lambda^T b + \sum \text{bound multipliers} < 0.$$

Thus |lambda^T b|>||A^T lambda||_1. Splitting A into its two candidate blocks gives exactly the support inequality above. This proves that, whenever the bounded exact elimination finishes, the engine returns either a primal overlap witness or a strict rational separating contrast. It does not depend on a floating optimization status or a generic convex-separation assertion.

`verify_joint_certificate` checks the displayed affine equality and all nuisance bounds, or independently recomputes the displayed support gap. Zero generators, rank-deficient sets, point observations and touching sets are included.

3.3 Complete finite panel optimization

`minimum_joint_panel` sorts every subset of the declared finite experiment pool by total cost, then cardinality and index order. It checks all target-distinct candidate pairs on each projected joint family. The first all-separated panel is globally cheapest in that finite pool. Earlier failed panels retain overlap certificates, and the successful panel retains strict contrasts. If no panel works, all subsets have been exhausted. Joint separation is not a weighted set-cover property: two individually ambiguous readings can jointly separate.

For exact observation sets these statuses are `optimal` and `impossible`. For outer enclosures they are instead `optimal_for_outer_enclosure` and `not_certified`: a physical optimum or physical impossibility does not follow from optimizing over conservative sets. The result carries that scope explicitly.

The default subset budget is 4,096. The elimination budget is 20,000 generated rows, including all positive-negative pairs before duplicate reduction. Either budget can

cause `GeometryBudgetExceeded`; that exception means no complete verdict was obtained. Exponential elimination and subset selection remain real limitations. Structural zero nuisance columns are eliminated internally and their displayed overlap coordinates are restored as zero, preserving the original generator/source contract.

3.4 Persistent thermal nuisance pilot

`affine_protocol_family` derives the exact affine observation map for the existing `AffineCandidate` dynamics. Its centers are the old simulator's values. For every independently reset experiment and every initial cell, it propagates a unit initial state through the zero-input linear dynamics. This yields the initial-box generator columns. The heater contribution belongs to the center; it is not accidentally added to the sensitivities. A persistent additive output offset and fresh sensor errors occupy the explicitly labelled extra columns. There is no uncertain multiplicative gain approximation in this adapter.

The retained four-by-four plate pilot compares the original left- and right-blocked-bond models using the left and right center sensors after word (2,0). Each experiment costs seven units under the original cost convention; their separately reset joint panel costs fourteen. Each candidate has a persistent offset bounded by 1/10 and independent sensor errors bounded by 1/10,000. The two candidate left-minus-right central contrasts are +1/32 and -1/32. The persistent offset cancels, and each world's sensor-error support in this contrast is 1/5,000. The strict gap is

$$\frac{1}{16} - \frac{2}{5000} = \frac{621}{10000} > 0.$$

Each scalar interval overlaps, so the box outer approximation cannot identify the pair using this pool. Replacing the persistent source by independent output offsets produces a genuine overlap witness and invalidates the alleged joint distinction. This adversarial comparison guards the modelling assumption that creates the benefit. The source model is synthetic and source-pinned; these uncertainty bounds are declared assumptions, not measured calibration.

3.5 Independent checks and replay

The tests use an independently written active-vertex enumeration oracle for small bounded polyhedra. It enumerates square active systems, solves them by Gaussian elimination, then checks the original equalities and bounds; it never calls production elimination. Twenty-four random pair instances agree, and twelve three-candidate families compare the complete subset optimum against independent pair feasibility. Separate tests cover false correlation, endpoints, malformed shapes, zero generators and explicit budget refusal. The thermal map is independently checked by substituting a nonzero reset cell, offset and sensor-error assignment into the original simulator.

Run `PYTHONPATH=src python scripts/joint_thermal.py --check`. The receipt stores the full family, all selected and cheaper rejected panel certificates, source provenance, exact costs, maximum rational bit lengths and work limits.

Chapter 4

NX03: whole original populations of bond faults

The module `fault_population.py` replaces a hand-picked list of hypotheses by an exact constrained population of original Boolean masks. It implements all three NX03 units without altering the historical six-candidate constructor. Configuration multiplicities and physical labels remain part of the contract.

4.1 Exact populations and prefix completion counts

For ordered distinct bond IDs e_1,...,e_m, the original population is

$$\Omega = \{\theta \in \{0,1\}^m : R\theta \leq b\}.$$

`FaultPopulation` accepts integer inequalities, an optional cardinality cap, an optional source-model digest and a completion-state budget. General signed integer rows are allowed, so coupling constraints such as theta_1+theta_2<=1 and theta_3+theta_4<=theta_2 are not reduced to independent cardinality limits. The domain is explicitly Boolean; arbitrary integer coordinates or floating constraint coefficients are rejected.

For a prefix p of length t, subtract its contribution from each original constraint. The memoized state consists of t and this residual right-hand side. The smallest possible remaining contribution of row r is the sum of its negative coefficients. If that lower bound exceeds the residual bound, the prefix has no completion. At t=m every surviving state is a feasible original mask. Otherwise

$$C(p) = C(p0) + C(p1).$$

SNG-NX03-01. The completion recurrence, with the stated rejection rule, counts every original feasible Boolean mask exactly once. This follows by partitioning completions according to their next bit; the lower-bound rejection cannot discard a feasible assignment. Memoization identifies equal residual subproblems but adds their original multiplicities through the recurrence.

Lexicographic rank adds C(p0) when the next bit is one. Unrank compares the requested index with C(p0), follows the zero branch when smaller and subtracts that count before taking the one branch otherwise. Induction on remaining length proves that these maps are inverse. Enumeration is implemented by unranking all integer indices; its own mask-count budget is separate from the completion-state budget. Conditioning retains the complete original masks that satisfy a caller predicate. A count is

not a probability and a uniformly sampled compressed state is not a uniformly sampled fault.

The worked three-bond at-most-one population has masks 000,001,010,100. Prefix zero has three completions and the final mask has rank three. More generally, summing counts over every prefix at a fixed depth partitions the original population and returns its total count.

4.2 Source-consistent physics of simultaneous faults

`fault_candidate` takes an existing exact `AffineCandidate`, an original mask, canonical undirected bonds and the declared nominal bond conductance k. It rejects repeated or reversed duplicate bonds, invalid endpoints and a selected edge whose two off-diagonal conductances do not equal k. Each selected bond updates

$$A_\theta = A_0 + k \sum_i \theta_i (e_{u_i} - e_{v_i})(e_{u_i} - e_{v_i})^T.$$

The off-diagonal entries lose k and both endpoint diagonal entries gain k. Thus symmetry and each row sum are preserved. Removing only off-diagonal entries would incorrectly add an ambient loss term. Distinct simultaneous faults may share endpoints; their diagonal increments add and do not remove a bond twice. Every result retains the source heater and sensors. Its identifier contains the full original mask, even if another configuration happens to give identical readings.

`plate_bonds` specifies row-major canonical undirected adjacency. The retained 12-by-12 pilot uses the explicitly frozen list

$$(0,1),(0,12),(1,2),(1,13),(2,3),(2,14),$$
$$(3,4),(3,15),(4,5),(4,16),(5,6),(5,17).$$

At most two blocked bonds therefore gives 1+12+66=79 original models. The full plate has 264 undirected bonds and the analogous population count is 1+264+34,716=34,981. This larger number is a count only, not a claim that all models and all long protocols have been simulated. The physical tiny check uses the two-by-two plate, whose four bonds yield eleven configurations.

4.3 Complete declared-pool diagnosis

`pilot_fault_candidates(12)` returns the population together with all 79 `AffineCandidate` models. No graph symmetry or duplicate reading merges the location identities. A source digest hashes the exact nominal matrix, heater and sensors. The diagnostic receipt retains every mask, its model identifier, every reading and every candidate-specific error bound, so the ordinary NX01 decoder can return all compatible original configurations.

The retained protocol pool is fixed before inspecting results: words (2,0) and (2,0,…,0) of length twelve, each observed with the four original sensors. Its eight experiments define a complete finite optimization problem. The exact zero-noise readings produce sixty behavioral classes among the 79 masks, so exact location identification is impossible within that pool. The receipt lists each original-mask class and every uncoverable target pair. This is not an all-future equivalence claim: later words or extra sensors may split those classes. With independent error 1/10,000, all pairs are uncoverable in

this particular distant-fault pilot, illustrating the distinction between exact arithmetic separation and robustly useful measurement.

The optimizer's infeasible status is a completed finite-pool proof, not a search-budget failure. The broader problem of finding a sufficiently powerful physical experiment language remains separate. The fixed-pool negative result therefore completes the declared pilot while explicitly retaining the reason to expand the sensor/protocol language.

4.4 Independent checks and resource contract

Every two-by-two mask is compared against a separately written undirected-flux update on a nonuniform rational state. This checks all eleven models, all row sums, matrix symmetry, nonnegativity and preservation of the original heater and sensors. Two removed bonds sharing endpoint zero are checked explicitly. Tiny coupled populations agree with literal Boolean products; all rank/unrank round trips and every prefix-partition sum are checked. The 79-model receipt independently propagates selected masks through both short and twelve-step words using the neighbour-flux oracle, without importing the new matrix assembler.

Completion counting has exponentially many possible residual states for general Boolean inequalities. The default limit is 100,000 states; enumeration has a separate 100,000-mask limit. Exceeding either raises a work refusal and cannot be interpreted as an empty population. The 79-model optimizer uses the existing bounded exact candidate-box engine, and its status is retained verbatim.

Run `PYTHONPATH=src python scripts/fault_population_diagnostics.py --check`. The receipt includes source provenance, source-model digest, all masks, prefix partition, raw readings, original behavioral-class multiplicities, independent flux spots, optimizer evidence and rational bit-length statistics.

Chapter 5

NX04: certified severity families and independent calibration

`thermal_parameters.py` supplies exact rational sample physics, sparse symbolic finite-horizon observations, and a bounded complete-domain separation checker. It completes the small NX04 scope. The parameter boxes are declared modelling assumptions; the synthetic source does not empirically calibrate them.

5.1 Physical coordinates and admissibility

The parameter ordering is conductance k, loss lambda, heater gain g and specified-bond severity alpha. The only accepted severity convention is alpha=0 nominal and alpha=1 fully blocked. For v=e_u-e_v,

$$A_f = (1-\lambda)I - kL + k\alpha vv^T, \qquad b = gb_0.$$

The fault bond retains conductance k(1-alpha). `ThermalParameterDomain` requires k>=0, lambda>0, g>=0, 0<=alpha<=1 and 4k+lambda<=1 over the entire declared box. A box crossing that last constraint must first be intersected or split; the current implementation rejects it rather than pretending the full invalid box is covered. The sufficient stencil condition makes every matrix entry nonnegative. Since all conductance terms have zero row sum, every row sum is 1-lambda; partial blockage preserves this contract.

`parametric_fault_sample` constructs the exact matrix for any rational point in the admissible domain. At alpha=0 it reproduces the nominal source matrix; at alpha=1 it reproduces `fault_candidate` full removal. This endpoint equivalence is exact, including heater and sensors. Rational intermediate severities are checked through independent edge-flux evolution, not interpolation between the two endpoint answers.

5.2 Symbolic maps and interval enclosures

`Polynomial` stores a sparse map from nonnegative multiexponents to rational coefficients. Addition combines equal monomials and multiplication convolves them. `panel_parameter_map` propagates the source stencil symbolically, with separately reset zero initial states for every requested word. Every coordinate of a finite-horizon response is therefore represented exactly as a polynomial in the four parameters. The default horizon is four and the coefficient budget is 100,000 terms across an evolved state. Exceeding either is a refusal to compile a larger problem, not a physical conclusion.

For a closed rational box, each univariate monomial factor is bounded exactly at its endpoints, also including zero for an even power across a sign change. Interval multiplication encloses each multivariate monomial, and summing these intervals encloses the whole polynomial. The resulting enclosure can be wider than the actual image because separate monomials can share variables. It is nevertheless a valid deterministic outer bound. This distinction is retained throughout the separation proof.

An independent two-step expansion verifies all symbolic coefficients. If the first input is two, the second zero, the source starts at zero and b is the unscaled heater, then

$$x_2 = 2g\{(1-\lambda)b - kLb + k\alpha vv^T b\}.$$

For any sensor c, the response is

$$2g\{(1-\lambda)c^T b - k\sum_{ab\in E}(c_a - c_b)(b_a - b_b) + k\alpha(c_u - c_v)(b_u - b_v)\}.$$

The test derives this polynomial separately and compares the complete sparse coefficient table with symbolic propagation, beyond merely checking samples.

5.3 Full-domain contrast certificates

`ParametricFaultFamily` specifies a polynomial observation vector and compact box. Left and right hypotheses have independent parameter assignments. Within one hypothesis those same parameters persist across all separately reset experiments. This is essential: forcing equal calibration between hypotheses would solve an easier problem, whereas independently redrawing calibration at each observation would throw away valid information.

For a contrast w and candidate-specific polynomial response Phi, first combine the coordinates symbolically into w^T Phi. This can cancel nuisance dependence before enclosure. If its left and right enclosures are [l_a,u_a] and [l_b,u_b], and each raw reading has independent error bounded by epsilon_i in either world, the support error in their contrast difference is

$$2\sum_i |w_i|\epsilon_i.$$

A positive value of

$$\max(l_a - u_b, l_b - u_a) - 2\sum_i |w_i|\epsilon_i$$

certifies separation on that entire product cell. Failed cells are bisected at a rational midpoint of their widest parameter coordinate. The certificate stores a binary tree of splits; its children share the midpoint boundary and together cover their parent without gaps. Strict margins cover those boundaries as well. `verify_family_certificate` reconstructs every split and replays every leaf enclosure and gap. A full separated result requires every leaf to pass.

SNG-NX04-01. If that checker accepts a separated certificate, the two complete declared observation-error families are disjoint. Proof follows by induction over the covering tree and the support inequality on each leaf. The proof does not rely on density of samples or on eventual subdivision convergence.

An overlap certificate instead supplies two complete parameter assignments and a common possible observation. The checker evaluates both polynomial maps and all error inequalities directly. Such witnesses may be supplied explicitly; the solver also

tests cell midpoints. It is not a complete algebraic intersection solver. A cell budget produces `unresolved` with every remaining cell retained. An actual zero-gap touching case is treated as overlap when its rational witness is available. Missing a difficult overlap witness never becomes a separation.

5.4 Scalar calibration example

For x'=a(alpha)x+gu with a=1/2-alpha/4 and 1<=g<=2, the two-reading transcript after pulse then wait is

$$(y_1, y_2) = (g, (1/2 - \alpha/4)g).$$

The nominal and alpha in [1/2,1] final-reading intervals overlap. In particular, nominal g=1 and faulty (g,alpha)=(4/3,1/2) both produce final value 1/2. The retained overlap certificate checks those independent nuisance assignments. Jointly, contrast y_1/2-y_2 equals alpha*g/4: it is zero nominally and at least 1/8 in the fault family. With error delta per reading, the two-world contrast error is at most 3delta. Thus delta<1/24 guarantees separation. The receipt replays a full-domain certificate at delta=1/100. No noisy division is needed.

There cannot be a fixed positive robust margin uniformly across alpha>0 all the way down to zero for a continuous finite panel: the faulty response converges to the nominal response. The declared positive-minimum-severity bands are part of the problem, rather than a technical afterthought.

5.5 Four-location pilot and interpretation

The retained four-by-four pilot uses bonds (4,5), (6,7), (1,5) and (5,6), with the original symmetric center sensor and pulse-then-wait words of horizons one through four. It covers the closed bands {0}, [1/2,3/4] and [3/4,1], using known k=1/8, lambda=1/100 and g=1. These singleton calibration coordinates are reported explicitly. The symbolic engine supports wider boxes, but no wider physical calibration guarantee is inferred from this pilot.

The two reflected exterior bonds have overlapping complete-domain families under that symmetric observation language. Several other pairs also overlap because the allowed severity can mimic the measured effect. Pairs involving the interior central bond obtain strict full-domain separation on the two positive bands. Every reported overlap has explicit rational parameters and every strict region has a covering certificate; samples independently replay source flux but do not establish the continuum claim.

Tests include a nonmonotone polynomial whose whole-domain interval touches the other family but whose two rational subcells prove strict separation. Tampering with the covering boundary is rejected. Independent scalar overlap, touching and intentionally unresolved fixtures exercise all three output semantics. Run `PYTHONPATH=src python scripts/thermal_severity.py --check` for the deterministic source-pinned replay, retained rational polynomials, covering trees and budgets.

Chapter 6

NX05: exact adaptive policies across independent resets

This track concerns a finite candidate population, finite scalar experiment pool and positive rational costs. Every selected experiment is independently reset, and its error is adversarial within its candidate-specific closed interval. Persistent nuisance learning and a continuing physical trajectory are different contracts handled by NX02 and NX06.

6.1 Exact finite abstraction of continuous observations

`ResetExperimentFamily` retains original candidate IDs, target labels, costs and an experiment-by-candidate table of closed intervals. A belief S is a tuple of possible original candidates. An observation y in experiment e updates it to

$$S_{e,y} = \{c \in S : l_{ec} \leq y \leq u_{ec}\}.$$

`observation_cells` collects every endpoint among these intervals. Each endpoint is a separate singleton cell. Consecutive endpoints bound an open cell, and the two exterior rays are retained as explicit model-rejection regions. Rational midpoints represent bounded open cells; one unit beyond the extreme endpoint represents each ray. Membership is constant on every open cell because interval membership can change only at an endpoint. Direct comparison at the endpoint itself gives the separate closed-boundary membership.

For [0,1] and [1,2], y=1 is compatible with both candidates. A branch compiler that omits this singleton would falsely certify guaranteed identification. Point, duplicate, nested and disjoint intervals are all covered by the same construction. Empty-consistency cells do not enter adversarial worst-case cost, but runtime replay returns `model_rejected` when an actual reading lands there.

6.2 Proper-policy theorem and Bellman recurrence

The error contract is memoryless across resets: conditional on a candidate, any value in each test's interval remains possible regardless of earlier numeric readings. Thus the set S is a sufficient belief. A terminal belief has one target label, even when several original configurations share that target.

Suppose a finite guaranteed policy chooses a test with a possible branch that retains

all of S. Its subtree on that branch must itself solve the same S problem, because no candidate or persistent nuisance has been eliminated. Use that subtree immediately, omitting the first test. It remains valid for all candidates and costs strictly less on every path through the deleted test. Repeating this deletion yields a policy whose every nonterminal child is a strict subset of its parent belief.

SNG-NX05-01. A finite worst-case optimum, if one exists under the declared memoryless interval contract, is attained by a proper strictly shrinking policy. Its depth is at most K-1 for K original candidates. Consequently the complete recurrence is

$$V(S) = 0 \quad (T \text{ constant on } S),$$

$$V(S) = \min_{e:\, S_{e,Q} \subsetneq S \ \forall Q} \left[c_e + \max_{Q: S_{e,Q} \neq \emptyset} V(S_{e,Q}) \right].$$

An experiment with a same-S branch is excluded, not assigned an optimistic finite value. If every experiment is excluded or reaches an impossible child, no finite guaranteed policy exists. The proof would fail for persistent gain or calibration, where numeric observations can constrain hidden parameters while leaving the candidate ID set unchanged; this solver does not claim that broader setting.

`minimum_reset_policy` memoizes exact beliefs, with a default 100,000-belief budget. Each candidate interval contributes at most two endpoints, so there are O(K) cells per experiment before equal child beliefs are consolidated. The number of possible beliefs remains exponential. A budget refusal is inconclusive and cannot be reinterpreted as impossibility.

6.3 Certificates, costs and independent replay

Every node records its complete original belief, selected experiment, exact endpoint cells, all nonempty children and worst-case subtree cost. Leaves retain the target and every remaining original ID. The certificate also stores all completed Bellman values. `verify_reset_policy` recomputes the continuous cell partition, checks every possible branch, rejects cycles, checks target-pure leaves and recomputes the exact maximum path cost. It also recomputes the finite Bellman optimum before accepting an optimality claim. Runtime `replay_policy` returns an identified target, a partial transcript awaiting more readings or explicit model rejection; an empty candidate set is not a successful diagnosis.

The four-candidate worked example has one test dividing AB from CD and two branch-specific tests that identify B or D. The adaptive policy costs two. Every fixed two-test panel leaves a pair ambiguous, while all three tests suffice. The one-test lower bound follows because each individual test has an ambiguity cell of size at least two. The tests independently enumerate finite trees to depth K-1 on twenty-five random rational interval instances and obtain the same optimum or impossibility result. That oracle directly evaluates endpoint inequalities and does not call the production cell compiler or Bellman routine.

6.4 Thermal comparison on identical contracts

The primary retained experiment uses the unchanged six source-derived 12-by-12 candidates and all 112 original protocols. Initial radius and sensor noise remain 1/100, with original per-candidate radii, costs, heater words and sensors. The static and

adaptive solvers therefore optimize the same physical and uncertainty contract. Both return exact worst-case optimum twenty. The adaptive engine does not claim an improvement where a single existing test already attains the minimum.

A second comparison uses the first eight original configurations from the 79-mask four-by-four pilot, twelve predeclared short protocols, zero reset radius and sensor error 1/100,000. Both static and adaptive exact-location identification are impossible within that declared pool. All original mask IDs are retained, and the failed ambiguity branches are part of the evidence. This negative result prevents an adaptive planner from appearing to solve a measurement-language limitation by merely choosing its tests in a new order.

Costs are worst-case sums of actual reset experiments. No prior is supplied, so no expected cost or branch probability is claimed. Repeated adversarial bounded-noise tests cannot be averaged as if their errors were random Gaussian samples. The receipt records the full policy, Bellman values and static result for both comparisons. Run `PYTHONPATH=src python scripts/adaptive_thermal.py --check`.

Chapter 7

NX06: experiments retaining physical history and original identity

This track provides an exact finite-world bounded-horizon planner and an exact small rational affine-conditioning backend. It distinguishes continuing a physical trajectory from buying independent reset experiments. It also records the original target even when actions erase the current distinction.

7.1 Original-world finite semantics

`HistoryWorld` consists of an original ID, immutable target and current state. `HistoryBelief` stores all possible such worlds and the action/observation transcript. `HistoryModel` specifies legal action names, per-world transition relations, measurement maps, positive exact action costs and reading costs. Transitions may be nondeterministic. An action missing from any currently possible world is infeasible, rather than a measurement that happens not to distinguish the candidates.

`advance_belief` applies the transition relation to every possible world while preserving its original ID and target. Equal current states with different original IDs remain separate. Duplicate paths reaching the same original ID, target and current state may merge because their future finite semantics are identical and no probabilistic path multiplicity is being inferred. `condition_belief` intersects with the exact displayed measurement equation. An empty intersection is model rejection, not identification.

For the erase fixture, original states zero and one have targets zero and one. Erasure maps both current states to zero. The correct belief still contains two original targets, and all later identity or erase actions retain the ambiguity. An observation before erasure identifies the target. The test independently enumerates every four-action word and compares literal original-world evolution with the belief updates; a current-state-only implementation would fail it.

7.2 Complete bounded-horizon policy search

`history_policy_search` considers every legal control action and every available measurement at each belief. A control has one successor belief containing all possible transitioned worlds. A measurement branches on every displayed possible output.

The adversary selects the most expensive possible branch; the solver chooses the cheapest action or measurement. Terminal beliefs have a single original target, and nonterminal zero-horizon beliefs are impossible.

SNG-NX06-01. The returned value is the exact minimax value among policies whose combined controls and readings have length at most the declared horizon. Proof is induction on that finite horizon: the first step must be one of the enumerated legal actions or measurements, and each successor is solved exactly at the preceding horizon. This is not an unbounded-history completeness claim. Positive costs do not themselves supply an infinite-history stopping theorem.

The cache key includes the complete current original-world possibilities and remaining horizon. The default state ceiling is explicit. The certificate checker regenerates every successor, verifies every possible observation branch, checks original-target leaves and maximum path cost, and recomputes the bounded optimum before accepting the optimality claim. Cyclic purported trees are rejected. No successful sample path substitutes for all-branch validation.

7.3 Exact rational affine conditioning

For small continuous worlds, `AffineHistoryWorld` stores

$$x_t = c_t + G_t z, \qquad H_t z \le h_t.$$

The latent labels distinguish persistent gain/calibration and each reset-local initial quantity. This representation is a constrained affine image, not an ordinary zonotope posterior. `latent_box_world` initializes exact [-1,1] bounds. For a known candidate matrix A and nominal heater b, with an optional affine heater-generator matrix B, control u gives

$$c_{t+1} = Ac_t + bu, \qquad G_{t+1} = AG_t + Bu.$$

This is exact when the uncertain heater dependence is affine in the same latents. The module does not silently linearize an uncertain transition matrix times an uncertain state, which would generally be nonlinear.

An observation y with sensor C and independently fresh bounded error epsilon adds the two inequalities

$$CG_t z \le y - Cc_t + \epsilon, \qquad -CG_t z \le -y + Cc_t + \epsilon.$$

Eliminating the fresh sensor-error variable into this strip is exact because each error may independently take any value in its closed interval. The persistent latent variables and all earlier strips remain. The rational polytope solver from NX02 returns a feasible latent witness or exact Farkas infeasibility certificate, subject to its row budget.

`affine_reset` replaces the current state while preserving original identity and all prior constraints on persistent latents. Fresh reset-local coordinates may be appended with new labels and independent box inequalities. Reusing a fresh label is rejected. Old latent coordinates may cease to affect the current state but are not confused with a new draw or with a changed original target.

7.4 Shared-gain example and robustness

For x'=a x+g u, a in {1/2,3/4}, g in [1,2] and x_0=0, the pulse-then-wait transcript is (g,ag). Observing y_1=6/5 and then y_2=3/5 leaves only a=1/2. The exact condition-

ing certificate preserves the gain learned at the first reading. A final-only value 9/10 is compatible with both candidates at different gains, so an independently redrawn gain or a current-output interval alone would lose the distinction.

The contrast y_2-y_1/2 is zero for a=1/2 and at least 1/4 for a=3/4. Error delta in each reading gives support at most 3delta/2 in either candidate world. Thus delta<1/12 is sufficient for robust joint separation. The receipt stores a strict rational contrast at delta=1/100 and a final-only overlap witness, each checked under independent nuisance values between hypotheses. Noncommuting pulse/wait order and a fresh reset-local variable are independent test fixtures.

7.5 Fair four-by-four thermal pilot

Three original models are used: nominal, left blocked bond and right blocked bond. All control transitions for the complete language of at most two inputs from {0,2} are built and independently checked by undirected neighbour flux. Left and right center sensors each cost two; waiting costs one; a heating step costs two including elapsed and heating; initial setup costs two. Initial state and measurement error are exactly zero for this finite-policy comparison.

The exact shared-history optimum is seven action/reading units plus the single setup cost two, total nine. The separately reset panel optimizer is supplied every word in precisely the same at-most-two-control language and the same two sensors. Its optimum is fourteen. The comparison does not grant the history planner a longer word or more powerful sensor. The saving comes from retaining the heated physical state between the two readings and paying setup/heating once. No measured laboratory savings or robustness outside the declared exact model are claimed.

A separately retained fixed schedule pulses once, waits twice and reads both sensors at every time. Its six readings cost eighteen when the trajectory is shared, versus forty-two for reproducing each reading as an independent reset experiment. This is a cost-accounting comparison, separate from the optimized two-control policy. The complete intermediate trajectories match the same source-independent flux oracle.

Run `PYTHONPATH=src python scripts/history_thermal.py --check`. Evidence includes the bounded policy, full original-world states and every branch, reset/cost convention, intermediate readings, shared-gain overlap/separation certificates, source digest and rational arithmetic/work limits.

Chapter 8

Capability purchase and the intervention language (NX07)

The accepted decision problem is finite. A capability is an explicitly priced sensor or actuator, an experiment has an immutable set of requirements, and each original candidate retains its identifier. Installation cost is paid once. Runtime cost is paid for each independently reset diagnosis. All costs below are exact accounting units. They are not measurements of hardware price, elapsed seconds, energy, or deployment demand.

For candidate indices i, j, let T_i be the requested target label and let an experiment return closed intervals $I_{ei} = [r_{ei} - \rho_{ei}, r_{ei} + \rho_{ei}]$. Its pair mask contains (i, j) exactly when

$$T_i \neq T_j, \qquad |r_{ei} - r_{ej}| > \rho_{ei} + \rho_{ej}.$$

Equality is insufficient: closed intervals touching at an endpoint admit a common observation. Models with the same target need not be distinguished. This coarsening never removes their original labels from the source population.

8.1 Activation theorem and exact optimizer

For capabilities K, experiment requirements R_e, one-time prices a_k, runtime prices c_e, and positive integer deployment horizon N, the objective is

$$\min_{z,x} \left[\sum_k a_k z_k + N \sum_e c_e x_e \right], \qquad x_e \leq z_k \quad (k \in R_e), \qquad \sum_{e:(i,j)\in S_e} x_e \geq 1 \quad (T_i \neq T_j).$$

Enumerate every installed subset $J \subseteq K$. Experiments with requirements outside J are unavailable. On the remaining experiments, the dynamic program maintains the least runtime cost for each covered-pair mask. Processing an experiment has two branches: skip it or include its mask and add its runtime cost. Induction on the processed experiment prefix proves that each state contains precisely the minimum cost of every panel producing that mask. The full mask therefore gives the minimum runtime cost under that installation. Taking the minimum of installation plus N times runtime over all J proves global optimality **within this finite catalogue**.

Empty masks can be omitted because their nonnegative costs cannot improve coverage or objective. The solver retains tied panels up to its explicit tie-reporting limit; `t`

`ies_truncated` reports any loss of tie enumeration without changing the proved optimal value. A node budget that interrupts installation or mask enumeration returns `optimality_unresolved` or `resource_limit`, never `optimal`.

A panel is a set of separate reset experiments. Combining simultaneous sensor outputs into one shared physical run requires a different joint-noise and cost contract. No such saving is inferred here.

8.2 Tiny independent acceptance oracle

Three candidates have states $(0,0),(1,0),(0,1)$. Sensors s_1, s_2 cost two to install and one to read; they measure individual coordinates. Sensor s_3 costs five to install and one to read and measures $x_1 + 2x_2$. Every reading has radius $1/10$. At $N = 1$, installing s_1, s_2 and installing s_3 both cost six and both are retained. At $N = 2$, their costs are eight and seven. Tests independently enumerate every installation bit vector and every experiment bit vector; that oracle does not call the dynamic program.

8.3 Synthetic thermal catalogue and all-word blindness

The 4-by-4 source-derived plate exposes four sensor purchases and one optional asymmetric heater. Every experiment uses one selected heater with a scalar amplitude word. The adapter does not overload scalar amplitudes as heater identifiers, and it does not simulate two simultaneous independently addressed heater channels. New heater and sensor installations are synthetic assumptions. The original six-candidate fixture is left intact.

For reflected blocked-bond candidates, the retained permutation P satisfies

$$A_b = PA_aP^{-1}, \qquad B_b = PB_a, \qquad x_{0,b} = Px_{0,a}, \qquad C_bP = C_a$$

under the original symmetric heater and symmetric corner/center sensors. If $x_b(t) = Px_a(t)$, then the chronological update gives

$$x_b(t+1) = PA_ax_a(t) + PB_au_t = Px_a(t+1).$$

Induction proves equality for every finite amplitude word, not merely the stored horizons. The receipt checks the exact permutation identities. It separately retains an actual local-sensor protocol that distinguishes the reflected candidates and recomputes its outputs by neighbour-flux stepping. Violation of sensor invariance alone is not used as a reachability proof.

8.4 Expanded fault population: a useful null result

The NX03 adapter supplies exactly 79 original configurations, including nominal, single removed bonds and two removed bonds among twelve declared bonds. At the declared horizon-six catalogue, four configurations are merged into the same response classes as others: there are **75 observable classes**. The exact identity-target optimizer therefore reports `unseparable`; the receipt lists the original indistinguishable pairs.

A second, explicitly labelled target is the maximal distinction this finite catalogue can express: each configuration is labelled by its complete catalogue response vector. Its Pareto frontier compares installation cost and per-diagnosis cost at $N \in$

$\{1, 2, 10, 100\}$. Every original fault ID and mask remains reconstructable in the receipt. This quotient frontier is not sold as original-fault identification or an all-word impossibility theorem. Longer horizons, additional sensors and different permitted protocols may split these classes.

The replay script is `python scripts/nx_decision_recovery.py --track NX07 --check`. Its retained file includes the complete source catalogue, target semantics, exact prices, all reported optima, finite null witnesses and original-mask reconstruction. Integrity hashes bind source bytes; independent enumeration and exact replay establish the finite optimization claim.

The optional asymmetric heater is a substitute installed actuator. In this synthetic catalogue its injection vector replaces the original heater vector, so it bypasses the original `heater_half` source fault instead of assuming that fault also halves newly installed hardware. Experiments on the original heater retain the original gain fault. This is an explicit hardware assumption of this finite demonstration.

Chapter 9

NX08: stable target recovery

This note covers the mathematics behind `src/singularity/target_recovery.py`. The tests are in `tests/test_target_recovery.py`, and the retained receipt is `results/target_recovery.json`, which `scripts/target_recovery.py --check` replays. The setting is finite-dimensional, linear, rational and deterministic. Noise lies in a declared ellipsoid. The note makes no statistical, calibrated-physical or nonlinear-global claim.

The question is this. When sensors determine a requested target exactly in the absence of noise, how much can bounded measurement error move the recovered target? Two sensor layouts of equal rank can answer this very differently, so rank alone cannot rank designs.

9.1 1. Objects and conventions

- **State.** A state is a vector x in Q^d. The admissible states are x = x_c + Qz, where Q is a rational d×k matrix of full column rank and z is unrestricted. A zero-dimensional admissible set (k = 0) is allowed.
- **Observations and target.** The observations are y = Hx + η, with H an m×d matrix. The target is t = Gx, with G a p×d matrix. The known centres Hx_c and Gx_c are subtracted. Write H_0 = HQ and G_0 = GQ.
- **Noise.** η satisfies η^T R^{-1} η ≤ ε^2. Here R is the positive-definite ellipsoid shape matrix, with default I, which is the convention in `INTERCHANGE_CONTRACT.md`. Write W = R^{-1} and $\|\eta\|^2_W$ = η^T W η.
- **Target error.** Target error is $\|e\|^2_P$ = e^T P e for a positive-definite P, with default I.
- **Forms.** A = H_0^T W H_0 and B = G_0^T P G_0 are positive semidefinite (PSD) k×k matrices. For α ≥ 0, S(α) = αA − B.

Definition. The squared target amplification is

$\kappa^2 = \min\{\alpha \geq 0 : S(\alpha) \succeq 0\}$, and $\kappa^2 = +\infty$ when no such α exists.

The set of admissible α is closed and upward-closed, so the minimum exists when the set is nonempty. When some z has $H_0 z \neq 0$, κ^2 equals sup $\|G_0 z\|^2_P$ / $\|H_0 z\|^2_W$ over such z. When $H_0 = 0$ and $G_0 = 0$, $\kappa^2 = 0$. This is the convention for "no admissible variation of the target", and it avoids a division by zero.

Invariance. κ* depends only on three things: the admissible subspace image(Q), the target metric P and the noise metric R. Replacing Q by QT for an invertible T maps z to T^{-1}z and leaves every ratio unchanged. An invertible state coordinate change x' = Tx sends H to HT^{-1}, G to GT^{-1} and Q to TQ, so H_0 and G_0 do not change.

No state norm appears anywhere. Changing observation or target units does change R or P, and therefore κ*. This is why units and metrics are stored in every receipt. The test `test_basis_invariance_and_whitening` checks basis invariance. It also checks that R = diag(4, 9, 1) is identical to dividing the observation rows by 2 and 3.

9.2 2. Exact theorems

Theorem 1 (kernel criterion). The following are equivalent: (a) $\kappa^* < \infty$; (b) $\ker H_o \subseteq \ker G_o$; (c) $G_o = LH_o$ for some p×m matrix L.

Proof.

- (b)⇔(c) is the row-space statement: every row of G_o is a combination of rows of H_o exactly when every vector annihilated by H_o is annihilated by G_o.
- (a)⇒(b): suppose some n has $H_o n = 0$ and $G_o n \neq 0$. Then n^T S(α) n = $-\|G_o n\|^2_P < 0$ for every α, so no α works.
- (b)⇒(a): let Z be a basis of row(H_o), the orthogonal complement of ker H_o = ker A. Write any z as Zw + n with n ∈ ker $H_o \subseteq$ ker G_o. Then z^T A z = w^T A_r w and z^T B z = w^T B_r w, where A_r = Z^T A Z is positive definite and B_r = Z^T B Z.
- S(α) $\succeq$ 0 exactly when α ≥ λ_max(A_r^{-1} B_r), so κ^{*2} is finite. □

If the kernel condition fails, the code returns `unobservable` with an exact hidden-target witness: a primitive integer z with $H_o z = 0$ and $G_o z \neq 0$. In words, the admissible states x_c and x_c + Qz produce identical noiseless readings but different targets.

Theorem 2 (no decoder beats εκ*). Let D be any map from observations to target estimates, linear or not. Then sup over admissible x and $\|\eta\|_W \leq \varepsilon$ of $\|D(Hx + \eta) - Gx\|_P$ is at least ε·κ*. If the kernel condition fails, the error is unbounded.

Proof.

- Take z with $\|H_o z\|_W \leq \varepsilon$.
- The state x_c + Qz with noise $-H_o z$ and the state x_c − Qz with noise $+H_o z$ both produce the reading Hx_c. Both noises are admissible.
- Their targets differ by $2G_o z$. Whatever D outputs, it is at distance at least $\|G_o z\|_P$ from one of the two targets.
- Taking the supremum over z gives εκ*. If z is a hidden-target witness, scaling z makes the gap arbitrarily large while the readings stay identical. □

`adversarial_witness` builds exactly these two states and noises, and replays them through a supplied decoder.

Theorem 3 (the minimum-norm decoder attains εκ*). Assume the kernel condition. Let Z be as above, H_Z = H_oZ and

L* = G_o Z (H_Z^T W H_Z)^{-1} H_Z^T W.

Then $LH_o = G_o$, and $\|L\eta\|_P \leq \kappa^*\|\eta\|_W$ for every η. The estimate $\hat{t}$ = Gx_c + L(y − Hx_c) therefore has error at most εκ.

Proof.

- **Decoder identity.** For z = Zw + n we have $H_o n = 0$, so L*H_oz = G_oZ(H_Z^T W H_Z)$^{-1}$(H_Z^T W H_Z)w = G_oZw. This equals G_oz because $G_o n = 0$.
- **Gain bound.** Decompose η = H_Z w + η_⊥ with η_⊥ W-orthogonal to image(H_o). Then L*η = G_oZw, and $\|\eta\|^2_W = \|H_Z w\|^2_W + \|\eta_\perp\|^2_W \geq \|H_o Z w\|^2_W$.
- Hence $\|L^*\eta\|_P = \|G_o Z w\|_P \leq \kappa^*\|H_o Z w\|_W \leq \kappa^*\|\eta\|_W$. □

L* is the rational form of $G_0H_0^+$ in the whitened geometry. It never forms a floating pseudoinverse.

Arbitrary decoders. Any decoder L with $LH_0 = G_0$ has gain at least κ^2, by Theorem 2. It can be strictly worse, because it may amplify noise that lies off the observation image. *`target_identifiability`* also returns a row-space decoder, which expresses each target row in the independent rows of H_0. That decoder is valid, but it is not optimal. In the retained two-row example, its exact squared gain is in [13.090169…, 13.090170…], while κ^2 is in [3.0962912…, 3.0962919…]. A decoder identity therefore never certifies a noise guarantee by itself. `certify_decoder_norm` certifies $\beta^2 W - L^T P L \succeq 0$ for the chosen L. `recover_target` refuses a certificate that fails to verify.

Single target row. When p = 1, B_r has rank one, so $\kappa^2 = g Z A_r^{-1} Z^T g^T \cdot P_{11}$ is rational. Any nonzero column of A_r^{-1}B_r is an exact maximizer, and the bracketing routine always returns *`exact_sharp_constant`* in this case. With two or more target rows, κ^2 is generally an algebraic number of degree up to min(rank H_0, p).

9.3 3. Exact PSD certificates with zero pivots

Strict positive-definiteness tests are the wrong tool here. These include DynaComp `stability.positive_definite` and the LDL inside `experiment_design.RationalMetric`. Invisible nuisance directions force $S(\alpha)$ to be singular, and $S(\kappa^{*2})$ is always singular. `psd_certificate(S)` performs symmetric elimination in the given order.

- **Positive pivot.** It is eliminated, and the Schur complement of a PSD matrix is PSD.
- **Zero pivot.** It is accepted only if its whole remaining column is zero. Every 2×2 principal minor $s_{kk} s_{ii} - s_{ik}^2$ of a PSD matrix is nonnegative, so a zero diagonal entry forces a zero row. The pivot then contributes $D_k = 0$ and an identity column of L.
- **Negative pivot, or zero pivot with a nonzero entry s_ik.** The routine builds an exact vector v on the remaining coordinates with $v^T C v < 0$. For a negative pivot, v = e_k. Otherwise, v = te_k + e_i with $t = -(|c_{ii}| + 1)/(2c_{ik})$, which gives $-(|c_{ii}| + 1) + c_{ii} < 0$. The routine then lifts v through the eliminated positive-pivot block by solving $S_{II} z_I = -S_{IJ} v$, so that $z^T S z = v^T C v < 0$. The lifted vector is re-evaluated exactly before it is returned.

Soundness. The certificate is $S = L \operatorname{diag}(D) L^T$, with L unit lower triangular and $D \geq 0$. This implies PSD with no further argument. `verify_psd_certificate` and the receipt verifier check only this identity. They never rerun the elimination.

Completeness. It follows from the two facts above. Section 8 lists the independent oracles that check it.

9.4 4. Rational brackets for κ^{*2}

`bracket_target_amplification(problem, tolerance, max_steps)` returns one of four statuses:

- `unobservable`, with a hidden witness;
- `exact_sharp_constant`, where lower = upper, with an attaining witness and a PSD certificate;
- `bracketed_sharp_constant`, where upper − lower ≤ tolerance;

- `resource_limit`, a valid bracket that is still wider than the tolerance after `max_steps` PSD tests.

It never reports infinity as a number. The routine has four parts.

1. **Initial upper bound.** It uses trace(A_r^{-1}B_r). Every eigenvalue is nonnegative, so this is at least λ_max. S(trace) is then certified, which must succeed.
2. **Lower witnesses.** It evaluates exact ratios of candidate directions: coordinate directions of row(H_0), exact columns of A_r^{-1}B_r, and a few power-iteration directions rounded to small denominators. Rounding only affects which direction is tried. The ratio is always computed exactly, and any exact ratio is a valid lower bound.
3. **Bisection.** It tests the simplest rational in the middle third of the bracket. PSD success lowers the upper bound. Failure yields z with z^T S(mid) z < 0, that is, ratio(z) > mid, which raises the lower bound to ratio(z).
4. **Snapping.** Let c be the simplest rational in the bracket. If S(c) is PSD and some null vector v of S(c) has v^T A v > 0, then v^T B v = c·v^T A v, so $\kappa^{*2} = c$ exactly. Both facts are verified exactly. This is how the $\tau = 3$ forecast target closes at exactly 5/2.

9.5 5. Sensor layouts (NX08-B)

A `SensorOption` has three parts: one or more rows, a positive installation cost, and a positive-definite noise-shape block. A layout stacks the chosen options in index order and uses a block-diagonal R. The noise contract is the joint ellipsoid η^T R^{-1} η ≤ ε^2 over all installed rows. It is not an independent per-sensor box. A box is a different support-function problem that this module does not compute.

Monotonicity lemma. Adding a sensor adds the PSD term H_s^T R_s^{-1} H_s to A. So S(α) for a superset is at least S(α) for the subset, κ^{*2} cannot increase, and the kernel condition is preserved.

Consequences:

- The decision "$\kappa^{*2} \le \kappa_{max}^2$" is exactly the LDL test of S(κ_{max}^2) for one layout. No bisection is needed.
- In the pruned search, a node is pruned when its cost already exceeds the incumbent.
- A node is also pruned when the node plus all remaining options is infeasible. By monotonicity every completion then fails.
- A feasible node is a leaf, because every extension costs strictly more.

Ties are broken by (cost, number of sensors, sorted indices). The rank objective `cheapest_identifying_layout` and the conditioning objective `cheapest_conditioned_layout` are separate functions.

Retained six-option catalogue. The target is (0,1,0). Costs are normalized installation units.

κ_{max}^2	optimum (pruned = exhaustive = brute force)	cost	κ^{*2} of optimum
kernel condition only	cheap_x1 + cheap_x1_weak_x2	2	5000
5000	cheap_x1 + cheap_x1_weak_x2	2	5000

κ_{max}^2	optimum (pruned = exhaustive = brute force)	cost	κ^{*2} of optimum
5, 12, 100	pair_x2x3_x3	3	5
2	pair_x2x3_x3 + cheap_x3	4	9/5
1, 3/2	direct_x2	5	1
2/3	direct_x2 + pair_x2x3_x3 + cheap_x3	9	9/14
1/2	infeasible	—	—

The cheapest identifying layout has rank 2, the same rank as {cheap_x1, direct_x2}. Its amplification is 5000 instead of 1. At a demanded noise radius ε, its target error guarantee is $\sqrt{5000}\cdot\varepsilon \approx 70.7\varepsilon$, so it is unusable at any tolerance tighter than that. The full 64-layout table and its brackets are in the receipt. Every one of the 54 identifiable layouts closes at an exact rational, because the target is a single row.

9.6 6. Forecast and thermal integration (NX08-C)

For a fixed, known linear transition F, the problem has three parts:

- **Observations.** Rows C_j F^t of the initial state.
- **Target.** G_τ = G F^τ.
- **Known inputs.** A known input contributes a subtracted centre.

Uncertainty in F itself is a different problem, and this module does not bound it.

Synthetic forecast. The state is (position m, velocity m/step, sensor bias m, decaying mode m), with F = [[1,1,0,0],[0,1,0,0],[0,0,1,0],[0,0,0,1/2]] and the sensor reading position + bias + mode.

- Without a bias constraint. The target is unobservable. The hidden witness (1,0,−1,0) means that position +1 with bias −1 leaves every reading unchanged.
- With bias fixed at its calibrated centre and observations at steps 0–3. The position forecast has exact κ^{*2} = 1069/46, 247/23, 149/46, 17/23, 149/46, 247/23 for τ = 0,...,5. It is smallest inside the observation window and grows when extrapolating.
- Two-row target. The (position, velocity) target gives algebraic values bracketed to 10^{-6}. One exception is τ = 3, where the value is exactly 5/2.
- Adversarial replay. At every horizon the minimum-norm decoder's error equals the lower-witness ratio times the noise. The bound is therefore tight up to the bracket width.

Thermal plate target. The model is the exact nominal 4×4 plate from `thermal_diagnostics.build_plate_candidates(4)`, with zero heater input. The receipt binds the model's SHA-256 digest and the generating call. All quantities are in K_model excess-temperature units, and time is in model steps.

- Admissible initial states. They are spanned by four modes: centre, edge ring, corners and the left–right antisymmetric mode.
- Target. The target is the plate-mean temperature after 4 steps.
- Observability. With all four sensors at step 0 alone, the target is unobservable.
- Chosen layout. corner_mean and center_mean read at steps 0, 1 and 2. The antisymmetric mode is in ker H_0, so it is invisible. It is a harmless nuisance,

because the target is symmetric. The exact $\kappa^{*2} \approx 0.69912$ is rational, as for any scalar target.
- Decoder comparison. The row-space decoder has squared gain ≈ 5.1447 on the same data, which illustrates the unnecessary off-image amplification.
- Thermal sensor catalogue. Each catalogue option is one sensor read at steps 0–2, with costs 1, 1, 2 and 2. The cheapest layouts are:
  - corner_mean alone (cost 1, $\kappa^{*2} \approx 156.17$) for $\kappa_max^2 = 200$;
  - corner_mean + center_mean (cost 2) for $\kappa_max^2 = 1$ or 3;
  - corner_mean + center_mean + left_center_mean (cost 4) for $\kappa_max^2 = 1/2$.

When the bounds are conservative. The bound $\varepsilon\kappa^*$ is the exact worst case over the declared ellipsoid, and the adversarial states attain it. It is conservative in three cases. First, it is conservative for a decoder other than L*. Second, it is conservative for noise that is not adversarial: a typical random draw is far smaller than the worst case. Third, the model or ellipsoid may be a conservative envelope of the physics. A bracket that is not exact adds at most its width.

Link to NX09. NX09 (fidelity selection) can call `bracket_target_amplification` or `cheapest_conditioned_layout` on each model fidelity's (H, G, Q) to price measurement design by target error. When it does, any fidelity discrepancy must be added as a separate bounded term in the same target units. This module does not establish calibrated physical performance. It also does not establish nonlinear global identifiability: a bounded or curved admissible set can behave differently from the unrestricted affine theorem used here.

9.7 7. Public API (for NX09, NX25, NX27, NX29)

Function	Purpose
`TargetRecoveryProblem(observations, target, admissible_basis=None, observation_metric=None, units=None, *, centre, target_metric, state_dimension, admissible_dimension)`	Validated exact problem. Inputs keep their order.
`target_identifiability(problem)`	Returns `identifiable` with a row-space decoder L ($G_0 = LH_0$), or `unobservable` with a hidden witness. Also returns the rank, the pivot rows and a basis of $\ker H_0$ (the nuisance directions).
`certify_target_amplification(problem, alpha2)`	`certified_upper_bound` with an LDL certificate, or `refuted` with a witness z whose ratio exceeds alpha2.
`bracket_target_amplification(problem, tolerance, max_steps)`	`AmplificationBracket`; see the statuses in Section 4.
`minimum_norm_decoder`, `certify_decoder_norm`, `bracket_decoder_norm`, `recover_target`	Optimal decoder, decoder gain certificate and certified estimate. `recover_target` gives the estimate together with the error bound $\beta^2\varepsilon^2$.
`adversarial_witness(problem, z, decoder=None)`	Theorem 2 replay.
`target_recovery_receipt`, `verify_target_recovery_receipt`	`singularity.target-recovery/1`. Rationals are stored as `[numerator, denominator]`, and matrices carry explicit shapes. The receipt also stores the units, the R/P semantics, the problem and Q digests, the LDL factors, the witnesses and the decoders. The verifier uses only matrix products and Bareiss determinants.
`SensorOption`, `layout_problem`, `cheapest_identifying_layout`, `cheapest_conditioned_layout`, `conditioning_frontier`, `layout_table`	NX08-B. At most 12 options.

`linear_forecast_problem`, `thermal_target_proble m`, `matrix_power` NX08-C adapters.

The admissible dimension is limited to 16 and catalogues to 12 options, as the packet requires. The receipt name `singularity.target-recovery/1` is new. It is not the P03 `relational-height/1` format or the `singularity.certificate/1` format, and the verifier rejects either schema string.

9.8 8. What is proved, computed and open

- **Proved here (human proofs).** Theorems 1–3, PSD certificate soundness and completeness, the monotonicity lemma and the correctness of pruning, and the rationality of κ^{*2} for a single target row.
- **Exact finite computation.** The oracle population and every retained example are in `results/target_recovery.json`:
  - 600 tiny symmetric matrices, 539 of them PSD and 376 of those with a zero pivot, where the LDL routine and an all-principal-minors oracle agree;
  - optimum agreement among the pruned, exhaustive and brute-force catalogue searches, where the brute force uses raw rows and minors;
  - 14 targeted receipt tamperings, all rejected;
  - 10 receipts replayed.
- **Not covered.** Box or per-sensor noise and other support-function geometries; bounded or curved admissible sets; uncertain transitions F; stochastic or calibrated noise models. None of this has formal (Lean) coverage.

Chapter 10

Experiment, rank and executable arithmetic selection (NX09)

A computational envelope and a physical observation uncertainty play different roles. The centre produced by an approximate evaluator need not equal the exact full-model reading. Its certified computational radius encloses that difference. Sensor error surrounds the actual physical observation. With all terms in the same output units and at the same chronological sample,

$$I_{e,c,f} = \widehat{r}_{e,c,f} + [-\rho_{\text{phys}} - \rho_{\text{comp}}, \rho_{\text{phys}} + \rho_{\text{comp}}], \qquad \rho_{\text{comp}} \le \rho_{\text{source}} + \rho_{\text{transformation}} + \rho_{\text{reduction}} + \rho_{\text{execution}}.$$

Each summand is nonnegative and is retained individually in source evidence. Triangle inequality proves this composition. It does not prove sharpness, statistical independence, calibration of a synthetic source, or attainability of every envelope endpoint. Initial-state/process uncertainty is included in the declared source domain; adding it again as an unexplained radius would double count it.

10.1 Typed binding and supported arithmetic

`BoundedReading` carries centre, physical radius, computational radius, source IDs, units, horizon and domain ID. `EvaluatorOption` additionally carries candidate and sensor identities, actual retained rank, executable arithmetic protocol, setup key, exact setup/run costs and a bound receipt. The constructor rejects orphaned IDs, stale horizons, unit/domain mismatch, unsupported executable protocols, negative radii and conflicting shared setup bindings.

`DynaCompAdapter` is optional and loaded only when requested. It hashes the names and bytes of every Python module in the selected upstream package. Relative imports run under an isolated package name, so the core has no import-time third-party dependency. The handoff contains the pinned source tree. Every replay rechecks its digest; changing source bytes invalidates an old adapter binding.

The implemented upstream evaluator routes are exact rational reduced dynamics and DynaComp's **integer-exact-dot, one RN-even binary64 rounding per row** executor. This is not NumPy, BLAS, an arbitrary sequential float loop, GPU arithmetic or user-selectable p-bit arithmetic. Generic precision remains unsupported. The typed menu reserves the separate sequential protocol identifier, but this adapter refuses it until a native contract is explicitly supplied.

For every candidate/sensor/fixed-input domain the adapter rebuilds `uniform_enrichment`, replays its exact sharp-box certificate, and retains E, G, H, D. Requested rank

is a budget; actual retained rank is authoritative because an exact low-rank response may close early. Exact reduced dynamics are

$$z_0 = Ex_0, \qquad z_{t+1} = Gz_t + Hu_t, \qquad \hat{y}_t = Dz_t.$$

The binary64 option then calls and replays `certify_execution` on these same matrices and this same domain. Its centre is extracted from the actual certified bit executor, not converted from an unrelated Python float. Reduction and execution bounds at the queried output time are summed. Source and transformation errors are zero only because these examples directly use the declared discrete rational source without an additional source transformation.

10.2 Finite optimizer and setup theorem

For each physical experiment, choose either no experiment or one compatible evaluator for each candidate. The robust pair mask follows the same strict interval-separation rule as NX07. A plan's objective is

$$\sum_{s \in \text{activated setups}} a_s + N \sum_{e \in \text{panel}} \left(c_e + \sum_c b_{e,c,f_{e,c}} \right).$$

The current accounting model recomputes the selected evaluations per diagnosis. Setup costs are paid once per exact setup key. Two options may share a key only if their setup cost and complete setup-binding fingerprint agree. Binding includes the source, output projection, domain, arithmetic and upstream bytes; rank alone is not a cache key.

The search enumerates skip/assignment branches for every experiment. At a partial node, all future costs are nonnegative, so a cost exceeding an incumbent cannot improve the objective. A full target mask is feasible and may terminate that branch because extra evaluations cannot lower cost. Induction on the finite branch tree proves that exhaustion gives the minimum objective among the supplied finite menus. Search exhaustion returns an unresolved status with an incumbent when available. This is not a global theorem that the chosen reduction rank is best among every possible subspace.

10.3 Independent evidence and reference plans

The tiny example has centres zero and $1/2$, physical radii $1/20$, cheap computational radii $1/4$, and accurate radii $1/20$. The cheap close experiment cannot certify separation because $1/2 \leq 3/5$. A wider physical test with centre difference one can use the cheap evaluator. Tests enumerate every evaluator assignment and panel by independent plain loops and check exact costs, strict equality refusal, target coarsening, stale source rejection and shared-setup consistency.

The thermal receipt checks two actual ranks, exact and binary64 execution, two candidates and every vertex word in $\{0,1\}^2$: 32 independently bounded executions. Each full-model output is recomputed by conservative neighbour-flux stepping, and every deviation lies within the retained envelope. Rank-two requests that close at rank one are represented honestly.

The diagnostic menu uses the nominal and heater-half candidates, two physical horizons, candidate-specific evaluator assignments and explicit setup prices. It retains

deployment plans for 1, 2, 10 and 100 diagnoses and compares full-exact and same-rank references. These prices are declared accounting units. The relation

$$S_1 + NQ_1 < S_2 + NQ_2$$

is an exact repeated-query break-even test; it is not inferred from one warm benchmark. The complete menu, source-bound upstream packets and selected envelopes are retained for replay. A loose bound means `not_certified`, not that the physical candidate is refuted.

Run `python scripts/nx_decision_recovery.py --track NX09 --check` and `python -m unittest tests.test_fidelity_design`.

10.4 Retained measured setup and query work

`results/nx_decision_benchmarks.json` separately records actual source-bound certification/setup, executor preparation, cold total and repeated warm query medians from `perf_counter_ns`. It uses the same two candidates, sensor, input word and accepted projection packets as the decision demonstration. Full exact reference queries call the original sparse simulator. Rank-one/rank-two exact and certified binary64 routes are all measured; no timing becomes an exact rational price in the optimizer.

For observed setup S, warm-query time Q and direct-query time D, the recorded first strictly cheaper repeated-query count is

$$N_{\text{break}} = \lfloor S/(D-Q) \rfloor + 1 \quad \text{if } Q < D,$$

and is absent when there is no positive runtime saving. The chosen rank-one binary64 evaluator can require hundreds of diagnoses to amortize certificate setup on this machine, while exact reduced evaluation amortizes sooner. These are machine-specific measured comparisons, including setup, rather than a universal acceleration theorem. The declared-price plan separately records its exact break-even, which is already favorable at the first diagnosis in this finite menu.

The same receipt includes eight vertex checks of one nondegenerate two-input box $[0,1]^2$ at two actual rank budgets. Upstream uniform reduction and binary64 execution are certified on the entire box; each vertex is independently stepped through the full rational source. This is distinct from the fixed-input domains used by the optimizer.

Actual upstream options are replayed at optimization entry, not merely checked for matching labels. Replacing a certified computational radius by zero while keeping its source IDs would otherwise create an unsupported separation. A dedicated regression rejects that forgery. Handwritten toy menus remain explicitly declared mathematical envelope assumptions; they are not promoted to physical or upstream numerical certificates.

Chapter 11

All-horizon stopping certificates (NX10): response tails

Status. NX10-A remains complete. NX10-B and NX10-C now have separate acceptance tests and retained decision receipts in `results/nx_10_decision_recovery.json`; the original response-tail receipt is preserved. The all-horizon claims below are restricted to the declared fixed protocol, reset and scalar reading semantics.

11.1 Setting

A step experiment applies a fixed input from a declared reset state:

$$x_{k+1} = Ax_k + b, \qquad y_k = cx_k + e.$$

When $I - A$ is invertible, the unique fixed point is $x^* = (I - A)^{-1}b$, and

$$y_k = y_\infty + cA^k(x_0 - x^*), \qquad y_\infty = cx^* + e.$$

Contraction is tested on A itself, never on the affine lift, which always has eigenvalue one.

11.2 Theorem (response tail)

Suppose:

- P is symmetric positive definite;
- $\gamma^2 P - A^\mathsf{T} P A$ is positive semidefinite;
- $\rho^2 \geq \gamma^2$ and $\rho < 1$.

Then for every $k \geq T$,

$$|y_k - y_\infty| \leq K\rho^{k-T}, \qquad K \;\geq\; \|c\|_{P,*}\,\|x_T - x^*\|_P,$$

where $\|v\|_P^2 = v^\mathsf{T} P v$ and $\|c\|_{P,*}^2 = cP^{-1}c^\mathsf{T}$.

Proof. The semidefinite condition says $\|Av\|_P^2 \leq \gamma^2\|v\|_P^2 \leq \rho^2\|v\|_P^2$ for every v. Induction gives $\|A^j v\|_P \leq \rho^j \|v\|_P$. By Cauchy–Schwarz in the P-inner product, $|cw| \leq \|c\|_{P,*}\|w\|_P$. Apply this to $w = A^{k-T}(x_T - x^*)$, using $y_k - y_\infty = cA^{k-T}(x_T - x^*)$. The constant K is a rational upper enclosure of the square root, computed by `sqrt_up`. □

In plain words: once a metric certifies that the error shrinks by a fixed factor each step, every future reading stays inside an explicit band around its limit. A finite prefix plus this band therefore describes every horizon, not just the computed ones.

11.3 Exactness and certification

The certificate is exact in the following senses.

- **Semidefinite test.** The condition is tested exactly by clearing denominators and running fraction-free Bareiss elimination on the integer matrix $sg^2P_I - tA_I^{\mathsf{T}}P_IA_I$.
- **Proposed metrics.** When P is not supplied, it is proposed by exact dyadic Smith doubling and then accepted only by that exact test.
- **Refusals.** Marginal or unstable matrices, such as eigenvalue ± 1 or a diagonal block with value one, are refused with `not_schur_stable`. A failed proposal is reported as `proposal_failed`, not as instability.

11.4 Evidence

`tests/test_horizon_design.py` compares the bound with independent forward iteration and exact closed forms:

- $A = 1/2$: $y_k = 2(1 - 2^{-k})$, with $K = 2$, which is sharp;
- $A = -1/3$: $y_k = \frac{3}{2}(1 - (-\frac{1}{3})^k)$;
- the non-normal matrix $\begin{pmatrix} 1/2 & 10 \\ 0 & 1/2 \end{pmatrix}$. Its free response $y_k = 10k\,2^{1-k}$ rises from 0 to 10 before decaying, and the certified band covers that transient.

The tests also cover tamper rejection: $\rho^2 < \gamma^2$, a failed contraction, a P that is not positive definite, a metric certified for a different matrix, and an altered serialized ρ. `scripts/horizon_design.py --check` retains the steady outputs, ρ, K and the bound check for the first 40 readings.

11.5 Limitations

- **Experiment type.** Only fixed, constant-input step experiments with independent resets are covered.
- **Excluded inputs and histories.** Adaptive or switched inputs and shared physical histories are out of scope.
- **Conservatism.** For non-normal A, ρ can be much closer to one than the spectral radius, which makes $K\rho^k$ conservative.
- **Formal status.** None of this is formalized in Lean.

11.6 NX10-B: decision-level proof and independent scalar examples

A checked prefix and a proved future bound perform different jobs. For minimum elapsed cost, the first strict separating index is globally cheapest in that fixed family because `IndexCost.per_index` is positive. Every earlier index has been checked exactly. An external incumbent accounting cutoff can only be used when that incumbent is already known feasible; the bare price is not itself a physical feasibility proof.

For maximum certified margin, write

$$m(k) = |\delta(k)| - \rho_a(k) - \rho_b(k).$$

After exact states at index T, the tail engine bounds all later deviations by $U_{\text{after}}(T)$. Together with a lower bound on the declared future radius envelopes,

$$\sup_{k>T} m(k) \le |\delta_\infty| + U_{\text{after}}(T) - \inf_{k>T}(\rho_a(k) + \rho_b(k)) =: F(T).$$

If $F(T)$ is no greater than the best checked prefix margin, the global optimal value is attained by a recorded prefix index. A strict inequality also excludes later ties. If $F(T) - m_{\text{best}} \le \varepsilon$, the retained finite index is epsilon-optimal even when no finite exact maximum exists. Its receipt says `supremum_bracket`, not attained optimum.

The first independent scalar example has steady outputs both equal to one and difference magnitude $2^{-k} - 3^{-k}$ for $k \ge 1$. Its successive difference is

$$-\tfrac{1}{2}2^{-k} + \tfrac{2}{3}3^{-k} < 0 \quad (k \ge 1),$$

so the exact global maximum is at index one. At $k \ge 6$, the independent inequality $2^{-k} + 3^{-k} < 1/50$ proves late-time impossibility with radius $1/100$ under each candidate. The tests verify both the optimal finite index and the late-time stopping receipt.

A second example has difference $1 - 2 \cdot 2^{-k} + 3^{-k}$. It is below one at every finite index and tends to one. The retained response is therefore an epsilon-optimal finite experiment with explicitly unproved attainment. No finite-prefix heuristic is used to assert the supremum.

The optional pair quotient is derived from a finite exact source system and checked against that source for enough terms to invoke Cayley–Hamilton. For source dimensions n_a, n_b and quotient order r, the difference between source and quotient sequences obeys a monic recurrence of order at most $n_a + n_b + r$. That many consecutive exact zeros imply all-index equality. A fitted recurrence without this source-derived identity check would be insufficient. In the reflected blocked-bond symmetric sensor example, the checked quotient has order zero and certifies exact equality for the fixed protocol family.

Radius envelopes are declared upper bounds for uncertainty. A `never_separating_after` conclusion means that no later reading meets this certified separation test. Physical overlapping-box impossibility is asserted only through the separately attainable sensor component and the receipt's `indistinguishable_all_indices` flag. A loose computational or process upper bound alone does not establish actual physical ambiguity.

11.7 NX10-C: source-derived thermal blocks

The additional receipt considers at most four candidate pairs on the synthetic 4-by-4 plate and two fixed chronological blocks, $(1, 0)$ and $(1, 1, 0)$. Each block compiler records phase and uses its exact affine block map. Every recorded short output is independently checked by conservative neighbour-flux stepping. Tests also exercise every within-block phase on the nominal, heater-half and two reflected blocked-bond candidates.

For these symmetric nonnegative source stencils, a row-sum bound suggests $P = I$ and norm factor at most $99/100$ per update. The implementation still performs the exact matrix contraction acceptance test. Block factors use $(99/100)^L$, with

the squared metric convention retained. Individual stability is never promoted to arbitrary-switching stability.

The receipt retains feasible finite decisions and inconclusive tails. Every result states actual prefix and metric effort. The comparison has no calibrated seconds, continuum-error claim, arbitrary-input optimum, multitime-panel guarantee or adaptive-policy guarantee. It accompanies the finite pool with a separately proved fixed-family stopping decision.

Replay the new decisions with `python scripts/nx_decision_recovery.py --track NX10 --check` and `python -m unittest tests.test_horizon_decisions`.

Chapter 12

Source-bound distant response queries (NX11)

For one fixed finite linear state system, initial state and scalar output,

$$s_n = CA^n x_0, \qquad S(z) = C(I - zA)^{-1}x_0 = P(z)/Q(z), \qquad Q(0) = 1.$$

This coefficient index is physical discrete time (or a repeated-block index). It is not the graph-height volume index. Nothing in this adapter asserts rationality for a general positive-q graph-height series.

12.1 Actual upstream reuse and indexing

The optional source-pinned adapter calls DynaComp `generating.from_state`, which derives a denominator by exact Faddeev–LeVerrier arithmetic and cancels common factors. It retains the normalized rational GF packet, its autonomous realization, initial prefix and a nonzero minimal Hankel determinant. Replay first checks the source matrices and their hash, then recomputes the GF from those matrices, then checks the packet. A recurrence fitted to a sample prefix would not suffice.

A unit impulse in $x_{t+1} = Ax_t + Bu_t,\ y_t = Cx_t$, with zero reset, yields $y_0 = 0$ and $y_{n+1} = CA^nB$. The extra factor z is therefore real. The autonomous sequence CA^nx is not silently reinterpreted as an impulse transfer sequence. Direct feedthrough, when present in a source interpretation, must be represented explicitly. Tests retain an exact delayed impulse example and zero/finite-transient cases.

For held input u, the source-state compiler uses the affine lift

$$H_u = \begin{pmatrix} A & Bu \\ 0 & 1 \end{pmatrix}, \qquad x_{\text{lift},0} = \begin{pmatrix} x_0 \\ 1 \end{pmatrix}, \quad C_{\text{lift}} = (C, 0).$$

The lift is used for sequence generation, not submitted as a Schur-stable matrix. A repeated block is composed chronologically using upstream `fastforward.compose`. Observation phase changes both the output row and its offset. The receipt records block length and phase, and independent neighbour-flux stepping checks every block index 0 through 32 for two real 4-by-4 candidate sources.

12.2 Difference cancellation and all-index equality

Compile both exact fractions and subtract. The common numerator/denominator cancellation is performed symbolically upstream. A zero reduced numerator proves equality at every index under the declared reset/input/output semantics. An invisible third mode disappears in the retained tests. A pair of two-mode responses sharing the $1/2$ mode reduces from a four-state block sum to two scalar modes. This is fixed-family equality, not every-input equivalence.

For $A = \mathrm{diag}(1/2, 1/3)$, $x = (1, 1)$, $C = (1, -1)$,

$$S(z) = \frac{z/6}{1 - 5z/6 + z^2/6}, \qquad s_n = 2^{-n} - 3^{-n}, \qquad \det \begin{pmatrix} 0 & 1/6 \\ 1/6 & 5/36 \end{pmatrix} = -1/36.$$

Independent closed-form arithmetic checks all indices 0 through 32. The recurrence convention is chronological: $s_{n+2} = \frac{5}{6} s_{n+1} - \frac{1}{6} s_n$.

12.3 Resource contract and finite experiment integration

Binary powering uses logarithmically many stages, but rational numerator and denominator bit lengths can grow linearly with the index. The public query performs a conservative source-derived bit preflight and enforces an explicit final output-bit budget. A refused query is a resource limit. It is not a zero reading and not evidence of a physical ambiguity. The two-mode query at 10^{100} is refused as an explicit rational value.

A separately labelled modular query requires every rational denominator to be invertible modulo the declared modulus. Matrix powering then returns only a residue. The receipt's huge-index check modulo 101 is independently compared with modular powers of 2 and 3. Such a residue is not a real observation and cannot certify a physical noise margin.

`remote_panel_pool` exposes exact queried readings at a supplied finite set of indices to the existing panel optimizer. It preserves the source candidate ordering separately from the backend's numeric candidate count. Its objective includes declared reset and elapsed costs. The result is optimal in this finite menu, not all horizons; NX10 supplies separate all-horizon stopping arguments.

Compilation/setup and repeated queries are different costs. The retained accounting example uses setup 100, compiled query one and direct query ten. One compiled query costs 101 versus ten, so setup dominates. The first strict break-even is twelve queries because $100 + N < 10N$ exactly when $N > 100/9$. This is an explicit rational accounting illustration, not a benchmark claim. A separate benchmark report records actual compilation, cold query, warm queries and output bits without polluting deterministic scientific receipts.

12.4 Portable replay

`response_receipt` uses canonical integer numerator/positive-denominator pairs. `replay_remote_receipt` rejects extra top-level fields, noncanonical rationals, incompatible source bindings, modified fractions, malformed indices and unknown arithmetic modes. Full source-state or recursive pair-difference evidence travels with the receipt. The upstream source digest binds executable bytes; actual recomputation es-

tablishes the identities. Public exact inputs reject booleans, floats and unchecked numeric strings.

Run `python scripts/nx_decision_recovery.py --track NX11 --check` and `python -m unittest tests.test_remote_queries`.

The retained actual timing example is deliberately unfavorable to distant GF evaluation at index 128: source-pinned per-query replay is more expensive than independently stepping this two-mode scalar source. The measured break-even is therefore absent. The report also records an index-one case where compilation plus checked query greatly exceeds direct stepping. This is an honest setup-dominance result; the logarithmic algebraic powering theorem alone does not imply a wall-clock speedup at these small horizons.

Physical maps are part of the source binding. A repeated block of length L and phase j has map $t(k) = Lk + j$. Progression sampling composes maps exactly. A finite pool rejects mixed maps or incompatible sample conventions and prices the actual physical index as reset plus elapsed price times $Lk + j$, rather than silently charging only the block counter.

Chapter 13

Complete constrained protocol populations (NX12)

A physical protocol consists of an original chronological symbol word and a terminal sensor identity. Different words or sensors remain different original experiments even if their readings coincide. The completion compiler compresses counting, not identity or multiplicity.

A deterministic finite automaton supplies states q, start q_0, transition δ and accepting states. Each symbol has a physical scalar amplitude, strictly positive rational elapsed/action cost and nonnegative integer resource increments. A sensor has its own cost and installation requirements. The protocol budget also includes reset cost, a finite horizon and a switch limit. The first symbol does not count as a switch.

13.1 Completion theorem

Scale every rational price by their exact common denominator D, refusing a declared scale budget if necessary. A completion state is

$$\sigma = (q, B, \mathbf{b}, h, a_{\text{last}}, s, \ell),$$

where B is remaining scaled cost, $\mathbf{b}$ remaining resources, h remaining horizon, s remaining switches and ℓ elapsed length. For each feasible symbol, consume its actual resources and one horizon step. The terminal contribution counts legal installed sensors only when the grammar accepts and the empty/nonempty convention permits termination:

$$C(\sigma) = t(\sigma) + \sum_{a:\text{feasible}} C(\sigma \cdot a).$$

Induction on h proves the count. Every word has exactly one DFA path; rejecting duplicate (q, a) transitions prevents counting ambiguous derivations as separate words. Missing transitions make words illegal. Strictly positive symbol cost and finite horizon exclude zero-cost cycles. Zero heater amplitude still consumes one positive-cost elapsed step.

Ranking uses a fixed order: legal terminal sensors in input order, followed by symbol branches in alphabet order. Each branch owns a consecutive interval of completion ranks. Descending the unique interval proves `rank(unrank(i))=i`; summing earlier intervals proves the converse for every legal original protocol. Conditioning on a prefix changes only the completion start state. Conditioning on terminal sensor restricts

the terminal term, without deduplicating words.

The number of reachable dynamic states depends on the numeric resource and scaled-cost ranges. Complexity is pseudo-polynomial, not polynomial in arbitrary budget bit length. Explicit materialization checks the exact root count before producing at most the requested number of members. A resource refusal does not imply an empty scientific population.

13.2 Independent tiny acceptance

Alphabet amplitudes are zero and two. Nonempty words have length at most three, total amplitude at most four and at most one adjacent switch. The legal words are two of length one, four of length two and five of length three, eleven in total. Two terminal sensors yield **22 experiments**. Reset one, elapsed cost one, heat charge half the amplitude and sensor cost one give only the words 0, 2 and 00 at total cost four, hence **six experiments**. Independent Cartesian-product tests enumerate all words and sensors without using completion recursion, check every rank, and compare twenty randomized tiny deterministic grammars.

13.3 Physical adapter and exact complete search

The protocol adapter maps symbol IDs to physical amplitudes explicitly. A sensor subset requires an explicit map to the original model's sensor channels; filtering the sensor catalogue must not silently rename a center sensor as channel zero. Each protocol is separately reset. Every output in the retained bounded 4-by-4 population is compared with a conservative neighbour-flux update independent of the production simulator.

A short complete population is passed to the existing exact panel backend and independently enumerated panel oracle. The larger bounded language retains protocols omitted by a matching-horizon step/pulse/alternating subset. That comparison does not alter or relabel the original 112-experiment baseline.

For the cheapest robust scalar observation, sort all legal original experiments by exact total price and rank. Evaluate in that order. A failed experiment retains a concrete target pair whose closed reading intervals overlap. The first feasible price is minimal because every cheaper original protocol has been explicitly excluded. All protocols tied at that price are checked. Once price exceeds the incumbent, later sorted protocols cannot improve it. This cost bound is the only search pruning rule; nominal linear dependence and duplicate nominal readings are not used as robust pruning justifications.

The search still reports the complete root population count. It may conclude that no scalar terminal reading distinguishes every requested target pair; that is distinct from a panel or adaptive-policy impossibility. The initial demonstration uses the named nominal/heater-half pair, plus a complete short panel comparison.

13.4 Portable completion evidence

The `singularity.protocol-population/1` certificate includes the alphabet, DFA, resources, installed sensors, exact scale, root state/count and the reachable completion DAG. Every edge decreases horizon. A local checker verifies each terminal count and

sum-of-children equation, so hashes are not substituted for arithmetic proof. Rank witnesses reconstruct original words and sensor identities. Grammar counting establishes population completeness; a separate source binding and output replay establish physical readings.

Run `python scripts/nx_decision_recovery.py --track NX12 --check` and `python -m unittest tests.test_protocol_population`.

The fair retained comparison uses the nominal/heater-half pair, the same radius $1/1000$, the same grammar/resources/costs and the same allowed terminal sensors. The complete population has 108 original experiments, versus 28 in the matching step/pulse/alternating subset. Both the cheapest robust scalar experiment and the exact finite panel optimum cost three in this instance. Completeness adds 80 legal original protocols without improving this optimum. The short independent exhaustive panel check remains separate from this larger exact backend comparison.

Chapter 14

Typed observability: finite type graphs, graph adapters and cheapest witnesses (NX13)

14.1 What this note covers

The existing observation engine (`observability.py`) works in one state space. Every action there is a square matrix on that space. Some graph operations do not fit that picture. Introducing a terminal, forgetting one, or adding the edge between the two terminals changes what the boundary looks like. A single ambient matrix can hide an illegal composition. Adding the terminal edge twice is the clearest case: the algebra silently computes a multigraph.

This note extends the engine to a finite directed graph of state types. It has three parts:

1. **NX13-A.** The typed algebra contract: closure, quotient invariance, per-type minimality and the cheapest-witness property, with proofs.
2. **NX13-B.** One finite boundary grammar for graph assembly that genuinely closes at a fixed volume budget. It is checked against height enumeration on the actual assembled graphs.
3. **NX13-C.** Portable certificates that replay without the closure computation. Cheapest witnesses are compared with exhaustive legal-path search under a cost bound.

The four categories are kept separate. Sections 2–4 are **proved project results** about the typed algebra and the grammar's formulas. Section 5 records **exact computations over declared finite populations**. Section 7 lists what remains **outside scope**.

Code: `src/singularity/typed_observability.py` (algebra and portable format) and `src/singularity/typed_contexts.py` (graph adapter). Tests: `tests/test_typed_observability.py` and `tests/test_typed_contexts.py`. Receipt: `scripts/typed_contexts.py` and `results/typed_contexts.json`.

14.2 1. Objects and conventions

A **typed operator system** consists of:

- a finite set of types B, with type b carrying $V_b = \mathbb{Q}^{d_b}$ ($d_b = 0$ is allowed);
- actions $a : s(a) \to t(a)$ with exact matrices M_a of shape $d_{t(a)} \times d_{s(a)}$ and costs $c_a > 0$;
- observations at type b: rows $L \in \mathbb{Q}^{1 \times d_b}$ with costs $c_L \geq 0$.

A zero-dimensional space's shape comes from its declared type. An empty tuple is accepted only where the type dimensions say the matrix has no rows. A 2×0 matrix is `((), ())`, never `()`.

Words are chronological. A word $w = (a_1, \dots, a_m)$ is legal when $t(a_i) = s(a_{i+1})$ for every i. It starts at $s(a_1)$ and ends at $t(a_m)$. It sends a state x to $M_{a_m} \cdots M_{a_1} x$, so the latest matrix acts on the left. An experiment (w, L) is legal when L lives at the end type of w. Its row and cost are

$$\rho(w, L) = L\, M_{a_m} \cdots M_{a_1} \in \mathbb{Q}^{1 \times d_{s(a_1)}}, \qquad c(w, L) = c_L + \textstyle\sum_i c_{a_i}.$$

The empty word is legal at the observation's own type. Reading an observation at another type is rejected. So is a word whose consecutive types do not match.

Write W_b for the span of all rows of legal experiments starting at b.

14.3 2. The typed algebra (NX13-A)

14.3.1 Algorithm

`typed_closure` keeps a priority queue of actual experiments, ordered by cost. It starts with all observations, read with the empty word. It pops the cheapest experiment e, which starts at type b. If $\rho(e)$ is independent of the rows already retained at b, the algorithm retains e. It then pushes the child $(a \cdot e)$ for every action a with $t(a) = b$. The child's row is $\rho(e)M_a$, it starts at $s(a)$, and its cost is $c(e) + c_a$. Prepending is correct because the action a happens before the experiment e. Dependent rows are discarded and never extended.

The algorithm outputs retained witness rows B_b (a $r_b \times d_b$ matrix whose rows are actual experiments), and unique coordinates A_b and T_a with

$$L = A_L B_b \;\; (L \text{ at } b), \qquad B_{t(a)} M_a = T_a B_{s(a)},$$

where T_a has shape $r_{t(a)} \times r_{s(a)}$. Neither A nor T need be square. The kernel $N_b = \ker B_b$ is reported as an exact basis.

14.3.2 Lemma 1 (monotone pops, termination)

Every child costs strictly more than its parent, and children are pushed only when the parent is popped. So the popped costs never decrease. Each retained row raises the rank at its type by one. Hence at most $D = \sum_b d_b$ rows are retained. Only retained rows have children, so at most $1 + D \cdot |A|$ batches of pushes happen and the algorithm stops.

14.3.3 Theorem 1 (typed closure and cost filtration)

For every legal experiment (w, L) starting at b, the row $\rho(w, L)$ is a linear combination of retained rows of B_b whose costs are at most $c(w, L)$. Consequently $\mathrm{row}(B_b) = W_b$ for every type.

Proof. Use induction on the length of w.

- **Length 0.** The experiment $(\,(), L)$ was pushed initially. When it was popped, it was either retained (and is its own combination) or it lay in the span of rows already retained at b. Those rows were popped earlier, so by Lemma 1 they cost at most c_L.
- **Prepending an action.** Let $w = (a) \cdot w'$, where w' starts at $t(a)$. By induction $\rho(w', L) = \sum_k \lambda_k \beta_k$, a sum over retained rows β_k at $t(a)$ with $c(\beta_k) \le c(w', L)$. Then $\rho(w, L) = \sum_k \lambda_k \beta_k M_a$. Each $\beta_k M_a$ is the row of a child that was pushed, with cost $c(\beta_k) + c_a \le c(w, L)$. When that child was popped, it was retained or dependent on rows of cost at most its own. Substituting gives the claim.

Thus $W_b \subseteq \mathrm{row}(B_b)$. Conversely each retained row is a legal experiment, so $\mathrm{row}(B_b) \subseteq W_b$. Both inclusions matter. An elimination residual is not an experiment, and the code never stores one as a witness. □

The same argument works with the packet's simultaneous iteration $W_b^{k+1} = W_b^k + \sum_{a:s(a)=b} W_{t(a)}^k M_a$. That chain is increasing. If it is stationary at one step for every b, it stays stationary. Each strict step raises the total rank, so it stabilizes after at most D steps. **Words of length at most D therefore span every W_b.** The independent oracle in the tests relies on this bound.

14.3.4 Theorem 2 (quotient invariance)

For every action, $M_a N_{s(a)} \subseteq N_{t(a)}$. The observable state $z_b = B_b x$ updates exactly by $z_{t(a)} = T_a z_{s(a)}$, and every legal reading factors through it: $\rho(w, L)x = A_L\, T_{a_m} \cdots T_{a_1}\, z_{s(a_1)}$.

Proof. For $x \in N_s$, $B_t M_a x = T_a B_s x = 0$, so $M_a x \in N_t$. The update is the identity $B_t M_a = T_a B_s$ applied to x. The reading formula follows by applying the update along the word and then using $L = A_L B_{t(a_m)}$. So the quotient dynamics on $V_b/N_b \cong \mathbb{Q}^{r_b}$ is well defined, and it is also rectangular across types. □

14.3.5 Theorem 3 (per-type linear minimality)

Let $P_b : V_b \to U_b$ be linear maps such that every legal reading from b factors through P_b, meaning $\rho(w, L) = f_{w,L} \circ P_b$ for some linear $f_{w,L}$. Then $\ker P_b \subseteq N_b$ and $\dim \mathrm{im}\, P_b \ge r_b$. The witness compression $P_b = B_b$ attains this bound.

Proof. If $P_b x = 0$, then every legal reading vanishes on x. By Theorem 1 every row of B_b is such a reading, so $B_b x = 0$. Rank–nullity gives $\dim \mathrm{im}\, P_b = d_b - \dim \ker P_b \ge d_b - \dim N_b = r_b$. Theorem 2 shows that B_b itself makes every reading factor. □

This minimality is linear and per type. It says nothing about nonlinear compressions, and nothing about compressions that may discard some future readings.

14.3.6 Theorem 4 (cheapest exact witness from the certificate alone)

Suppose a certificate satisfies the replayed identities, and each nonzero coordinate obeys the filtration $A_L[k] \ne 0 \Rightarrow c(\beta_k) \le c_L$ and $T_a[k, m] \ne 0 \Rightarrow c(\beta_m^s) \le c(\beta_k^t) + c_a$. Then for any $\delta \in V_b$, the first retained row of B_b in cost order with $\beta\delta \ne 0$ is a cheapest legal experiment separating states whose difference is δ. If no retained row pairs nonzero with δ, no legal experiment does.

Proof. Rerun the induction of Theorem 1 using only the two coordinate identities and their filtrations. Every legal row ρ is a combination of rows of B_b with cost at most

$c(\rho)$. If $\rho\delta \neq 0$, some basis row in that combination has $\beta\delta \neq 0$ and $c(\beta) \leq c(\rho)$. The rows of B_b are ordered by cost, and each one is itself a legal experiment. □

Theorem 4 holds for replayed certificates. It needs neither the queue nor any trust in the code that produced the certificate. It covers **exact** distinction only. A dependent row can have a much larger discrepancy than a cheaper one under noise. Span pruning therefore says nothing about maximum margin. The ordered-contexts note gives a margin law that needs its own argument.

14.3.7 Worked example, checked by hand

Let type a have $V_a = \mathbb{Q}^2$ with no observation, and type b have $V_b = \mathbb{Q}$ with $L_b = (1)$. Let $M_f = (1\ 1) : a \to b$ cost 2 and $M_g = (1, 0)^{\mathsf{T}} : b \to a$ cost 3.

- The first pop is L_b at b (cost 0). It is retained, and the child $f \cdot L_b$ is pushed with row $(1)(1\ 1) = (1, 1)$ at a and cost 2.
- That child is retained. Its child $g \cdot (f \cdot L_b)$ has row $(1, 1)(1, 0)^{\mathsf{T}} = (1)$ at b and cost 5. It is dependent.

So $B_a = (1, 1)$, $B_b = (1)$, $N_a = \text{span}\{(1, -1)\}$, $N_b = 0$, A_a is 0×1, $A_b = (1)$ and $T_f = T_g = (1)$. The states $(2, 0)$ and $(0, 2)$ at a agree forever. The states $(2, 0)$ and $(0, 3)$ are separated by (f, L_b) at cost 2. Reading L_b directly at a is rejected. The unit test `test_packet_example_by_hand` asserts exactly these numbers.

14.4 3. A finite boundary grammar (NX13-B)

14.4.1 Types and coordinates

Fix a volume budget $R \geq 0$ and a rational $0 < q < 1$. A placed graph is a finite labeled simple graph with an ordered tuple of at most two distinct terminals. Its terminal kernel $K_G(\mathbf{b})$ sums $x^{\text{interior volume}} q^{E_G}$ over interior heights. The energy is the **full** graph energy, so a terminal–terminal edge is included. The four types are:

type	terminals	coordinates	dimension at $R{=}2$
`closed`	none	$Y[v]{=}[x^v]L_G$, $0{\leq}v{\leq}R$	3
`one`	(a)	$Y[(s),v]{=}[x^v]x^s K_G(s)$, $s{\leq}v{\leq}R$	6
`two`	(a,b), **not adjacent**	$Y[(s,t),v]{=}[x^v]x^{s+t} K_G(s,t)$	10
`two_adj`	(a,b), **adjacent**	same coordinates	10

These are raw labeled counts at q. There is no normalization and no automorphism quotient. `two` and `two_adj` have the same coordinates; the edge status is legality metadata. Every type observes the local coefficients $[x^v]L$ of the current graph, $v = 0, \ldots, R$, at cost 0.

14.4.2 Actions and their exact formulas

Each formula is proved by summing over the heights of the vertices whose role changes. Every other sum is unchanged.

action	typing	formula	why
`open`	closed→one	$Y'[(s),v]=Y[v-s]$	new isolated terminal: $K'(s)=L_G$
`close`	one→closed	$Y'[v]=\sum_s Y[(s),v]$	the terminal height joins the interior volume
`intro`	one→two	$Y'[(s,t),v]=Y[(s),v-t]$	new isolated terminal b: $K'(s,t)=K(s)$
`intro_adj`	one→two_adj	$Y'[(s,t),v]=q^{(s-t)^2}Y[(s),v-t]$	the new edge ab contributes $(s-t)^2$
`pendant`	one→one	$Y'[(s),v]=\sum_j[x^j]K_{\text{leaf}}(s)\,Y[(s),v-j]$	fresh leaf: $K'(s)=K(s)\sum_h x^h q^{(s-h)^2}$
`join`	two→two_adj	$Y'=q^{(s-t)^2}Y$	adds the edge ab
`cut`	two_adj→two	$Y'=q^{-(s-t)^2}Y$	removes the edge ab, which the type guarantees is present
`forget_a`	two, two_adj→one	$Y'[(t),v]=\sum_s Y[(s,t),v]$	a joins the interior and b becomes the terminal
`forget_b`	two, two_adj→one	$Y'[(s),v]=\sum_t Y[(s,t),v]$	symmetric to `forget_a`
`extend_a/b`	two, two_adj→two	existing `OrderedMoveSystem` operator	new terminal adjacent only to the old terminal
`swap`	two→two, two_adj→two_adj	$Y'[(s,t),v]=Y[(t,s),v]$	reorders the terminals
contexts	two→two, two_adj→two_adj	existing ordered context operator	a fresh interior and no terminal edge

The default contexts are an asymmetric one-sided leaf `leaf_a` and the shared neighbour `shared`. The context and extension matrices are reused unchanged from `terminal_moves.OrderedMoveSystem`, and the code checks that their coordinate order matches. Action identifiers on `two_adj` carry the suffix `@adj`, for example `extend_a@adj`.

14.4.3 Proposition 5 (the grammar closes at fixed budget)

Every output coordinate of volume v is a combination of input coordinates of volume at most v. Consequently, for every legal word of any length, the product of the typed matrices applied to the truncated state of a placed graph equals the truncated state of the assembled graph.

Proof. Read off the table. `open`, `intro` and `intro_adj` use input volume $v-s$ or $v-t$. `pendant` and the contexts use $v-j$. Every other action keeps v. The ordered-contexts note proves the extension and context cases. The formula for each action is exact on the untruncated kernel. Truncation keeps it exact because no output ever reads an input of higher volume. Induction on word length finishes the proof. □

14.4.4 Proposition 6 (legality is sound for assembly)

If a word is legal from the type of a placed graph, then sequential assembly never creates a duplicate edge or a loop, and the actual type of the assembled graph equals the typed end type.

Proof. Use induction over the word. `open`, `intro` and `extend` add a fresh vertex adjacent to at most one current terminal, so the two terminals are non-adjacent afterwards. `intro_adj` makes them adjacent through a fresh edge. `join` is legal only on `two`, where the terminals are non-adjacent, so the new edge is not a duplicate. `cut` is legal only on `two_adj`, where the edge exists. Contexts are glued through fresh interior copies and have no terminal edge (the `OrderedContextAttachment` check), so they keep the

adjacency status and cannot repeat an edge. `swap` and the forget actions add no edges. □

Proposition 6 does not follow from any matrix identity. Squaring the `join` matrix gives the factor $q^{2(s-t)^2}$. That is the population of a multigraph with a doubled terminal edge, and the tests confirm it with an independent multigraph enumeration. The type split `two`/`two_adj` is what rejects this word. This is the "edge-status changes need boundary metadata" failure mode named in the packet, resolved here by one bit of type information.

14.4.5 Refused extensions

- **Three or more terminals.** Allowing `intro` from `two` would require an unbounded family of types. The grammar refuses it. This is an implementation boundary, not a theorem that such operations are impossible.
- **Joins of two unknown blocks.** These are bilinear and stay outside the linear setting. Gluing a fixed context is linear, and those operations are included.
- **State-dependent guards and minimum-zero normalization.** Neither is modelled.

14.5 4. Certificates and costs (NX13-C)

The portable format is `singularity.typed-operator-closure/1`. It is a new schema. The single-space format `singularity.operator-closure/1` is unchanged, and nothing is renamed or migrated. A bundle records the following:

- each type's identifier and dimension;
- each action's ordered source and target, its matrix with explicit `rows` and `columns`, and its cost;
- each observation's type, row and cost;
- per type: the witness experiments (observation, start type, chronological word, row, cost), the observation coordinates, the nullspace and the pivots;
- per action: the coordinates T_a with explicit shape.

Rationals are canonical `[numerator, denominator]` pairs. An optional `provenance` field holds the graph adapter's volume convention, terminal labels, context graphs, coordinate lists and assembly rule. That field is descriptive only. Algebraic replay does **not** prove the adapter; Section 5's enumeration does.

`replay_typed_bundle` decodes strictly. It rejects booleans, floats, noncanonical rationals, shape disagreements, unknown fields, order mismatches and oversize inputs. `load_typed_bundle` also rejects duplicate JSON keys and NaN or Infinity. Replay then calls `verify_typed_certificate`, which rechecks every identity, every witness, cost order, pivots, kernels, the inclusion $M_a N_s \subseteq N_t$ and the filtration. Replay never calls `typed_closure`. `scripts/typed_contexts.py --replay` disables both the closure and the adapter module while it runs, and a unit test disables the closure the same way.

14.5.1 Proposition for the operator chapter (proposed text)

> **Proposition (typed observation closure).** Let a finite typed operator system have rational matrices $M_a : V_{s(a)} \to V_{t(a)}$, positive action costs and nonnegative observation costs. There are realizable witness experiments B_b, with $\operatorname{rank} B_b \le \dim V_b$, and rational matrices A_b, T_a with $L_b = A_b B_b$ and $B_{t(a)} M_a = T_a B_{s(a)}$, such that (i) every legal chronological experiment

row lies in row B_b; (ii) $\ker B_b$ is the space of states that no legal future experiment detects, and it is carried into $\ker B_{t(a)}$ by each M_a; (iii) every linear compression preserving all legal future readings has rank at least $\operatorname{rank} B_b$; and (iv) if the coordinates respect the cost filtration, the first witness pairing nonzero with a difference is a cheapest exact distinguishing experiment. The identities, and hence (i)–(iv), can be checked from the certificate alone.

14.6 5. Exact finite evidence

These are exact computations over declared finite populations. They are not proofs of the infinite quantifiers, which Sections 2–3 supply.

- **Adapter against enumeration.** The receipt uses 13 base placements: P3 closed; K1 closed; P3 at its middle, at an end, and K1 with one terminal; P3 endpoints in both orders; 2K1; P4 at $(0, 2)$; P3 adjacent in both orders; K2; and a triangle edge. Every legal word of length at most 4 at $R = 2$, $q = 2/3$ (27,439 words) and every legal word of length at most 2 at $R = 3$, $q = 1/2$ (610 words) gives a matrix state that **equals the full state of the actually assembled graph**. That state is computed by weak-composition enumeration (`kernel_by_enumeration`, or `local_coefficients` for `closed`). The unit tests repeat length 3 at $R = 2$ (4,087 words) and length 2 at $R = 3$.
- **Cross-check with existing machinery.** The `forget` and `intro` matrices agree with `charges.forget_terminal` and `separator_compiler.introduce_terminal`, followed by `charges.kernel_from_charges`.
- **Noncommutation.** On actual graphs:
  - (`extend_a`,`shared`) ≠ (`shared`,`extend_a`) on P3 endpoints;
  - (`swap`,`leaf_a`) ≠ (`leaf_a`,`swap`) on P4 $(0, 2)$;
  - (`join`,`extend_a@adj`) ≠ (`extend_a`,`join`).

  The swap/leaf pair agrees on the automorphic P3 endpoints even though the matrices do not commute. This is a reminder that a reading depends on the base as well as the word.
- **Closure.** At $R = 2$, $q = 2/3$ the full grammar has ranks 3/6/10/10, so every type is fully observable. The symmetric sublanguage `shared`, `join`, `cut`, `forget_a`, `close`, `open` has ranks 3/3/7/7. Certificates for both, and for two rectangular fixtures, are retained and replayed. One fixture has a rank-zero type, a zero-dimensional type, a type with no outputs, cycles, rectangular maps and noncommuting actions.
- **Cheapest witnesses.** Every reported witness equals the minimum found by exhaustive forward search over all legal experiments within the cost bound. The bound is 8 for the fixtures and 3 for the graph pairs, giving up to 1,185 legal experiments per pair. Each graph witness also reads differently on the two actually assembled graphs. The automorphic P3 endpoint pair has no witness, and the exhaustive search also finds none within the bound. The unbounded statement follows from Theorem 4.
- **Tampering.** Reversing a witness word (`("u","f")` becomes illegal; a graph witness with a legal reversal gives a different row), changing a cost, row, coordinate or type order, or replacing Fractions with ints all make verification fail.

14.7 6. Reproduction

```
PYTHONPATH=src python -m unittest tests.test_typed_observability
↪  tests.test_typed_contexts -v
PYTHONPATH=src python scripts/typed_contexts.py --check
python scripts/typed_contexts.py --replay results/typed_contexts.json
```

14.8 7. Limitations

- The grammar is finite and declared: at most two ordered terminals, one bit of edge status, fixed R and fixed rational q. The type graph is not derived from an arbitrary separator schedule.
- Ranks and witnesses are specific to the declared R, q and grammar. Full rank at $R = 2$ says nothing about other budgets or parameters. Equal truncated states do not imply equal full series or isomorphic graphs.
- Minimality is linear and per type (Theorem 3). Cheapest means exact nonzero distinction under the declared additive costs (Theorem 4). It is not robust or maximum-margin optimality.
- State-dependent guards, nonlinear charge joins, infinite type collections, three or more terminals, minimum-zero normalization and physical or thermal semantics are excluded.
- No proof-assistant formalization exists.

Chapter 15

NX14: resource-bounded parameter-module certificates

The original `parametric.py` already computes polynomial observation modules. This addition supplies portable module-generation evidence, a small replay route, explicit chart guards, and a bounded next-budget attempt. It does not replace that implementation or promote isolated rational ranks to parameter-wide theorems.

15.1 Objects and the two stabilization questions

For polynomial observations $L_j(q)$ and generators $M_i(q)$, put

$$N_0 = \langle L_j \rangle_{\mathbb{Q}[q]}, \qquad N_{r+1} = N_0 + \sum_i N_r M_i.$$

A module chain can change without changing rational-function rank. For example, the full-rank submodule generated by $(q,0),(0,1)$ is properly contained in the module generated by $(1,0),(0,1)$. Both have rank two over $\mathbb{Q}(q)$. Hence the constructor compares complete Hermite bases, rather than stopping when the number of basis rows stops increasing.

At stage r, let I_r contain the original observation rows and the preceding basis multiplied by each generator. The receipt stores a basis B_r, a polynomial matrix F_r, and another polynomial matrix C_r, with

$$B_r = F_r I_r, \qquad I_r = C_r B_r.$$

The first identity proves every proposed basis row is generated by real experiments. The second proves no experiment in the stage was lost. Once $B_r = B_{r-1}$, every original observation belongs to this module and every generator preserves it. Induction on word length therefore proves it contains every future experiment. These are exact polynomial identities; no sampled evaluation is used in this implication.

Proposition NX14.1. If all displayed identities replay, the last two bases coincide, and the original system has the declared dimensions, then the final basis generates exactly the complete polynomial observation module.

Proof. The stage-zero identities give equality with the original observation module. Each subsequent pair gives equality with the stated sum of observations and transforms, so induction identifies every stored stage. Equality of adjacent stages implies closure under right multiplication by every generator. The final module contains all

finite words, and the forward identities show all its rows remain polynomial combinations of those words. Both inclusions follow. □

The constructor records elementary polynomial row operations while computing Hermite form. The verifier does not run the Hermite constructor: it multiplies the supplied polynomial matrices and checks these two inclusions directly. Generic rank is checked separately. Tiny determinant tests use a separate sparse-polynomial Leibniz calculation, so they do not merely compare two calls to Hermite form.

15.2 Divisors, exceptional parameters, and actual experiments

If the final basis has generic rank r, form its maximal determinantal divisor

$$D_r(q) = \operatorname{monic} \gcd_{|J|=r} \det B_{[:,J]}(q).$$

The receipt wrapper only enumerates these minors when their number is below its declared limit. For a univariate polynomial matrix, specialization has rank below r exactly when D_r vanishes. This statement concerns all maximal minors. A determinant of one selected generic experiment basis can vanish at a perfectly observable parameter.

The permanent false-exception fixture uses $a = q - 1/2$, observations $(1, 0), (0, a)$, and $M = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. The original observation determinant is a, but $(1, 0)M = (0, 1)$. The complete module is the entire two-dimensional polynomial module and $D_2 = 1$. There is no exception at one half.

The true-exception fixture has $L = (1, a)$ and $M = \operatorname{diag}(1, 2)$. Every word gives $(1, 2^k a)$. Consequently

$$\operatorname{rank}(q) = \begin{cases} 1 & q = 1/2, \\ 2 & q \neq 1/2, \end{cases} \qquad D_2 = q - 1/2.$$

These fixtures have certified charts containing actual observation/word pairs. Each chart records the selected coordinate minor and its determinant. The chart is valid precisely where that determinant and the declared denominator are nonzero. The atlas additionally checks whether the gcd of its chart determinants equals the module divisor. When that equality holds, the charts cover the entire regular locus. It is not enough that every displayed sample point works. In the two permanent fixtures the coverage equality does hold. Exceptional ranks are computed exactly, and replay recomputes the complete exceptional list rather than trusting an omitted entry. Algebraic isolating intervals are refined against the exact rational domain endpoints before inclusion is decided. A common algebraic factor of the divisor and denominator is excluded as a pole, just as a rational pole is excluded. The regression with a domain endpoint equal to an initial isolation bound prevents accidentally omitting an interior irrational root.

A module basis may contain polynomial combinations that are not single physical experiments. The schema keeps this distinction explicit. Its `chain` rows certify algebraic generation; `charts` store real witness words. Costs of the latter belong to the original experiment system and are not inferred from polynomial coefficients.

15.3 Rational-function guard

The fixture with numerator generator $N = \mathrm{diag}(1, 2)$, observation $(1, 1)$, and common denominator $h(q) = 2q-1$ represents $M = N/h$. At every $q \neq 1/2$, a length-k word is multiplied by the nonzero scalar $h(q)^{-k}$. Thus pointwise row spans agree with those of the numerator language on the guarded domain. The excluded point is an undefined pole, not an observed rank drop. Direct rational specializations at one quarter and three quarters have rank two; attempting one half raises an explicit error.

This scalar-denominator argument permits the stated rank comparison. It does not identify the polynomial modules over the unlocalized ring, and it does not authorize treating poles as admissible experiments.

15.4 The next graph budget and its precise stopping point

Item	Retained result
Standard graph budget	$R{=}5$
Interface dimension	34
Symbolic wrapper dimension limit	22
Symbolic result	`budget_refusal`, dimension preflight
Exact specialization at $q{=}1/3$	rank 34
Exact specialization at $q{=}1/2$	rank 34
Parameter-wide rank at $R{=}5$	not established by this receipt

This is the bounded refusal permitted by NX14-B, rather than a concealed incomplete computation. It identifies the next bottleneck: a full module/divisor calculation at dimension 34 with explicitly larger resource allowances. The graph matrices were actually compiled and the two rational closures were computed. Their full ranks are finite specialization evidence, not an interval theorem and not an all-budget cyclotomic factor theorem.

`parameter_module_receipt` exposes dimension, degree, coefficient-bit, module-step, maximal-minor and wall-clock limits. The deterministic resource record retains achieved polynomial degree, coefficient bits and module steps; machine-dependent time and memory are not serialized as reproducible mathematical data. The wrapper's refusal state has no rank field. The distinction between `complete`, `module_complete_factor_partial`, and `budget_refusal` is part of the API contract. A partial factorization never supplies a complete exceptional atlas.

15.5 Reproduction and compatibility

Implementation: `src/singularity/parameter_receipts.py`. Tests: `tests/test_parameter_module_receipts.py`. Receipt: `results/parameter_module_receipts.json`. Run `python scripts/parameter_module_receipts.py --check`. The tests alter a module generator, a factor, and a witness word and require rejection. Rank-zero systems and the denominator guard are retained. The existing `parametric.py` public meanings remain unchanged.

The accepted finite scope is the replay architecture, the two complete interior-exception atlases, the guarded rational example, and the documented R5 bounded attempt. Multi-parameter determinantal varieties, larger-budget factor classification, and a universal factor theorem remain separate questions.

Chapter 16

NX15: actual graph populations versus formal invisible vectors

This work combines exact necessary filters, two complete bounded graph constructors, and a broad structural exclusion obtained from the newly proved threshold inverse. No graph realizing the surviving general frontier is reported. A nonzero vector in a formal nullspace is not an adjacency matrix, and agreement through a finite budget is not equality of full series.

16.1 Which difference is being sought?

For an ordered two-terminal block, use the existing raw kernel convention

$$K_G(s,t;x,q)=\sum_{h_I\geq 0} x^{|h_I|}q^{E_G(s,t,h_I)}.$$

The total interface volume is $s+t+|h_I|$. Symmetrization adds the two ordered terminal states once each when they differ; it does not average or divide height multiplicities by automorphism counts.

At fixed rational pinning, the retained terminal-edge candidate at total volume 38 has states $(1,18),(6,17),(10,15)$, all on the circle $s^2+t^2=325$. Their required interior volumes are 19, 15, and 13. Write

$$y=(q^{289},q^{121},q^{25}),\qquad v=(y_3-y_2,\ y_1-y_3,\ y_2-y_1).$$

Then $\sum v_i=\sum y_iv_i=0$ by cancellation. At the edge-free frontier, total volume 11, the formal vector is $e_{(0,5)}-e_{(3,4)}$, with interior volumes six and four. These calculations reproduce the candidate frontiers. They do not construct graph pairs. The earlier rational interpolation hypotheses remain in force; the exceptional irrational parameters are not silently included.

For the declared symmetric context language, the existing quotient characterization implies that a frontier-only hidden difference must agree at every lower interface volume. Therefore any exact discrepancy already found in a lower kernel prefix excludes that pair from realizing the frontier vector. The search uses this necessary condition rather than computing enormous volume-38 populations.

16.2 Necessary charge-population restrictions

The charge table groups original interior assignments by volume j, terminal charge Q, and energy. Every coefficient is a nonnegative integer. If the interior has m vertices, its total population in slice j is

$$\sum_{Q,e} a_{j,Q,e} = \binom{j+m-1}{m-1} \quad (m > 0).$$

For zero interior vertices it is one at slice zero and zero otherwise. A charge produced by j units of height must be a sum of j available terminal-neighborhood patterns from $\{0,1\}^2$. The necessary relaxation checks this semigroup condition, nonnegative energies, positive stored multiplicities, and exact slice population totals. All 64 labeled four-vertex graphs pass the filter through interior slice three. A deliberately altered multiplicity and an impossible zero-slice charge are rejected.

These checks are only necessary. They do not ensure that every slice arises from one common graph, that interior-edge energies are compatible, or that an integer charge table has an adjacency realization. This distinction makes the filter useful for rejection while preventing it from manufacturing false witnesses.

16.3 Two complete bounded exclusions

Every census is stratified by known graph order. Isomorphism reduction fixes the terminal set, allows its swap, and permutes only interior labels. This reduction selects one representative block per isomorphism class. Every kernel still counts all original labeled height assignments; no population is divided by automorphisms.

Declared family	Labeled constructor objects	Terminal-swap block classes	Independent profiles through volume four	Result
All labeled trees on 2–5 vertices, terminals 0 and 1	145	24	2,395	Every distinct class separates in the required prefix
Both terminals dominate an arbitrary labeled interior on 0–4 vertices; no terminal edge	76	19	3,004	Every distinct class separates in the required prefix

The tree constructor enumerates every Prüfer word. Its counts per order are n^{n-2}. An independent edge-subset constructor tests every $(n-1)$-edge set for connectedness and confirms equality of the actual labeled edge sets, not just agreement of totals. The second constructor enumerates every interior adjacency bit mask, yielding $\sum_{m=0}^{4} 2^{\binom{m}{2}} = 76$ objects. It is complete by direct binary encoding of every allowed interior edge.

For every representative, the bag-message terminal kernel is compared with independent weak-composition enumeration. The fixed-parameter symmetric signatures are then compared within each known-order stratum. No nonisomorphic pair survives volume four. Since a nonzero frontier difference at volume eleven must vanish at every lower interface volume under the existing quotient theorem, none of these pairs can

realize it. This is a complete finite exclusion with an explicit constructor and lower-prefix certificate, not a claim about all trees, all dominating interiors, or cross-order ambiguities.

Proposition NX15.1. At $q = 1/2$, no pair of distinct terminal-swap isomorphism classes within either displayed known-order family realizes the edge-free volume-eleven frontier vector.

Proof. Constructor completeness gives the stated exhaustive object sets. Independent bag and composition kernels agree for every class. Their retained volume-four signatures are pairwise different within each order stratum. Every proposed frontier realization would have equal required lower kernel coefficients, including this prefix, contradicting one of those differences. Isomorphic representatives have identical complete kernels and cannot supply a nonzero difference. □

16.4 Structural exclusion for threshold interiors

There is a much larger family where enumeration is unnecessary. Let both nonadjacent terminals dominate an unknown threshold interior H of known order m. The existing closure operator is unit lower triangular and converts its pinned sequence $A_{H,j}$ to the raw series of any supported single closing context. FR05 proves that, at rational $q = A/B$, this sequence through

$$R_* = \min\{R \geq 2 : B^{2(R-1)} > m\}$$

reconstructs the promised threshold graph. Therefore every such single closure distinguishes nonisomorphic threshold interiors by volume R_*.

Proposition NX15.2. For a fixed rational pinning and an integer budget $R \geq 2$, two dominating blocks whose promised threshold interiors have equal known order

$$m < B^{2(R-1)}$$

cannot have equal supported closure prefixes through R unless their interiors are isomorphic.

Proof. Invert the known unit-triangular context map at the same budget. The resulting pinned coefficient prefix meets FR05's strict cutoff inequality. Apply its inverse theorem, which uses only the known order, pinning and prefix. Both interiors reconstruct to the same canonical threshold creation word. Hence the blocks are isomorphic, and their entire ordered interfaces coincide. □

At one half, the volume-eleven ambiguity is therefore incompatible with these dominating threshold interiors whenever $m \leq 2^{20} - 1$. At volume 38 the analogous order bound is $2^{74} - 1$. The enormous second number is an exact structural bound, not a completed graph census or a promise that coefficients of that size can be processed cheaply. This is a new application of FR05 to the surviving interface ambiguity, not a second implementation of its reconstruction algorithm.

The class promise is essential. A successful arithmetic or charge check does not prove threshold membership. The theorem does not eliminate general balanced blocks, general trees, or arbitrary interiors with the same pinned series.

16.5 Resource contract and replay

The supporting module is `src/singularity/realizable_interfaces.py`; the receipt is `results/realizable_interfaces.json`; run `python scripts/realizable_interfaces.py --check`. Constructor bounds, original labeled counts, isomorphism-class counts, pinning, language, lower prefix, target frontier and independent profile work are retained. Profile work is estimated as $\binom{R+n}{n}$ per representative before the independent population engine is called. A limit refusal is an arithmetic resource error, never a fabricated negative result.

The search states distinguish `complete_bounded_exclusion` from `incomplete_budget`. A surviving lower-prefix class would remain incomplete until its whole frontier condition and independent glued observations were checked. The current two complete exclusions need no such positive-pair check because no nonisomorphic lower-prefix survivor exists. The general realization questions remain open outside the precise families above.

Chapter 17

Incremental connected-support compilation (NX16)

Status of this note: units NX16-A, NX16-B and NX16-C, on branch `claude/exciting-brown-95vyq3`. The note keeps four categories separate. **Literature** is background that is cited and not reproved. **Proved** marks project statements proved here. **Exact finite** marks computations over a declared finite population. **Open** marks what remains a research target.

Code: `src/singularity/incremental_locality.py`. Tests: `tests/test_incremental_locality.py`. Receipt: `scripts/incremental_locality.py --check` and `results/incremental_locality.json`.

17.1 1. Objects

Take vertices `0, ..., n-1`, symmetric nonnegative integral edge weights `w_uv`, and nonnegative integral pinnings `lambda_v`. For a height vector `h` of nonnegative integers put

$$E_w(h) = \sum_{u<v} w_{uv}(h_u - h_v)^2 + \sum_v \lambda_v h_v^2, \qquad F_w(x,q) = \sum_{h \geq 0} x^{|h|} q^{E_w(h)}.$$

A zero weight is the same thing as a missing edge. Every energy is a nonnegative integer, so each coefficient `[x^m]F_w` is a finite sum, namely a polynomial in `q` with nonnegative integral exponents and nonnegative integral counts. Rational weights are allowed only through a declared energy unit. `from_rational_weights` multiplies every weight and pinning by the least common multiple `L` of their denominators and records `energy_unit = 1/L`. The exponent `k` then means energy `k/L`. Negative weights are outside this contract.

The **positive-weight graph** `G_w` has an edge `uv` exactly when `w_uv > 0`. The **decoration** of a vertex is `d_v = sum_u w_uv + lambda_v`, its weighted degree plus its pinning. Expanding the square gives

$$E_w(h) = \sum_v d_v h_v^2 - 2\sum_{u<v} w_{uv} h_u h_v.$$

With `w = 1` on the edges of a graph `H` and `lambda_v = lam` constant, `F_w` is the `F_lam(H)` of `locality.py`: `lam = 0` gives `L_H` and `lam = 2` gives `A_H`. The implementation checks this equality against `locality.pinned_log`.

Now give each vertex its own variable `x_v`. For a vertex set `T`, `Z_T` is `F_w` with `x_v = 0` for `v` outside `T`. Equivalently, `Z_T` sums over heights that vanish outside `T`, with the full energy. In the expanded form this means vertices in `T` keep their **ambient** decorations `d_v`. Define

$$\Phi_S(w) = \sum_{T \subseteq S} (-1)^{|S|-|T|} \log Z_T(w),$$

and then specialise every `x_v` to the common volume variable `x`.

17.2 2. The support expansion (proved)

Lemma 1. `Phi_S` is exactly the part of the multivariate series `log F_w` whose monomials use every vertex of `S` and no other vertex.

Proof. `F_w` is a formal power series in `x_0, ..., x_{n-1}`, with coefficients in `Z[q]` and constant term `1`, so `log F_w` is well defined. Write `Psi_A` for the part of `log F_w` whose monomials have support exactly `A`. Setting `x_v = 0` outside `T` is a ring homomorphism. It commutes with `log` and kills exactly the monomials whose support is not contained in `T`. Hence `log Z_T = sum_{A subset T} Psi_A`. Möbius inversion on the Boolean lattice gives `Psi_S = sum_{T subset S} (-1)^{|S|-|T|} log Z_T = Phi_S`. □

Corollary 2. `Phi_S = O(x^{|S|})`: each monomial of `Psi_S` has every exponent `x_v`, `v in S`, at least one.

Lemma 3 (disconnected supports vanish). If `S` is not connected in `G_w`, then `Phi_S = 0`.

Proof. Write `S = A ⊔ B` with both parts nonempty and no positive edge between them. Take `T ⊆ S` and a height vector `h` supported in `T`. The cross terms `w_ab h_a h_b`, with `a ∈ A` and `b ∈ B`, vanish because those weights are zero. Every other term involves only `T∩A` or only `T∩B`. So the energy splits, and `Z_T = Z_{T∩A} Z_{T∩B}`. Then `log Z_T = f(T∩A) + g(T∩B)`. An alternating sum over `T ⊆ S` of a function of `T∩A` alone factors through `sum_{T_B ⊆ B} (-1)^{|B|-|T_B|} = 0`, because `B` is nonempty. The same holds for `g`, using that `A` is nonempty. □

Lemma 4 (what Phi depends on). `Phi_S` depends only on the decorations `d_v` for `v ∈ S` and the internal weights `w_uv` for `u, v ∈ S`. It is unchanged by any bijection of `S` that preserves both.

Proof. For `T ⊆ S`, the expanded energy of a height supported in `T` involves only these quantities. A relabelling that preserves them permutes the heights and preserves volume and energy. □

Theorem 5 (support expansion). For every `R ≥ 0`,

$$\log F_w \equiv \sum_{S \text{ connected in } G_w,\ |S| \le R} \Phi_S(w) \pmod{x^{R+1}}.$$

Proof. By Lemma 1, `log F_w` is the sum of `Psi_S` over nonempty `S`. The supports with `|S| > R` contribute nothing modulo `x^{R+1}` by Corollary 2. The disconnected ones contribute nothing by Lemma 3. □

The unweighted form of this theorem is the existing SNG-F55 locality result. What is new here is the weighted, per-vertex-pinned statement. Its consequence is that a zero weight behaves exactly as an absent edge.

17.3 3. The union-graph difference identity (proved)

Compare two weightings `(w, lambda)` and `(w', lambda')`. Let `D` be the set of endpoints of every edge whose weight changes, together with every vertex whose pinning changes. Let `U` be the **union graph**: it has an edge `uv` whenever `w_uv > 0` or `w'_uv > 0`.

Theorem 6. Modulo `x^{R+1}`,

$$\log F_{w'} - \log F_w \equiv \sum_{\substack{S \text{ connected in } U \\ |S| \le R,\ S \cap D \neq \emptyset}} \big(\Phi_S(w') - \Phi_S(w)\big).$$

Proof. Each of `G_w` and `G_{w'}` is a subgraph of `U`. So a set connected in either graph is connected in `U`. By Lemma 3 we may therefore sum both expansions of Theorem 5 over all `U`-connected `S` with `|S| ≤ R`; the added terms are zero. Now suppose `S ∩ D = ∅`. Every changed edge has both endpoints in `D`, so no changed edge meets `S`. In particular, no internal weight of `S` changes. No weight incident to a vertex of `S` changes either, so the weighted degrees on `S` are the same. No pinning on `S` changes. By Lemma 4, `Phi_S(w') = Phi_S(w)`. Subtracting the two sums leaves only the supports that meet `D`. □

Why the three naive filters are wrong. Each failure below is exhibited by an exact test, `test_union_difference_identity_and_naive_filters`, on a path with 4 vertices at `R = 3`.

- New-graph supports only. After a deletion, a support that was connected only through the deleted edge has `Phi_S(w') = 0` and `Phi_S(w) ≠ 0`. This filter drops that term.
- Old-graph supports only. After an insertion, a newly connected support has `Phi_S(w) = 0`, and this filter drops it.
- Supports containing both endpoints of a changed edge. A singleton `{u}` contains only one endpoint, yet its decoration `d_u` changes, so `Phi_{{u}}` changes. This happens in the mandatory two-vertex example below. For a change to the pinning of a single vertex, “contains every endpoint” coincides with “meets `D`”.

17.4 4. Worked example (exact finite; regression)

Compare two isolated vertices with one edge between them, at zero pinning and `R = 2`.

- Isolated: `F_0 = 1 + 2x + 3x^2` and `log F_0 = 2x + x^2`.
- Edge: `F_1 = 1 + 2q x + (1 + 2q^4) x^2` and `log F_1 = 2q x + (1 + 2q^4 - 2q^2) x^2`.
- Each singleton contributes `q x + (q^4 - q^2/2) x^2`. The pair contributes `(1 - q^2) x^2`.
- At `q = 1/2`: `Delta log F = -x - 3x^2/8`, and `F_1 = 1 + x + 9x^2/8`.
- Applying the reverse update recovers the original compiled state exactly, including the support index.

All of these values are retained in `results/incremental_locality.json` under `contract`.

17.5 5. Cache soundness (proved)

A support's memo key is `(k, decorations, weighted internal edges)` after relabelling `S` to `0..k-1`. The relabelling minimises the edge list over orders that respect the vertex classes `(decoration, sorted incident internal weights)`. When the number of such orders exceeds `CANONICAL_LIMIT`, the sorted vertex labelling is kept instead. Either way, the key is a **complete description** of a decorated, weighted graph that is isomorphic to `S`. Two supports with equal keys are therefore isomorphic in the sense of Lemma 4, and they have equal `Phi`. Moreover, `Phi` is computed from the key alone (`_phi`), so a cache entry is correct by construction. Cache slots also carry `R` and the convention string.

Consequences:

- There are no false collisions across graphs. Sharing one cache between unrelated graphs and truncation depths is safe (`test_cross_graph_cache_isolation`). The packet's warning about a set-only fallback key does not apply, because the fallback key here is the full labelled description, not a vertex-set name.
- The labelled fallback loses only sharing (`test_labelled_fallback_keys_are_exact` forces it everywhere and gets identical logs).

17.6 6. Update algorithm and its locality (proved)

`apply_local_changes(comp, changes)` runs in five steps.

1. It validates every change against the current state before changing anything. A **stale** patch, whose recorded old value differs from the current one, raises `ValueError`. So does a duplicate target, and so does an out-of-range vertex. In all three cases the state is left untouched.
2. It enumerates the `U`-connected sets with `|S| ≤ R` that meet `D`. It does this by growing from the least marked vertex while avoiding earlier marked vertices (`connected_sets_meeting`). Adjacency is read only at members of the growing set.
3. It reads the old keys from the support index; a support missing from the index is one that was disconnected in the old graph. It then applies the changes and computes the new keys, or none for a support that is now disconnected.
4. It adds `Phi(new) - Phi(old)` to `log F`, with an absent side counted as zero, and updates the support index.
5. It optionally emits a `singularity.locality-update/1` record (section 7).

By Theorem 6 the result is exactly the recompiled logarithm. By Theorem 5 and the correctness of the index, the new support index equals the recompiled index.

Proposition 7 (local update cost). Let `Delta_U` be the maximum degree of `U` and `N` the set of vertices within `U`-distance `R-1` of `D`. Assume that dictionary operations take unit time.

- Every affected support lies inside `N`. A connected set of at most `R` vertices that meets `D` has every member within distance `R-1` of `D`.
- The number of affected supports is at most `|D| · sum_{k=1}^{R} (k-1)! Delta_U^{k-1}`. Here is the proof. Grow a connected `k`-set from a vertex of `D` by repeatedly adding a neighbour of the current set. At step `j` there are at most `j·Delta_U` choices, so there are at most `(k-1)! Delta_U^{k-1}` growth sequences, and every such set arises from one of them. Literature gives sharper bounds of the form `(c·Delta)^{k-1}`; they are not needed here.

- The work per affected support depends only on `R`, `Delta_U` and the decorations and weights inside `N`. That work is one key canonicalisation, costing at most `C ANONICAL_LIMIT · R^2` comparisons, and two cache lookups. On a cache miss, computing `Phi` enumerates `2^k` subsets and `sum_{j ≤ R} C(j + c - 1, c - 1 )` profiles per restricted block. The exponents are at most `R^2 · max d_v`. The coefficients of `Phi` are functions of the key alone.
- Updating `log F` costs `R+1` polynomial additions per changed support and `R+1` for the final merge. Their exponent range is bounded as above. The **bit size** of the coefficients of `log F`, however, does depend on the whole graph, because `log F` sums contributions from every support in the graph.

So, after setup, the number of supports touched, the polynomial operations and the cache traffic depend only on the decorated radius-`(R-1)` union neighbourhood of `D`. They do not depend on `n`. The only `n`-dependent quantities are the bit lengths of the global sums. The scaling receipt below measures both.

The update does not promise constant-time output of the full series. Formal exponentiation costs `R(R+1)/2` polynomial products, and it touches every coefficient. The operation count of that query is independent of `n`, but its operands carry `n`-dependent coefficient sizes.

17.7 7. Update records and replay

A `singularity.locality-update/1` record stores the following fields:

- the baseline specification (`n`, the positive weights, the pinnings, and the energy unit as `[numerator, denominator]`);
- the baseline logarithm;
- the exact changes, as `{kind, target, old, new}`;
- `affected_endpoints` (the set `D`);
- `declared_union_edges`, the edges of `U` among the vertices of the affected supports;
- the list of affected supports, each with its old and new contribution or `null`;
- `result_log`.

Polynomials use the existing `polynomials.polynomial_to_json` encoding, `[exponent, numerator, denominator]`.

`verify_local_update` replays a record without using the incremental code.

1. It rebuilds the old and new specifications, checking every old value.
2. It checks **completeness**. It recomputes `U` from the two specifications and takes the ball of radius `R-1` around `D`. It then forms every subset of at most `R` ball vertices, keeps those that meet `D` and are `U`-connected by a direct search, and requires the listed supports to be exactly these.
3. It recomputes every listed contribution from its definition, `support_contribution_by_definition`. That function enumerates heights supported in each `T ⊆ S` with the edge-difference energy and uses no memo and no factorisation. A side must be `null` exactly when the support is disconnected in that version. As a by-product, this also confirms that disconnected supports have zero contribution.
4. It checks that `result_log` equals the baseline log plus the listed corrections.
5. Optionally, it compares `exp(baseline log)` and `exp(result log)` with `direct_series`.

A digest would not be enough here, because a hash cannot show that an omitted

support was zero; step 2 is what does that. The baseline logarithm is trusted unless step 5 runs. The completeness check is exponential in the ball size, so it is budgeted.

17.8 8. Evidence

NX16-A (proved + exact finite). The tests check the following:

- The two-vertex example (section 4) holds exactly, including the invalidation of both singletons and exact reversal.
- Zero weight is absence: a `1 → 0` update yields exactly the compiled state of the graph without that edge.
- The compiler agrees with `direct_series` on 30 random weighted, pinned graphs with `n ≤ 5` and `R ≤ 3`. Every exponent is an ordinary nonnegative integer.
- With positive weights and no pinning, it agrees with `weighted.weighted_coefficients` on 12 cases. With unit weights, it agrees with `heights.local_coefficients` and with `locality.pinned_log` at `lam = 0` and `lam = 2`.
- Disconnected supports vanish, and `Phi_S = O(x^{|S|})`, both checked by definition.
- The union identity holds, and the naive filters fail as described in section 3.
- Rational weights are handled through a declared unit, and floats are rejected.
- Malformed input is rejected: `TypeError` for `bool`, `float` and `str`, and `ValueError` for bad values or shapes.

NX16-B (exact finite). The population is all `3^6 = 729` weightings of `K_4` with edge weights in `{0,1,2}` and zero pinning, at `R = 3`.

- Every single-edge edit was checked: 8748 edits, comprising 2916 insertions, 2916 deletions and 2916 weight changes. Each updated state, both log and support index, equals full recompilation. Each recompiled series equals direct enumeration over the box `[0,3]^4`.
- Pinning changes `0 → 2` and back were checked on all 64 unweighted 4-vertex graphs, with exact reversal.
- A sequential run on 5 vertices applied 14 random batches of edge and pinning changes, mixing in-place and copying updates. After every batch it compared against recompilation and direct enumeration, and it replayed each update record with the direct oracle. Reversing the whole sequence recovered the original state exactly.
- Stale, duplicate and out-of-range patches were rejected, with the state unchanged.
- Sharing a cache across 40 random graphs and different `R` changed no result. Supports with the same vertex set but different weights or ambient degrees receive different keys.
- Tampering with a record was detected: removing a support, swapping a contribution, altering the result log, a stale change, a missing union edge, a wrong `D`, or an altered baseline.

NX16-C (proved bound + exact finite measurement). The family is the ladder `P_2 × P_m`, with maximum degree 3 and at most 4 after the diagonal insertion, for `m = 16, 32, 64, 128, 256, 512` (`n` up to 1024), at `R = 4`. Four edits are made in column `m // 2`: a rung reweight `1 → 3`, a rail deletion, a diagonal insertion, and a pinning change `0 → 2`. Retained, separately for each `m`, are:

- **setup**: 319 to 11231 supports, which grows linearly with `n`; 15 distinct keys; 15 contributions computed;

- **storage**: support-index entries, which equal the supports, and 102 cache entries;
- **update**, with a cold cache after setup and with a warm cache. The counts are 55, 53, 69 and 35 affected supports. On a cold cache the edits compute 13, 4, 17 and 2 new contributions; on a warm cache, 0. The edits use 1395, 780, 1790 and 540 polynomial additions. **All of these counts are identical for every `m`**, and the script asserts this.
- **query**: 10 polynomial products in the formal exponential;
- **coefficient bits**: the largest numerator in the log grows from 11 to 16 bits, and the largest numerator in the series from 14 to 36 bits. This growth is consistent with the counts `C(n+R-1, R) ≈ n^4/24`. Denominators stay at 2 to 3 bits.

Every updated log equals full recompilation, whose support count grows with `n`. For the unit-weight edits it also equals `locality.pinned_log`. On the ladder with `m = 3`, every edit also matches `direct_series`. Wall-clock times are printed and not retained. On the development machine, a cold update took about 6 to 45 ms at every size, and a warm update about 2 to 6 ms. Full recompilation took about 40 ms at `n = 32` and about 300 ms at `n = 1024`.

Receipts. Two update records, on ladders with `m = 3` (oracle included) and `m = 5`, each make three simultaneous changes. They are retained in full. `--check` regenerates everything, compares the result with the retained JSON, and replays the retained records with `verify_local_update`.

17.9 9. Proposition for the manuscript (proposed text; chapters are not edited here)

> **Proposition (incremental locality).** Let `F_w` be the height series of a nonnegative integral weighted energy with per-vertex pinning. Let `D` be the set of vertices touched by a change of weights or pinnings, and let `U` be the union of the old and new positive-weight graphs. Then `log F` changes, modulo `x^{R+1}`, by the sum of `Phi_S(w') - Phi_S(w)` over the `U`-connected sets `S` with at most `R` vertices that meet `D`. Every such `S` lies within `U`-distance `R -1` of `D`, and there are at most `|D| sum_{k ≤ R} (k-1)! Delta_U^{k-1}` of them. A zero weight is an absent edge. Singletons in `D` must be recomputed even when they contain only one endpoint of a changed edge.

17.10 10. Limitations and open items

- The scope is nonnegative integral weights, or rational weights with a declared unit. Negative or signed energies, continuous weights, vertex deletion or insertion, and changes to `n` are not handled.
- The `n`-independence in Proposition 7 concerns support counts, polynomial-operation counts and cache traffic, and it assumes unit-time dictionary operations. Coefficient bit sizes of the global log and series grow with `n`, roughly `R lo g n` bits for the series. Wall times are measured on one machine and are not a proof.
- The bound in Proposition 7 is fixed-parameter: it is useless for large `R` or dense union graphs.
- Copying updates (`in_place=False`) and records that include the baseline cost `O( size)`. Only in-place updates without a record are local.
- The baseline logarithm in a record is trusted unless the direct oracle runs. Replaying completeness is exponential in the size of the radius-`(R-1)` ball and is

budgeted.

- Boundary-charge-aware updates (`charges.py`, `separator_compiler.py`), dynamic canonicalisation, and moment or cycle extensions under updates are left to later pushes.
- `locality.connected_supersets` copies the full adjacency on each call, which costs `O(n)`. This module therefore has its own local enumerator, which is tested against brute force. A lazy version of the shared function would let both modules use one enumerator; that is a suggested change to an existing file.

Chapter 18

NX17 — Exact low-rank physical fault transport

This release implements NX17-A/B/C for a declared finite-horizon, source-derived discrete thermal model. It establishes exact algebra and reuse costs, not continuum calibration or measured physical diagnostic performance. The reference source is `applications/thermal_diagnostics/source_fixture.json`; the receipt pins its SHA256. Graph-height populations and thermal state dynamics remain separate mathematical models.

18.1 Model and sign

A thermal candidate is the existing `AffineCandidate` contract

$$x_{t+1} = Ax_t + Bu_t, \qquad y_t = Cx_t.$$

For the plate, $A_0 = I - kL - \ell I$, $k = 1/8$, $\ell = 1/100$. For a physical bond with column $b = e_u - e_v$, removing conductance $\delta \in [0, k]$ means

$$A_f = A_0 + \delta b b^T.$$

The positive diagonal update and negative off-diagonal update preserve every row sum $1 - \ell$, preserve symmetry, and preserve the heater and sensor definitions. `BondFault` canonically stores the unordered original endpoints; orientation reversal does not change its meaning. Repeated declarations of one bond add severities, and their sum must not exceed the nominal available symmetric positive coupling. Zero severity is valid. The original fault ID remains an application-level population identity: no models are deduplicated merely because outputs coincide.

18.2 Formal Sherman–Morrison and Woodbury

Write $R_0(z) = (I - zA_0)^{-1}$. Formal inversion is valid because the constant matrix is the identity. The exact rank-one formula is

$$R_f = R_0 + \frac{z\delta R_0 b b^T R_0}{1 - z\delta b^T R_0 b}.$$

Consequently the sensor correction is

$$\Delta H(z) = \frac{z\delta(CR_0b)(b^TR_0B)}{1 - z\delta b^TR_0b}.$$

`bond_transfer_factors` retains all factors through the requested horizon, including the unit-constant scalar denominator. These are truncated formal series; they are not evaluations at an unguarded numerical z, and they are not claimed to be factored exact rational functions.

For selected bond columns U and their diagonal severity matrix D,

$$R_f = R_0 + zR_0UD(I - zU^TR_0UD)^{-1}U^TR_0.$$

This formula handles zero severities and dependent bond columns. The factor order is significant. In particular the inner matrix is $I - zGD$, not a casually reordered product.

18.3 Coefficient implementation and proof

The implementation uses a small Volterra system, coefficient-wise equivalent to the Woodbury inverse. Let

$$v_j(t) = b_j^TA_0^tB, \quad G_{ji}(t) = b_j^TA_0^tb_i, \quad c_{si}(t) = C_sA_0^tb_i.$$

Duhamel's identity follows by induction from $A_f = A_0 + UDU^T$:

$$A_f^t = A_0^t + \sum_{r=0}^{t-1} A_0^{t-1-r}UDU^TA_f^r.$$

Therefore, setting $w_j(t) = b_j^TA_f^tB$,

$$w_j(t) = v_j(t) + \sum_{r=0}^{t-1}\sum_i \delta_i G_{ji}(t-1-r)w_i(r),$$

$$C_sA_f^tB = C_sA_0^tB + \sum_{r=0}^{t-1}\sum_i \delta_i c_{si}(t-1-r)w_i(r).$$

Every right side uses only earlier w. Thus induction proves all retained impulse coefficients exactly. This is an independent reuse calculation, not simulation of another full faulted state matrix. The nominal heater powers and lazily requested nominal bond powers are shared across all queries.

For input words and a declared initial state,

$$y_T = \sum_{t=0}^{T-1} CA_f^{T-1-t}Bu_t + CA_f^Tx_0.$$

`protocol_from_impulse` implements the convolution and accepts the explicitly supplied initial-state contribution. An initial-state response is computed by requesting the same transport with `input_vector=x0`. Input amplitudes may be arbitrary exact rationals, including negative diagnostic pulses; physical implementability is an application restriction.

18.4 Acceptance evidence

Unit	Retained evidence
NX17-A	Two-node full/half/zero blockage, orientation reversal, exact impulse cancellation, source sign and heater/sensor invariance; direct sparse powers and an independent neighbor-flux oracle
NX17-B	Every pair of six small-plate bonds at two severities; short arbitrary input words, nonzero initial states, symmetry and row sums; excessive total severity rejected
NX17-C	Every one of the 264 distinct bonds of the 12-by-12 plate at horizon six; four sensor channels, all seven retained impulse coefficients, exact direct original-model replay; all 264 reflection comparisons on invariant sensors

The deterministic receipt is `results/fault_transport.json`; its benchmark is retained separately from the reproducibility-critical exact data. In one measured run the first population required about 1.47 seconds including nominal setup, compared with about 3.08 seconds for direct sparse impulse simulation. Recursive owned-cache storage was about 18.2 MB with shared Python object identities counted once. This is an object-graph memory measurement, not peak process RSS. Coefficient bit sizes and retained vector-cell counts are also recorded. The break-even calculation uses the measured setup and population-average query costs; it is machine-specific, and no universal speed theorem is asserted.

The cache has explicit horizon, selected-fault and retained-cell limits. These are resource refusals, not mathematical claims of non-identifiability. No uncertainty certificate is transported by this module: original interval/disturbance propagation remains the responsibility of the existing physical diagnostic contract. Individually stable static faults do not imply stability of adversarially switched faults.

Run `python scripts/fault_transport.py --check` to replay the whole population.

Chapter 19

NX18: exact recovery conditioning, physical noise, and coefficient bits

Exact injectivity and usable recovery accuracy are different properties. This module exposes a particular exact left inverse, propagates declared physical reading errors, verifies target-specific primal/dual optimization certificates, and compares real graph experiments at a common target. All statements use exact integer or rational arithmetic. Floating-point conditioning estimates are not substituted for proved bounds.

19.1 Sharp box-noise propagation

Let $y = Av$, and let R be a left inverse, $RA = I$. For raw reading error $|\eta_i| \le \epsilon_i$,

$$\hat{v} - v = R\eta, \qquad |\hat{v} - v| \le |R|\epsilon.$$

For a scalar target $g^T v$, the sharp independent-box error is

$$\sup_{|\eta_i| \le \epsilon_i} |g^T R\eta| = \sum_i |(g^T R)_i| \epsilon_i.$$

The upper bound follows from the triangle inequality. Equality is attained by choosing each independent error at its signed endpoint, $\eta_i = \epsilon_i \operatorname{sgn}(g^T R)_i$. A different sign pattern may be required for a different target; simultaneous attainment of every coordinate bound is not promised.

For

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 17/16 \end{pmatrix}, \qquad R = \begin{pmatrix} 17 & -16 \\ -16 & 16 \end{pmatrix},$$

the coordinate errors under unit box noise are exactly 33 and 32. The tests enumerate all four box vertices independently and attain both bounds. These are exact worst-case errors, not floating-point upper estimates.

If measurements are preprocessed by P, the correct raw reconstruction is RP, and the bound is $|RP|\epsilon$. Applying $|R|$ to an unchanged raw noise vector would silently omit amplification by preprocessing. Correlation can reduce the true worst case, but a correlated uncertainty set must be supplied explicitly; it is not inferred from similar-looking rows.

19.2 Physical sensor or mere rescaling?

Consider measurements of one coordinate with rows 1 and 100. If both are distinct physical sensors with error bound one, the second can reconstruct the coordinate with error $1/100$. If the second reading is only 100 times the first reading, its error bound is 100 and the reconstruction error remains one. Exact row-span pruning preserves distinguishability, but it cannot preserve this physical distinction when noise contracts differ.

The receipt records both examples. Costs and noise bounds are supplied per reading. The real graph comparison below charges each coefficient reading explicitly; no hidden setup-sharing optimization is assumed. A basis change in unknown coordinates similarly changes the forward map and target representation. It cannot improve a physical signal-to-noise guarantee merely by multiplying its coordinates.

19.3 Certified target-specific left inverses

For a scalar target, choose weights w solving $A^T w = g$. The best independent-box error is the linear program

$$\min_{w} \sum_{i} \epsilon_i |w_i| \quad \text{subject to } A^T w = g.$$

Its dual is

$$\max_{z} g^T z \quad \text{subject to } |(Az)_i| \le \epsilon_i.$$

A rational certificate consists of a primal feasible w, a dual feasible z, and equality of the two displayed objective values. Weak duality proves optimality without trusting the active-set search. The verifier checks precisely these rational equalities and inequalities. It does not invoke the optimizer. The constructor enumerates small basic supports and dual vertices under a declared active-set limit; a refusal is not an infeasibility theorem.

Proposition NX18.1. If the supplied primal and dual witnesses replay, their common value is the least possible box-noise amplification among all linear unbiased estimators of the declared target from those readings.

Proof. Every feasible estimator has expected noiseless output $g^T v$, since $A^T w = g$. For every primal/dual feasible pair,

$$g^T z = w^T A z \le \sum_{i} |w_i| \epsilon_i.$$

Equality of a particular pair reaches this universal lower bound. The box-error identity above identifies its operational worst-case error. □

The bounded design selector solves this problem separately for every member of a finite supplied menu. It then chooses the least declared total cost among members meeting the requested error. The retained example has a cost-one reading with error one, a cost-three reading with error $1/100$, and a cost-seven reading with error $1/200$. For tolerance $1/50$, the cost-three design is the exact optimum in that menu. This is not a theorem about unlisted experiments, shared panels, nonlinear estimators, or arbitrary correlated errors.

19.4 A real graph comparison at the same target

The common target is $(A_{H,0}, \dots, A_{H,3})$ for the dominating block over $H = K_2$, at $q = 1/2$. The edge-free and weighted probe matrices are first compiled on the **entire generic symmetric interface** through volume three. Their inverse wrappers therefore do not assume a dominating-interior prior while constructing the interface recovery map. Target rows are selected afterward at orbit $(0, 0)$, whose coefficients are exactly the displayed A_H.

Eighteen actual simple glued graphs are independently counted by weak compositions and direct energy evaluation. Their outputs match the closure operators. The old exact finite-difference and weighted restoration APIs recover the same target on these consistent exact outputs. The new wrappers declare a particular linear extension to inconsistent noisy readings. The old APIs correctly reject inconsistent overdetermined data, so they are not quietly assigned an unstated numerical noise policy.

Route	Raw coefficient readings	Volume budget	Declared total reading cost	Sharp target-coordinate-3 amplification under unit independent noise
Edge-free finite-difference probe family	24	3	144	343326323/6082944
Weighted terminal-edge restoration, weights 0 and 1	48	3	312	82769649/8012480
Ordered charge chart	4	3	4	1

This is a finite comparison under the supplied cost and sensor contracts. It does not imply that every charge chart is less noisy, that weighted recovery is always better, or that the displayed left inverses minimize full-state noise. In this particular dominating family the charge chart uses terminal state $(0, 0)$, so its forward matrix is the identity. Its advantage follows from the available physical reading, not from arbitrary rescaling.

Volume three precedes the first collision circle. The receipt separately records the two-node $R = 7$ collision moments at $(0, 5), (3, 4)$: finite-difference nodes τ_o and weighted-edge nodes $q^{(s-t)^2}$. Those inverse metrics concern isolated conditional moments; they are not advertised as raw whole-graph noise bounds. The full delayed wrapper explicitly refuses budgets with delay until nuisance-coordinate elimination is supplied. This prevents the common mistake of assuming all higher-volume unknown coefficients vanish merely because the target is truncated lower.

19.5 Uniform bounds and bit costs

For nodes z_i, $V_{ri} = z_i^r$. The coefficients of the Lagrange polynomial

$$\ell_i(X) = \prod_{j \neq i} \frac{X - z_j}{z_i - z_j}$$

form row i of V^{-1}. If a declared compact parameter domain satisfies $|z_i| \le M$ and $|z_i - z_j| \ge \gamma > 0$, its coefficient one-norm is at most

$$\|\ell_i\|_1 \le \frac{(1+M)^{m-1}}{\gamma^{m-1}}.$$

For equal reading error ϵ, this gives coordinate error at most $\epsilon((1+M)/\gamma)^{m-1}$. The permanent interval example has nodes 1 and $1+q$, with $q \in [1/4, 1/2]$. Exact inequalities give $M = 3/2$, $\gamma = 1/4$, hence the uniform bound 10. The proof is the displayed product inequality, not a parameter grid. Endpoints zero and one are not included by extrapolation.

A finite difference of order j has coefficient one-norm 2^j. This quantifies preprocessing amplification before any Vandermonde inverse. Its sufficient observation delay does not bound its necessary precision. The receipt separately records matrix and inverse numerator bits, denominator bits, signs, nonzeros and total encoding bits. Its encoding uses one sign bit, at least one numerator bit, and denominator bit length per cell. These reproducible bit counts are not Python memory usage or an asymptotic speedup theorem.

Implementation and tests are `recovery_metrics.py` and `tests/test_recovery_metrics.py`. Run `python scripts/recovery_metrics.py --check`. The accepted scope includes exact propagation, the bounded real-block comparison, one certified target design, and one genuinely uniform compact-interval bound. Sharp all-budget delayed precision, optimal delay, and general correlated uncertainty remain open.

Chapter 20

NX19: integer-compatible observational quotients

The rational observable quotient answers which directions can ever affect an experiment. It does not automatically describe which integer coordinates are attainable, how many original objects lie above one output, or which congruences a quotient transition must preserve. This companion supplies those arithmetic contracts using exact Smith certificates and finite original-coordinate populations.

20.1 Denominator clearing retains the original observation map

Given an exact rational observation matrix B, clear each row's denominators separately with a positive integer diagonal matrix S. The constructor retains

$$A = SB \in \mathbb{Z}^{r\times D}$$

and the individual row scales. Its image lattice is $A\mathbb{Z}^D$, not necessarily all integer points of the rational row-image space. A row scale changes representation; it does not rescale a physical noise bound or measurement cost unless the caller also supplies that change.

The certificate stores integer matrices U, V with determinant ± 1, and a rectangular Smith diagonal

$$UAV = \mathrm{diag}(d_1, \dots, d_s, 0, \dots), \qquad d_i > 0, \quad d_i \mid d_{i+1}.$$

The constructor uses elementary row and column additions, swaps and signs. Euclidean reduction clears the pivot row and column. If the resulting pivot does not divide a trailing entry, that row is added back to the pivot row and Euclidean reduction continues with a smaller pivot. When it divides every trailing entry, the algorithm advances. Explicit operation and entry-bit budgets stop coefficient growth honestly.

The verifier is smaller than this construction. It validates dimensions and integer types, multiplies UAV, checks every diagonal and zero entry, checks positivity/divisibility, and computes the two determinants exactly. It never calls the Smith constructor. Booleans are rejected as integer entries.

20.2 Image membership and a saturated fibre

For a proposed integral output y, write $u = Uy$. Unimodularity gives an exact integer change of coordinates $x = Vz$, so

$$y \in A\mathbb{Z}^D \iff d_i \mid u_i \ (i \le s), \qquad u_i = 0 \ (i > s).$$

When these conditions hold, put $z_i = u_i/d_i$ for $i \le s$, and set remaining coordinates to zero for one particular solution. Every integer solution is

$$x = x_0 + \sum_{j=s+1}^{D} t_j V_{[:,j]}, \qquad t_j \in \mathbb{Z}.$$

Proposition NX19.1. A replayed Smith certificate, the displayed divisibility test, and the displayed affine basis describe exactly the integral image and the full integer fibre.

Proof. Because U and V are integer unimodular matrices, both coordinate changes and their inverses preserve integer vectors. The equation becomes a diagonal system. Each nonzero diagonal coordinate has the unique stated integer solution; zero diagonal coordinates require zero transformed output and leave all corresponding z_j free. Transforming back gives all and only original integer solutions. □

The integer kernel is saturated: if kx is in the kernel for a nonzero integer k, then $kAx = 0$ in the torsion-free group $\mathbb{Z}^r$, so $Ax = 0$. The image need not be saturated. These statements are different.

For $A = (2, 1, 1)$, independently clearing denominators of a rational nullspace basis can give $(-1, 2, 0)$ and $(-1, 0, 2)$, whose integer span misses $(-1, 1, 1)$. The Smith fibre contains the missing vector. This permanent regression prevents a rational kernel from being mislabeled a complete integer kernel.

20.3 Original objects, finite completion counts, and multiplicity

An affine integer fibre can be infinite. Counting requires explicit inclusive bounds on every original coordinate. `BoundedFibrePopulation` retains the original tuple order and memoizes completion counts by coordinate position and residual output. Before branching, exact rowwise lower/upper bounds discard impossible residuals. The resulting recursion counts original labeled tuples without enumerating a Smith-coordinate box or sorting by a transformed order.

`count`, `contains`, `rank` and `unrank` all refer to those original tuples. Rank is zero-based lexicographic rank. `pushforward_population` enumerates a separately bounded original domain and sums multiplicities over outputs. Its total mass equals the original domain size by construction, and tests compare every fibre and every rank inverse with independent direct lexicographic enumeration.

For $A = (2, 2)$ and binary original states, the image is $\{0, 2, 4\}$ with weights $1, 2, 1$. Output one is impossible despite a rational solution. Output two has two original states, $(0, 1)$ and $(1, 0)$. The quotient identifier does not collapse their multiplicity. The retained JSON stores the affine description and the two bounded objects separately.

The exhaustive Smith corpus contains 837 matrices: every matrix with entries in $\{-1, 0, 1\}$ of shapes 1×3, 2×2, and 3×2. All 8,019 original states in the declared small boxes are independently checked for image membership. Additional population

tests cover rank zero, full rank, rectangular maps, nontrivial denominator clearing and the saturation regression.

20.4 Quotient updates and the image-lattice domain

Suppose an ambient transition M preserves $\mathbb{Z}^D$. This implementation requires every entry of M to be integral; a rational matrix can preserve the standard integer lattice only under that condition. A different invariant lattice needs a different explicit contract.

The transition descends to the observation quotient exactly when it preserves the observation kernel. The constructor checks this on the full Smith integer-kernel basis. It then constructs a rational output map Q and verifies directly

$$QA = AM.$$

The map Q may contain fractions. Its certified domain is $A\mathbb{Z}^D$, and every accepted update checks membership before application. For example, with $A = \mathrm{diag}(2, 1)$ and coordinate swap M,

$$Q = \begin{pmatrix} 0 & 2 \\ 1/2 & 0 \end{pmatrix}.$$

It sends the realizable output $(2, 3)$ to $(6, 1)$. An arbitrary integer vector $(1, 3)$ is rejected before its fractional update can be mistaken for a legal state. A rational ambient half-scaling is rejected for failure of standard-lattice preservation. A diagonal ambient update that does not preserve an observation kernel is rejected for a different reason.

Integer states and their multiplicities are preserved through these transitions. The packet does not authorize merging original seeds, graph labels, or height assignments after quotienting. An affine update can be handled only after adding and constraining a homogeneous coordinate; no unconstrained affine API is claimed here.

20.5 Portable contract and reproduction

The schema `singularity.integer-image/1` includes the canonical original rational matrix, its cleared integral matrix, row scales, U, V, diagonal, rank, original coordinate order and operation/bit resource counts. Malformed diagonals, nonunimodular transformations and forged identities fail replay. Changing a row scale without changing the original-map identity also fails replay. Zero-row input denotes a zero-by-zero map; a zero observation on a positive-dimensional state should be represented by an explicit all-zero row so dimension is retained.

Implementation: `src/singularity/integer_quotients.py`. Tests: `tests/test_integer_quotients.py`. Run `python scripts/integer_quotients.py --check`. Other tracks can consume `image_contains` as a necessary membership predicate without adopting Smith coordinates. In particular, arithmetic compatibility of an interface vector is not sufficient graph realizability: nonlinear slice identities and a common adjacency matrix remain additional constraints. Rational minimal dimension and faithful integer population representation are distinct optimization goals.

Chapter 21

NX20: finite quadratic-unit residue experiment automata

This adapter turns a declared arithmetic seed population into a finite exact experiment language. It preserves original seed IDs, distinguishes modular collisions from observational equivalence, and evaluates remote residues without constructing enormous Pell coordinates. It makes no claim to enumerate all global Pell seeds.

21.1 Certified arithmetic actions

A unit is specified by nonsquare $D > 0$ and integers a, b satisfying

$$a^2 - Db^2 = 1, \qquad U = \begin{pmatrix} a & Db \\ b & a \end{pmatrix}.$$

Multiplication sends (x, y) to $(ax + Dby, bx + ay)$. Direct expansion gives

$$(ax + Dby)^2 - D(bx + ay)^2 = (a^2 - Db^2)(x^2 - Dy^2) = x^2 - Dy^2.$$

Also $\det U = 1$, and the inverse is obtained by replacing b with $-b$. These identities certify integral norm preservation and invertibility modulo every positive modulus, including composite moduli. The constructor rejects square or nonpositive D, malformed units, floats and booleans.

For $D = 2$, $a = 3, b = 2$, the integer sequence beginning at $(1, 0)$ starts

$$(1, 0), (3, 2), (17, 12), (99, 70), ...$$

Every retained integer step is checked by direct quadratic multiplication. Modulo five its exact cycle is

$$(1, 0) \to (3, 2) \to (2, 2) \to (4, 0) \to (2, 3) \to (3, 3) \to (1, 0).$$

The cycle is computed, not copied as an assumed fixture. Bounded repeated integer multiplication and an independent cycle-remainder calculation check the remote implementation.

21.2 Which finite population is represented?

Input seeds form a nonempty mapping from stable string IDs to original integer pairs. Their provenance is retained verbatim as a scope declaration. The state constructor

begins with their actual reductions and closes under the declared legal actions. It does not replace that set with every solution of a modular norm equation.

Two projections remain explicit:

$$\{\text{original seed IDs}\} \longrightarrow \{\text{generated residue states}\} \longrightarrow \{\text{behavioral classes}\}.$$

For example, original seeds $(1, 0)$ and $(6, 0)$ collide modulo five but remain separate IDs with separate original norms. Their modular equality does not assert global orbit equality or even equality of the original norm equation. Conversely, original seeds $(1, 0)$ and $(4, 0)$ have equal initial y-sensor output modulo five but become distinguishable after one unit action: their readings are two and three. The cheapest witness has action cost one when that cost is declared as one.

The norm-congruence class can contain residues outside a chosen seed orbit. The adapter's closure is derived from the actual supplied seed images, and the tests retain this distinction. No global completeness conclusion follows from covering the finite closure.

21.3 Exact observations and chronological words

Each sensor reads x, y, or $x^2 - Dy^2$ modulo the declared modulus. Every output symbol becomes an indicator row. Thus equal readings correspond exactly to equal indicator vectors over $\mathbb{Q}$; numerical addition of residue values is not confused with finite-field equality.

For a finite transition f_i, the column-state matrix is $M_i e_s = e_{f_i(s)}$. A chronological word $(i_1, \ldots, i_k)$ applies i_1 first, giving

$$L_j M_{i_k} \cdots M_{i_1} e_s.$$

The adapter uses the existing observability closure and cheapest-distinguishing-experiment contracts. Independent finite-machine partition refinement computes the same all-future equivalence classes. The test begins with the sensor-output partition and repeatedly refines by successor classes until stable. This oracle uses transition tables directly, without rational row-space elimination.

A sensor reading only the norm residue is forever blind between two states in the same norm class, because every legal unit action preserves the norm. The permanent fixture records such a genuinely all-future equivalent pair. This is a theorem for the declared finite sensors and actions, not a statement that the original arithmetic objects coincide.

Observation costs and positive action costs are exact rationals. Different sensor costs survive compilation by assigning the cost to every symbol-indicator row of that sensor. The least-cost witness returned by the existing engine therefore remains a real sensor/action experiment. One-hot dimension and dense-matrix limits are checked before allocation; a residue space can contain m^2 states, so unrestricted dense compilation is not implied.

21.4 Commuting units and a restricted noncommuting extension

Units in the same quadratic field commute. Their matrices have the form $aI + bK$, where $K^2 = DI$, so multiplication is a commutative polynomial calculation. Adding

more such units cannot create order sensitivity.

Conjugation is a separately declared action

$$J = \mathrm{diag}(1, -1), \qquad JUJ = U^{-1}.$$

It preserves the norm but can leave a positive-coordinate seed convention. The API therefore requires `signed_states=True` before allowing it. For $(1, 0)$ modulo five, unit then conjugation gives $(3, 3)$, while conjugation then unit gives $(3, 2)$. The actual chronological column-state actions reproduce this difference. The receipt treats this as a signed norm-preserving language, not as a positivity-preserving Pell iteration.

No arbitrary reset, projection or translation is called a Pell symmetry. Unsupported partial actions are rejected rather than silently becoming identities. Extending the action alphabet requires a new legality contract.

21.5 Remote residues without giant integer coordinates

`remote_residue` accepts a nonnegative integer or a decimal digit string. To process a decimal prefix exponent k and next digit d, update

$$U^{10k+d} = (U^k)^{10} U^d \pmod{m}.$$

Small binary matrix powers perform each step. This avoids parsing a thousands-digit integer into Python's bounded decimal conversion path and avoids constructing the enormous exact Pell coordinates. Negative indices require an explicit inverse-action flag and use the certified inverse unit.

The retained remote index has 6,000 decimal digits, the pattern `9876543210` repeated 600 times. Decimal-stream matrix powering matches independent digit-stream reduction modulo the separately computed cycle length. The output is a residue pair. Nothing in the receipt claims that the corresponding full integer coordinates were computed cheaply.

21.6 Serialization, tests and completion boundary

`singularity.arithmetic-orbit/1` records the exact field/unit, modulus, original seeds and their norms, provenance, generated residue states, seed-to-state map, named actions, action and sensor costs, transition tables, signed-state contract and explicit scope. Replay reconstructs arithmetic transitions and requires exact equality with the serialized receipt; an altered original norm is rejected. The tests cover modulus one, composite moduli, negative residues, malformed units, original modular collisions, forever-equivalent states, costed witnesses, huge indices and conjugation chronology.

Implementation: `src/singularity/arithmetic_automata.py`. Tests: `tests/test_arithmetic_automata.py`. Run `python scripts/arithmetic_automata.py --check` for the retained exact evidence. Completion concerns the finite declared population and language. Arbitrary Pell seed completeness, global orbit identification, positive global branch conventions and duplicate removal across a future PerfectPower adapter require separate mathematical proofs.

Chapter 22

NX21 — Exact symmetry channels and costed repairs

The implementation supports verified finite rational representations, rational projectors, exact row-space closure, original-model permutation conjugacy, and finite-catalogue robust sensor repairs. It never substitutes a floating eigenspace for an exact channel certificate.

22.1 Two distinct blindness theorems

For a single state system let an idempotent P commute with every allowed operator M_i. If the actual output rows obey $LP = 0$, then for every legal word,

$$LM_{i_m} \cdots M_{i_1} P = LPM_{i_m} \cdots M_{i_1} = 0.$$

Thus the channel $\operatorname{im} P$ is permanently invisible under those operators and sensors. Reachability and whether the actual original population occupies that channel are separate questions.

For two candidate models the relevant theorem is instead intertwining. Suppose

$$A_b P = P A_a, \quad B_b = P B_a, \quad x_{0,b} = P x_{0,a}, \quad C_b P = C_a.$$

Induction gives $x_{t,b} = P x_{t,a}$ under every shared input word, and hence equal outputs at every time. Neither faulted matrix needs to commute with P. The sparse thermal adapter checks this theorem on the original candidate entries and a bijective original-state permutation. It uses the fixed zero initial state and actual named sensor rows.

22.2 Rational finite groups

`FiniteRepresentation` checks the entire finite multiplication table: identity, inverses, associativity, and every representation multiplication identity. It rejects duplicated group labels, incompatible dimensions, non-rational entries, and oversized groups. `verify_channel_decomposition` independently checks

$$P_i^2 = P_i, \qquad P_i P_j = 0 \ (i \neq j), \qquad \sum_i P_i = I, \qquad [P_i, M_a] = [P_i, \rho(g)] = 0.$$

For an involution $S^2 = I$, `reflection_channels` returns $(I + S)/2$ and $(I - S)/2$. Rational division is exact.

The corpus includes the rational regular representation of C_3, with a rank-one constant channel and rank-two augmentation channel. The augmentation representation is irreducible over $\mathbb{Q}$; over $\mathbb{C}$ it splits into two conjugate one-dimensional characters. The release does not identify these field conventions. A Klein-four corpus verifies four rational channels and repeated irreducible multiplicities in two regular copies.

22.3 A necessary caveat about summing channel ranks

Commutation alone does not make projected sensors physically available. For example $M = I$, a reflection that swaps two coordinates, and the one sensor $(1, 0)$, have full observable rank one, while separately projecting the sensor onto the plus and minus channels produces two nonzero rows. Claiming rank two for the actual sensor would be false.

`channel_observability` therefore computes the actual full closure and all projected-channel closures, then explicitly reports whether the dimensions add correctly. It does not infer equality from commutation. In the retained finite-group examples the available group operators generate the necessary channel separation, and full closure rank equals the sum. Original observation semantics are preserved.

22.4 Exact uncertainty transport

A basis change generally destroys independence of coordinate noise. For an original box $D[-1, 1]^n$, a transform T produces the zonotope

$$TD[-1, 1]^n.$$

`transformed_noise_set` retains the factor TD, not merely its coordinate bounds. For a Euclidean-ball model it retains the ellipsoid factor and Gram matrix

$$G = TD^2T^T.$$

For $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, unit radii produce Gram $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Replacing it by a unit ball would erase both scaling and correlation. The physical repair examples work directly with original sensor intervals and need no such change of coordinates.

22.5 Thermal reflection and cheapest repair

The original blocked-left and blocked-right candidates on the 12-by-12 source-derived plate are related by horizontal reflection. The heater is invariant; the corner mean and central mean sensor rows are invariant. Sparse conjugacy verifies the all-input all-time blindness theorem for those rows. Direct constant and alternating input words through horizon twelve provide independent finite checks of the identities' consequences.

The repair catalogue contains original mean sensors and one newly installed top-left point sensor, with declared total accounting costs and absolute reading error $1/1000$. Each experiment starts from a reset zero state. The cheapest target-separating repair in this catalogue is the existing left-center mean after two unit input steps, cost three. The two exact readings differ by $1/64$, exceeding the sum of their error radii $1/500$.

Every one of the 32 possible capability panels is independently checked by finite enumeration. The far-corner point sensor breaks reflection but gives equal readings at

this short horizon, because influence has not reached that location. It is retained as an explicit counterexample: symmetry breaking does not establish target observability.

A repair separates a target pair exactly when some installed reading has disjoint closed uncertainty intervals. `minimum_channel_repair` optimizes total positive rational catalogue cost and retains actual experiment words, sensor descriptions, errors and original candidate IDs. Its optimality is restricted to the declared finite catalogue, independent-reset semantics and interval error model. It does not claim minimal laboratory hardware over all imaginable actions.

Replay `python scripts/symmetry_channels.py --check`. The strict representation and projector constructors reject broken commutation or multiplication identities before a channel proof is accepted.

Chapter 23

NX22 — Finite-volume multi-frequency arithmetic probing

This release reconstructs original integer energy histograms from complete exact phase samples. It distinguishes this finite-volume theorem from periodic full-series identities and from finite-candidate phase identification.

23.1 Degree bound and complete transform

For a simple graph and a nonnegative integer profile of volume r,

$$Q_G(h) = \sum_v d_v h_v^2 - 2 \sum_{uv \in E} h_u h_v \le \Delta \sum_v h_v^2 \le \Delta r^2.$$

Thus every coefficient polynomial through volume R has degree at most $D = \Delta R^2$. For $M > D$, let ζ be primitive of order M in the ambient exact ring $\mathbb{Z}[t]/\Phi_M(t)$. Then

$$a_{r,e} = \frac{1}{M} \sum_{k=0}^{M-1} P_r(\zeta^k) \zeta^{-ke}.$$

Expanding P_r, the inner sum is M for equal exponent residues and zero otherwise. Since every exponent lies in $[0, D] \subset [0, M-1]$, residue equality is genuine exponent equality. This proves exact integer reconstruction including the original labeled-profile multiplicities.

All phase indices, including zero and nonprimitive phases, are embedded in the same order-M ring. The inverse sum must reduce to an integer multiple of the unit, with the integer exactly divisible by M. Negative reconstructed multiplicities or incorrect total profile mass are rejected. The mass checksum is

$$\sum_e a_{r,e} = \binom{r+n-1}{n-1}.$$

`phase_table` supports independent direct height-composition and periodic residue-box routes. The latter first forms a finite numerator in the same phase field, then expands its rational denominator. Their equality is tested before inverse reconstruction is compared with symbolic coefficient enumeration.

23.2 Aliasing is a theorem boundary

For one edge at volume two,

$$P_2(q) = 1 + 2q^4.$$

Order five recovers energies zero and four with multiplicities one and two. Order four instead produces residue histogram $\{0 : 3\}$. Every exact order-four phase sample is three: no arithmetic precision can undo the lost residue distinction. The unrestricted reconstruction API therefore requires the strict bound $M > D$. An explicitly named `allow_aliasing` diagnostic mode returns residue counts and makes no original-energy claim.

The certificate checker reconstructs the graph and recomputes its degree bound. Missing or duplicate phase labels, unknown fields, noncanonical cyclotomic tuples, a forged bound, nonintegral inverse results and mass failures are rejected. `verify_phase_reconstruction` additionally reruns the declared original-model sample route.

23.3 Three different design questions

Problem	Assumption	Certified conclusion
Complete histogram recovery	Every phase, exact samples, proven $M>D$	All original energy multiplicities through declared volume
Finite candidate identification	Explicit finite candidate IDs, phase-output tables, positive rational costs	Cheapest phase panel separating that declared population
Support-conditioned sparse recovery	An externally declared exact support set	Unique multiplicities on that support when the checked sample matrix has full column rank

The candidate panel optimizer enumerates every phase subset in its bounded catalogue. Candidates remain distinct original identities even if every available phase makes them observationally equal; in that case it reports infeasibility rather than deduplicating them.

For conditional sparse recovery, expand each cyclotomic sample over its rational basis. This gives an ordinary exact rational linear system in the unknown support masses. Full column rank proves uniqueness within the declared support, while consistency and nonnegative integrality validate the recovered masses. The algorithm neither discovers an unknown support nor proves compressed-sensing uniqueness across all sparse supports. In particular two support exponents congruent modulo M can never be distinguished by these phases, and rank testing rejects that case.

23.4 Retained census and costs

`results/phase_probes.json` contains all eleven labeled graphs of orders one through three at volume two, with direct and residue routes, explicit histograms and total original profile masses. A disconnected graph with an isolated vertex is also checked at composite order six, including its nonprimitive phases. The finite four-candidate panel uses five potential phases; exhaustive subset search finds a cost-two singleton phase that separates empty, edge, path and triangle candidates at volume one. The

single-edge order-five sparse example recovers counts one and two from a sample under declared support $\{0, 4\}$.

The complete transform does not promise a computational speedup. Direct enumeration costs $\binom{R+n}{n}$ profiles per phase; a residue route can cost M^n, and the ambient field dimension is $\varphi(M)$. Polynomial coefficients can instead be obtained from $D + 1$ distinct rational evaluations by ordinary Vandermonde interpolation. Neither alternative is universally better: phase arithmetic avoids numerical approximation but retains exact coefficient and storage growth. The receipt reports which routes actually ran rather than hiding their setup.

Run `python scripts/phase_probes.py --check`.

Chapter 24

NX23 — Population geometry with original multiplicities

The original objects are explicit finite labeled assignments, not the compressed feature classes. Their declared rational weights define a probabilistic model. All target probabilities, posterior risks and experiment objectives below depend on those weights; no uniform physical fault prior is inferred from a catalogue.

24.1 Pushforward masses and exact tilts

For original objects ω, nonnegative rational weights w_ω, integer feature tuples $f(\omega)$, and positive rational tilts t, define

$$m_z = \sum_{f(\omega)=z} w_\omega, \qquad Z(t) = \sum_z m_z t^z, \qquad p_t(\omega) = w_\omega t^{f(\omega)}/Z(t).$$

Signed bounded features give Laurent monomials. Positivity of every tilt and positive total original weight ensure a nonzero denominator. A zero-weight original object retains its identity but contributes no probability. `WeightedPopulation` preserves stable original IDs and an original-object rank/unrank map; it rejects duplicate IDs, noninteger features, negative weights and zero total mass.

`pushforward_mass` sums original weights. `joint_target_mass` instead keys by both feature tuple and target, retaining ambiguity when one feature class contains multiple targets. For rational tilts, normalization, means and covariance are exact fractions:

$$\mu_i = Z^{-1} \sum_\omega w_\omega t^{f(\omega)} f_i(\omega),$$

$$\Sigma_{ij} = Z^{-1} \sum_\omega w_\omega t^{f(\omega)} (f_i(\omega) - \mu_i)(f_j(\omega) - \mu_j).$$

This is the gradient/Hessian of $\log Z(e^u)$ without evaluating logarithms. Positive semidefiniteness follows because $a^T \Sigma a$ is an average of squared centered values. It vanishes exactly when $a \cdot f$ is constant on positive-weight support.

For two bits with feature their sum, the feature masses are $1, 2, 1$, not three unit masses. At tilt two,

$$Z = 9, \quad \mu = 4/3, \quad \text{Var} = 4/9.$$

The middle feature class contains both values of the first-bit target. Replacing original objects by one object per feature would change both the prior and the target problem.

24.2 Observation sufficiency and Bayes decisions

A `FiniteObservationChannel` declares a finite observation alphabet and a normalized nonnegative rational likelihood row for every original ID. If two originals share a compressed key but have different likelihoods, that key does not support the proposed observation semantics. `verify_channel_compression` rejects such a compression. A target-enriched key is allowed only when it actually resolves every relevant likelihood difference.

Bayes updating is direct original-object summation. An observation with zero probability raises `UnsupportedObservation`, explicitly rejecting the declared model instead of dividing by zero. At zero-one target loss, the optimal expected error is

$$R(e) = 1 - \sum_{y} \max_{\tau} \sum_{\omega : T(\omega) = \tau} p_t(\omega) K_e(y \mid \omega).$$

For each observation, choosing a target with maximal joint mass maximizes correct classification probability; summing these disjoint events proves the formula. The implementation uses exact rational arithmetic and independently checks the original-ID likelihood coverage.

Panels are declared conditionally independent given each original object. Their joint likelihood is the product of constituent likelihoods. This is an explicit assumption, not a consequence of bounded error. The exact design engine exhausts every catalogue panel within the positive rational cost budget and minimizes expected error, with cost and stable panel IDs as deterministic tie breakers. Resource guards bound channel count and joint observation outcomes.

24.3 Bayes and worst-case objectives genuinely differ

Consider three original states with different targets and prior masses $9/20, 9/20, 1/10$. One channel perfectly distinguishes the first two states but aliases the third with the first. Its Bayes error is $1/10$. Another reports the correct target with probability $17/20$, assigning $3/40$ to each other label; its Bayes error is $3/20$. At a one-experiment budget Bayes therefore selects the first channel.

For deterministic decision rules, the first channel has worst-original-state error one: whichever target is assigned to the aliased observation, the other aliased original is always wrong. The second channel has minimax error $3/20$. Exhausting all finite decision maps proves these values. `deterministic_minimax_risk` explicitly names this decision class; it does not claim optimization over randomized rules. It ignores prior weights, unlike Bayes risk.

This distinction is not an inconsistency. The first decision knowingly accepts failure on a low-prior state. A robust target guarantee requires preventing that failure, and cannot be obtained by silently interpreting expected error as a worst-case bound.

24.4 Integration and replay

The retained `FaultPopulation` has four named original bonds and at most two blocked bonds: eleven original masks. A synthetic weight $2^{-\#faults}$ is declared, not estimated. Feature compression by total fault count retains original multiplicity and target ambiguity; direct bond-inspection channels retain every original mask ID. The design is synthetic inspection semantics with declared accounting costs, not an empirical thermal-sensor likelihood.

The NX19 lattice example uses output $(2,2)x = 4$ inside $0 \le x_i \le 2$. The three original vectors $(0,2),(1,1),(2,0)$ have one compressed sum feature and three distinct first-coordinate targets. Their feature mass is three, and their joint target table has three entries. Original-coordinate rank/unrank and mass conservation are checked without treating Smith coordinates as new objects.

`results/population_geometry.json` retains the original masks and fibre members, mass tables, exact binary moments, posterior example, finite design certificates, Bayes/minimax disagreement and scope assumptions. Run `python scripts/population_geometry.py --check`.

Chapter 25

NX24 — Mechanism-certified finite prefixes

This release completes NX24-A/B/C for explicitly restricted families. It does not give a universal cutoff for arbitrary graph-height series. The new graph theorem is intentionally narrow: a single weighted edge with a known number of isolated vertices. Its constructive mechanism is an all-coefficient ideal-generation identity, rather than a guessed recurrence or temporary ideal stabilization. A supplemental binomial-ODE family demonstrates a common comparison recurrence for a genuinely nonrational series.

25.1 Unified contracts and existing baselines

`FiniteStatePrefixCertificate` defines the original sequence itself by exact rational matrices $f_n = cA^n s_0$. For two declared systems of dimensions d_1, d_2, put their transitions in a block-diagonal matrix and subtract their output rows. The difference is a scalar output of a $d_1 + d_2$-dimensional system. Cayley–Hamilton gives a constant recurrence of that degree, so agreement through index $d_1 + d_2 - 1$ controls every later coefficient. This conclusion concerns these declared linear-state families, not an external sequence lacking a proved adapter.

`PeriodicPrefixCertificate` wraps the existing periodic graph theorem without enlarging its scope. For the same order n and phase modulus M, the existing palindromic numerator gives the sufficient cutoff $\lfloor n(M-1)/2 \rfloor$. Direct original profile enumeration remains the coefficient oracle.

Certificates have a strict schema with reduced rational numerator/positive-denominator pairs. Unknown kinds, extra fields, zero denominators, guessed recurrence payloads and unsupported family names are rejected. The finite-state constructor bounds dimension. No constructor accepts finite samples as a mechanism. A repeated ideal, a parameter count, and an arbitrarily long zero prefix are deliberately rejected rather than interpreted as proof.

25.2 A new nonperiodic graph-family ideal theorem

Let G_k be one edge with $k \geq 0$ known isolated labeled vertices. Give the edge its ordinary parameter q. The original nonnegative-height series is

$$F_{k,q}(x) = \sum_{h \in \mathbb{N}^{k+2}} x^{|h|} q^{(h_1 - h_2)^2}.$$

Write the two endpoint heights as a common nonnegative minimum plus a signed difference. A zero difference has one orientation; a positive difference has two. Each common translation adds volume two. Hence

$$F_{k,q}(x) = \frac{1 + 2\sum_{d \geq 1} q^{d^2} x^d}{(1-x)^k (1-x^2)}.$$

This identity comes from a direct bijection of original labeled profiles. It is not inferred from observed coefficients. Its first coefficient is

$$f_1(q) = k + 2q.$$

For two parameters q, η, put

$$S_m(q,\eta) = \sum_{i=0}^{m-1} q^{m-1-i} \eta^i, \qquad q^m - \eta^m = (q - \eta) S_m(q,\eta).$$

The latter is a polynomial identity valid even when $q = \eta$: there is no forbidden division by a parameter difference.

Let $c_{r,d}$ be the nonnegative integer multiplicity obtained by multiplying the known denominator series; explicitly it counts the compatible common translations and isolated-vertex compositions. Then

$$f_r(q) - f_r(\eta) = 2(q - \eta) \sum_{d=1}^{r} c_{r,d} S_{d^2}(q,\eta).$$

Therefore every coefficient difference lies in the ideal generated by the first difference, and conversely the first difference is one of those generators:

$$\langle f_0(q) - f_0(\eta), f_1(q) - f_1(\eta), \ldots \rangle = \langle 2(q - \eta) \rangle \subset \mathbb{Q}[q, \eta].$$

This proves an explicit sufficient cutoff **one**, with polynomial witnesses for every future coefficient. `ideal_difference_quotient` evaluates those witnesses directly by telescoping, including the singular-looking diagonal $q = \eta$, without cancellation assumptions. This is the requested coefficient-generation mechanism; finite agreement beyond the cutoff merely checks its implementation.

25.2.1 Nonrationality and exact exceptional strata

For $0 < q < 1$, the numerator

$$B_q(x) = 1 + 2\sum_{d \geq 1} q^{d^2} x^d$$

is entire by the root test and has infinitely many nonzero coefficients. If $F_{k,q}$ were rational, multiplying by $(1-x)^k(1-x^2)$ would make B_q rational. An entire rational function is polynomial, a contradiction. Thus the new cutoff genuinely covers a nonperiodic nonrational graph-series family.

The endpoint parameter strata are handled exactly rather than discarded:

$$F_{k,0} = \frac{1}{(1-x)^k (1-x^2)}, \qquad F_{k,1} = \frac{1}{(1-x)^{k+2}}.$$

At zero only equal endpoint heights contribute; at one all original profiles have unit weight. The same first-coefficient generation theorem remains valid. Comparisons require the same declared k; comparing different known graph orders through this theorem would be an invalid extension.

This is a theorem about the weighted-edge parameter family, not a full-series completeness theorem for all graphs at one fixed positive parameter. It does not resolve the surviving symmetric-interface realizability problems.

25.3 Supplemental polynomial-index comparison recurrence

Let $H_a(x) = (1-x)^{-a}$ as a formal power series over $\mathbb{Q}[a]$. From the formal ODE

$$(1-x)H_a' = aH_a, \qquad H_a(0) = 1,$$

its coefficients satisfy

$$(n+1)h_{n+1}(a) = (n+a)h_n(a), \qquad h_n(a) = \frac{a(a+1)\cdots(a+n-1)}{n!}.$$

The recurrence follows by equating formal coefficients; the product formula follows by induction. It remains meaningful at negative integer parameters, where the sequence terminates, and at zero. For positive noninteger rational a, the series is nonrational.

For comparing different parameters, merely subtracting their two first-order recurrences would not control their difference. Instead elimination gives the **common** difference recurrence for $y_n = h_n(a) - h_n(b)$:

$$(n+1)(n+2)y_{n+2} - (n+1)(2n+a+b+1)y_{n+1} + (n+a)(n+b)y_n = 0.$$

To verify it, substitute the individual ratio formulas separately: each sequence is annihilated by this same second-order operator, because

$$(n+1)(n+2)\frac{(n+a)(n+a+1)}{(n+1)(n+2)} - (n+1)(2n+a+b+1)\frac{n+a}{n+1} + (n+a)(n+b) = 0.$$

The expression factors $(n+a)[(n+a+1)-(2n+a+b+1)+(n+b)] = 0$. The identical argument with b works, including vanishing numerator cases since the displayed cleared identity is polynomial.

The leading term $(n+1)(n+2)$ never vanishes for integer $n \geq 0$, for any parameter values. Thus $y_0 = y_1 = 0$ determines every later difference by induction, giving cutoff one. All singular-looking individual ratios at negative integer numerator parameters are harmless; only division by positive integer indices occurs. This supplemental formal-ODE family is explicitly separate from arbitrary graph-height populations.

25.4 Census, strict failures and resources

The retained graph corpus has $k = 0, 1, 2, 3$, parameters $0, 1/5, 1/3, 1/2, 2/3, 4/5, 1$, 196 ordered parameter comparisons, and direct original-profile coefficients through volume ten. Every all-coefficient ideal witness through that volume is checked independently of the comparison routine. The binomial corpus includes negative integers,

negative nonintegers, zero, positive nonintegers and positive integers, with 684 common recurrence identities through index eighteen and independent rising-factorial coefficients.

A delayed family $F_b(x) = bx^{100}$ agrees at every earlier coefficient although it is nonconstant. Likewise consecutive ideals may repeatedly equal $\langle a - a' \rangle$ before a later coefficient introduces $b - b'$. Such sample payloads do not match any accepted certificate mechanism. A parameter count cannot provide a missing transition identity. Canonical denominator parsing rejects denominator degeneration before any arithmetic.

Resource limits and refusal are separate from mathematical outcomes. The retained corpus is small; exact coefficient sizes can grow rapidly at larger rational denominators. No timeout or refused resource request certifies inequality or infinite ambiguity. The replay data records source digest, full parameter census, singular strata, mechanism and sufficient cutoff, so the work can resume without treating finite empirical agreement as a theorem.

Run `python scripts/prefix_certificates.py --check`.

Chapter 26

NX25 — Future targets require information about their drivers

This is an exact finite linear application layer with a separate measured ASOS pilot. The algebra does not validate a weather model. Source: `src/singularity/forecast_targets.py`. Exact evidence: results/forecast_targets_synthetic.json. Pre-fit configuration: applications/nx25_30/PILOT_FREEZE.json.

26.1 A target is a row, not an entire reconstructed state

For declared dynamics $x_{t+1} = Ax_t + Bu_t$,

$$J_\tau = g^\top A^\tau x_0 + \sum_{t=0}^{\tau-1} g^\top A^{\tau-1-t} B u_t.$$

Known inputs contribute the known offset. Unknown disturbances require extra state directions or a separate uncertainty population. The known_input_offset function evaluates forcing by forward dynamics; forecast_row compiles $G_\tau = g^\top A^\tau$. State schema, target interpretation, time interval and admissible domain must be declared.

For $x_0 = x_c + Zz$ and $y = Hx_0 + e$, exact recovery is equivalent to

$$G_\tau Z = LHZ \quad \Longleftrightarrow \quad \ker(HZ) \subseteq \ker(G_\tau Z).$$

The forward implication follows by applying the factorization to a null vector. Conversely, define the linear map $HZz \mapsto G_\tau Zz$ on the image of HZ. The kernel inclusion makes it well defined, and a linear extension gives L. Rational elimination constructs a decoder or an invisible direction $n = Zz$, $Hn = 0$, $G_\tau n \neq 0$. To turn a direction into two finite admissible states, the declared domain must actually contain their displacement.

The affine decoder is $G_\tau x_c + L(y - Hx_c)$, plus known forcing. Ambient full rank is unnecessary: when velocity is fixed to zero, position alone determines a future position even though the observation has ambient rank one. Centres are retained separately from admissible directions.

26.2 Cost and attainable error

For a scalar decoder with $w^\top HZ = G_\tau Z$ and deterministic coordinate-box noise,

$$\sup_{|e_i|\le\epsilon_i} |w^\top e| = \sum_i |w_i|\epsilon_i.$$

Aligned error signs attain this bound. The implementation minimizes it over all exact linear estimators, rather than taking the first available decoder. Splitting weights into nonnegative positive/negative parts gives a linear program. Some optimum has at most the equality rank in nonzero variables and an independent row support; opposite signs on the same coordinate can be cancelled without increasing cost. Enumerating independent supports therefore finds a global optimum, including zero noise radii.

The finite panel solver minimizes acquisition cost subject to a declared error cap, then error as a tie breaker. Without a cap it minimizes error first. Strict and closed thresholds are different contracts. An infeasible panel catalogue can fail through structural blindness or excessive noise; only structural failure necessarily supplies a null witness. Arbitrary correlated noise and shared setup costs are not silently substituted into this box contract.

Exact fixture	Result	Meaning
$A=\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$, $g=(1,0)$, $\tau=3$	$G_3=(1,3)$	Future position needs its persistent driver
Position alone	$n=(0,1)$	Velocities $1,-1$ conceal target gap six
Errors $1/10,1/5$ on position/velocity	$7/10$	Exact, attained worst-case error
Closed cap $7/10$	Two cost-one sensors feasible	Boundary included
Strict cap $<7/10$	Retained catalogue infeasible	Touching intervals do not become separated through rounding

Tests independently evolve states through horizons zero to eight, enumerate box vertices, and compare the panel solver with a direct two-dimensional determinant oracle over all 64 panels of six sensors. The oracle uses no elimination helper. The declared state box and known-input assumptions are retained in the synthetic receipt.

26.3 What the measured pilot establishes

The actual WIND continuation archive was recovered. Its BOS ASOS source receipt binds the original CSV to SHA-256 `a7def62ca977d281a51fbae632252dce6afefc8c45aed5afbb032f1372a2c21a`. The four-day public subset and source receipt are bundled. This is an ASOS station pilot, not a replay of the older 272-volume NEXRAD claim or a validated radar-to-ground transfer.

The target is reported sustained wind in knots at exactly +60 minutes. Training uses September 25–26, 2026, and training targets also stay before September 27. Evaluation uses September 27–28. Missing observations are not interpolated. Each policy uses two prespecified scalar features at declared equal feature cost and identical complete cases. Feature cost is bookkeeping, not a claim about equal instrument purchase prices.

Policy	Frozen features	Held-out MAE, knots	Held-out MSE
Direct target	current wind, wind 30 minutes earlier	2.18648	8.49312
Driver aware	current wind, 30-minute altimeter change	2.16858	8.14812

Policy	Frozen features	Held-out MAE, knots	Held-out MSE
Generic information	current altimeter, current direction	5.33963	35.54742

There are 192 training and 57 evaluation cases. Among 1,219 source reports, 258 lack an exact required lag/future timestamp and 705 lack a required field. Only two evaluation days support the descriptive day-block bootstrap. The small driver-policy gain is not evidence of a general operational advantage. Direction is a declared scalar feature, not a circular physical metric.

The optional NumPy runner fits standardized training-only least squares. The public standard-library checker independently fits rational normal equations and replays point scores to 10^{-9}. Bootstrap summaries are integrity checked. None of these numerical checks bounds atmospheric model error.

Offline replay: python3 scripts/nx25_30_replay.py –check.

Chapter 27

NX26 — Ambiguity, curvature and finite witness repairs

The nonlinear_fibres module implements bounded rational polynomial witnesses, not a complete semialgebraic optimizer. Limits are four variables, degree four and 32 monomials per polynomial. Domains contain finite rational boxes and original polynomial equalities/inequalities. Every witness is substituted into those original constraints.

27.1 A pair proves a lower bound

For compact D, observations H, constraints $C = 0$, and target T,

$$\mathcal{F}_y = \{x \in D : H(x) = y,\ C(x) = 0\}, \qquad A_T(y) = \sup_{x,x' \in \mathcal{F}_y} |T(x) - T(x')|.$$

A verified pair proves a target-diameter lower bound. It normally does not prove the diameter. Receipts keep target_gap, diameter_lower_bound and complete_diameter separate. A zero observational residual means exact rational equality.

For $H(x_1, x_2) = x_1,\ T = x_2^2,\ D = [-1, 1]^2$, states $(0, 0), (0, 1)$ have gap one even though the target gradient at the origin is zero. This special diameter is exactly one because $0 \le x_2^2 \le 1$, with both endpoints attained. The elementary upper bound is recorded separately; the generic witness verifier does not infer it.

For $H(x) = x^2,\ T(x) = x,\ D = [-1, 1]$, states $-1, 1$ share observation one and have gap two. Observation derivatives are nonzero at both states. Local invertibility therefore does not settle a global branch.

27.2 Exact quadratic constancy

Let $x = x_0 + Nz$ and $T(x) = g^\top x + \frac{1}{2} x^\top Q x$. Symmetrize Q, which preserves the polynomial. Substitution gives

$$T(x_0 + Nz) = T(x_0) + z^\top N^\top (g + Q x_0) + \tfrac{1}{2} z^\top N^\top Q N z.$$

On a parameter domain containing a relatively open neighbourhood, constancy is equivalent to

$$N^\top (g + Q x_0) = 0, \qquad N^\top Q N = 0.$$

Necessity follows because a polynomial vanishing on an open set has zero coefficients; sufficiency is immediate. A finite admissible set can make a nonzero polynomial accidentally constant, so a failed criterion without the open-domain hypothesis returns domain_analysis_required. Independent tests directly expand substituted polynomials and compare coefficients.

27.3 Curvature belongs to the fibre too

Write $K = (H - y, C)$. Along a constraint-preserving curve with tangent n and acceleration a,

$$DK\, n = 0, \qquad DK\, a + \left(n^\top \nabla^2 K_i\, n\right)_i = 0.$$

At a regular stationary point, $DT = \lambda^\top DK$, so

$$\frac{d^2}{ds^2} T(x(s))\Big|_0 = n^\top \left(\nabla^2 T - \sum_i \lambda_i \nabla^2 K_i \right) n.$$

The code checks the point lies on the original constraints, verifies Jacobian rank and tangency, solves the multiplier identity, and supplies a compatible acceleration. Singular points are refused as singular_stratum; regular nonstationary points are not forced into a stationary Hessian formula.

Fibre and target	Ambient target Hessian	Correct intrinsic variation
$y-x^2=0$, $T=y$, $(x,y)=(0,0), n=(1,0)$	zero	2
$x^2+y^2=1$, $T=x$, $(x,y)=(1,0), n=(0,1)$	zero	-1
Same circle, rational charts with tangents $(0,\pm 2)$	zero	-4 in either chart parameter

Direct parametrization $T(s, s^2) = s^2$ proves the parabola result independently. The two circle charts

$$(x, y) = \left(\frac{1 - u^2}{1 + u^2}, \frac{\pm 2u}{1 + u^2} \right)$$

obey $(1 + u^2)(T(u) - 1) = -2u^2$, giving second derivative -4. Parameter scaling matters, so tangent vectors stay in the receipt.

27.4 Finite coverage is not a global theorem

A sensor S with deterministic radius ϵ separates a witness pair only when

$$|S(x) - S(x')| > 2\epsilon.$$

The finite optimizer enumerates at most six sensors and 256 retained pairs. It returns the least declared cost and explicitly says its scope is retained finite pairs only.

With old observation $x^2 + y^2 = 1$ and target $x + y$, sensor x repairs $(-1, 0), (1, 0)$. Yet $(0, -1), (0, 1)$ still share the old observation and the repair reading while their target gap is two. Removing every sampled witness can leave an unseen branch. This is retained as a permanent regression.

No interval subdivision, numerical candidate finder or general diameter solver is claimed. The synthetic bounded pilot implements affine constancy, exact global

branch witnesses, curvature-aware regular analysis and a finite repair whose global limitation is demonstrated.

Chapter 28

NX27 — A target can close when a coarse state does not

The multiscale_transport module implements exact finite linear transfer and separately labelled measured temporal-aggregation evidence. The source graph-height series is not presented as a physical heat equation.

28.1 The defect is a failure of commutation

For fine/coarse steps A_f, A_c and projection E,

$$D = EA_f - A_cE, \qquad e_{t+1} = A_ce_t + Dx_t.$$

The identity $EV = I$ for a reconstruction map V does not imply $D = 0$. The four-cell reflecting diffusion pilot has $A_f = I - L_{\text{path}}/8$,

$$E = \frac{1}{2}\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \quad A_c = \begin{pmatrix} 15/16 & 1/16 \\ 1/16 & 15/16 \end{pmatrix},$$

and

$$D = \frac{1}{32}\begin{pmatrix} 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{pmatrix}.$$

At $x = (0, 1, 0, 0)$, projected fine evolution is $(7/16, 1/16)$, while coarse evolution is $(15/32, 1/32)$. Their difference is exactly $(-1/32, 1/32)$. Independent cell-flux updates reproduce every matrix entry.

Total mass closes exactly: $\mathbf{1}^\top A_f = \mathbf{1}^\top$. Pair averages do not. Thus target closure and full coarse-state closure are separate design questions. The exact_target_quotient function supplies a rational T only when $EA_f = TE$.

28.2 Domains and horizon bounds

For a correlated domain $x = c + Gz,\ |z_i| \le 1$, each defect component has exact range

$$(Dc)_j \pm \sum_i |(DG)_{ji}|.$$

Each endpoint is individually attainable. Different component extrema need not share one state. A box is the special case $G = \operatorname{diag}(r)$. On the unit four-box the infinity-norm defect bound is $1/8$, independently checked at all sixteen vertices.

With proved reachable-domain defect bounds δ_t and induced coarse operator bounds L_t,

$$b_{t+1} = L_t b_t + \delta_t$$

is a valid finite-horizon error bound by induction and the triangle inequality. The reachable_zonotopes function preserves initial generator correlations and appends independent per-step forcing generators. It does not replace one persistent disturbance by independent disturbances at every time. Tests compare actual trajectories against these enclosures, including bounded nonzero inputs.

Adding the single coordinate $D_1 x$ to the two pair averages yields an exact three-dimensional quotient:

$$\widetilde{A}_c = \begin{pmatrix} 15/16 & 1/16 & 1 \\ 1/16 & 15/16 & -1 \\ 1/128 & -1/128 & 5/8 \end{pmatrix}.$$

The enriched commutation defect is zero, so the old valid positive bound improves to zero for the represented state. This does not remove physical source-model error.

For target compatibility,

$$|J_f(x_T) - J_c(z_T)| \le \underbrace{|J_f(x_T) - J_c(Ex_T)|}_{\text{representation mismatch}} + L_J b_T.$$

The two terms are retained separately. Across three scales with $E_{fc} = E_{mc}E_{fm}$, the exact identity is

$$D_{fc} = E_{mc}D_{fm} + D_{mc}E_{fm}.$$

If the direct map differs from the composed map, that discrepancy is another representation term, not a hidden reduction residual. Units and time-step labels are required; actual unit conversion belongs in the numerical map.

28.3 Measured temporal-aggregation pilot

The recovered BOS ASOS source supplies a documented fine/coarse relationship: exact five-minute reports and nonoverlapping means of three complete reports. The frozen target is the next fifteen-minute mean. Separate scalar AR(1) models are fit on training observations only. Current/future samples do not cross the training/evaluation boundary.

Evaluation quantity	Retained value
Complete coarse blocks in source	110
Matched held-out target windows	14
Fine model vs measured target MAE	0.84612 knots
Coarse model vs measured target MAE	1.16476 knots
Coarse prediction vs fine prediction MAE	0.72246 knots
Maximum current last-sample/block-mean mismatch	2.66667 knots

The fine prediction is a numerical reference prediction, not truth. Both predictions are also compared with the actual measured window. Only two evaluation days are

represented. This is a small temporal scale-transfer benchmark, not a spatial atmospheric transfer theorem. Neither model has a validated uniform physical error bound. Public replay independently fits rational AR coefficients and checks all point metrics; raw source and aggregate hashes are retained.

Chapter 29

NX28 — Conserved totals can conceal a species target

The exact module reactive_transport.py implements a conservative two-cell, two-species toy, finite hypothesis-set design, and a separately sourced daily-ozone proxy pilot. The measured proxy cannot identify atmospheric reaction rates or a uniform physical model-error bound.

29.1 Exact conservative toy

Use

$$D = \begin{pmatrix} 7/8 & 1/8 \\ 1/8 & 7/8 \end{pmatrix}, \quad M = \begin{pmatrix} (3/4)D & 0 \\ (1/4)D & D \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}.$$

State order is (A_1, A_2, B_1, B_2). The source-derived model is a declared conversion $A \to B$ followed by mixing; it is not an asserted atmospheric mechanism. The exact identity $HM = DH$ proves that cellwise total concentrations evolve autonomously. All columns of M are nonnegative and sum to one, so every nonnegative initial state stays nonnegative and unweighted total mass is preserved.

The states $(1, 0, 0, 0)$ and $(0, 0, 1, 0)$ have the same total observations at every time, yet their total-B target gap at horizon T is $(3/4)^T$. Observing totals more often cannot recover the missing species fraction. A species-selective total-A sensor repairs this target; exact factorization and an invisible original-state direction are retained. Waiting shrinks the target gap here but does not create structural information.

The independent oracle performs reaction and cell fluxes separately, without assembling M. Tests compare every simplex vertex, verify $HM = DH$, preserve all mass balances, and check horizons zero through eight.

Conservation depends on units and stoichiometry. For the one-cell step

$$\begin{pmatrix} 3/4 & 0 \\ 1/2 & 1 \end{pmatrix},$$

one A produces two B. The unweighted species count is not conserved, but mass row $(2, 1)$ is. This prevents molar counts and physical masses from being interchanged.

29.2 Robust finite sensor design

Each experiment declares attainable readings for every finite hypothesis, a fixed deterministic noise radius and cost. The solver enumerates at most twelve experiments and minimizes cost under coordinate interval separation. Its negative status is infeasible_interval_separation: overlapping interval hulls alone do not prove equality of joint correlated reading sets.

The retained uncertain-rate example is stronger than a hull artefact. An A-initial hypothesis can yield species value $1/2$, while the B-initial hypothesis yields one. With radius $1/4$, both can actually report $3/4$. Nominal rate $1/4$ would instead give a larger gap and suggest successful separation. The explicit common reading shows why a nominal row rank is insufficient after reaction uncertainty. A smaller radius repairs the finite hypotheses. An independent subset oracle compares all panels for a three-label catalogue.

29.3 A measured atmospheric proxy, with narrow interpretation

Anonymous EPA AQS bulk downloads timed out. Before fitting any chemical measurements, the fallback was frozen to the R maintainer-distributed airquality benchmark. Source: https://github.com/r-devel/r-svn/blob/main/src/library/datasets/data/airquality.tab with immutable Git blob d39db693a2c5dd470ed48aa7c86b6a360527b164. The primary maintainer description is https://www.stat.ethz.ch/R-manual/R-devel/RHOME/library/datasets/html/airquality.html. The separately retained R COPYING file governs the distributed fixture; code and data provenance are not conflated.

The target is mean ozone in ppb at Roosevelt Island, 1300–1500 hours, for the next daily window. The original ozone source is the New York State Department of Conservation. The dataset contains 153 days, 116 observed ozone days and 37 missing values. It does not preserve instrument identity, detection limit or UTC offset. Other meteorological variables have different sites and windows and are not treated as simultaneous species readings.

The declared reduced open-box mechanism is loss to an unobserved sink plus constant external source,

$$c_{t+1} = \rho c_t + b, \qquad 0 \le \rho \le 1, \quad b \ge 0.$$

This map preserves nonnegative concentrations; the observed species alone is not conserved. Training-only exact least squares searches $\rho \in \{0, 1/100, \dots, 1\}$ and computes the optimal nonnegative b for each value. No latent sink concentration is scored as measured truth.

Frozen comparison	Result
Training dates	May–July 1973; both source and target remain in training
Evaluation dates	August–September 1973
Consecutive complete train/evaluation pairs	48 / 49
Fitted empirical retention	$47/100$
Reduced-model held-out MAE	$507611/23520 \approx 21.5821$ ppb
Persistence held-out MAE	$1133/49 \approx 23.1224$ ppb

This is an anonymous atmospheric reactive-transport proxy with measured target

truth and a deliberately reduced one-box map. The fitted retention is not an identified chemical reaction constant. Unknown forcing, seasonal effects and missing observations can dominate discrepancy. A near-optimal training-rate population is an explicitly heuristic loss-tolerance set, not a confidence interval. Its held-out coverage is reported rather than promised.

The public standard-library checker completely re-fits the exact benchmark and recomputes scores. It separately replays the conservative transport toy. Agreement with measurements and algebraic correctness remain different claims.

Chapter 30

NX29 — Identify the requested capability, then test whether it predicts

The exact movement_design module treats a declared bilinear response model. Its optional measured NFL pilot tests predictive usefulness separately. Identification, extraction stability and prediction are not interchangeable.

30.1 Target-specific design

With normalized conditions s, a,

$$\eta(s,a) = \beta_0 + \beta_1 s + \beta_2 a + \beta_3 sa, \qquad \phi(s,a) = (1, s, a, sa).$$

The requested capability is a row $h^\top \beta$, possibly obtained from an externally defined demand distribution. For observations $r = X_D\beta + e$,

$$h \in \operatorname{row}(X_D) \quad \Longleftrightarrow \quad \ker X_D \subseteq \ker h^\top.$$

A full profile rank requirement is unnecessary when the target uses only a known direction. Conversely, repeated rows do not reveal a missing interaction.

At conditions $(0,0), (1,0), (0,1)$, the invisible coefficient direction is $n = (0,0,0,1)$. Direct polynomial substitution proves $X_D n = 0$, while target $(1,1)$ has $h^\top n = 1$. Adding that fourth condition gives determinant one and identifies all coefficients. The restricted target $(1,1,0,0)$ is already identifiable from the original design.

For deterministic measurement box radii ϵ_i, a decoder satisfying $X_D^\top w = h$ has attained error radius $\sum_i |w_i|\epsilon_i$. The minimum_drill_repair function supports original measurement radii, candidate admissibility, costs and a finite error cap. It returns new drill names separately from the full estimator sensor order. As in NX25, its affine offset and raw-reading contract must be preserved.

Repair type	Structural rank	Error statement
Repeat an existing condition	unchanged	Requires an explicit dependence model
New interaction condition	can increase	Target-specific decoder and box bound replay
Full rank but weakly scaled condition	sufficient noiselessly	May amplify noise badly

Repair type	Structural rank	Error statement
Missing normalization or response definition	rejected	Units and model meaning cannot be inferred

The retained repeat model is independent symmetric two-point error, for which mean variance is ϵ^2/k. Its adversarial worst-case radius stays ϵ. Persistent calibration bias would not gain the same variance reduction. The API does not turn repeated bounded-error samples into unjustified $1/\sqrt{k}$ worst-case bounds.

30.2 Recovered official data and honest negative prediction

The actual restricted NFL source archives were recovered from the user's backups. Four consumed files match the pinned PerfectJet official adapter hashes exactly: combine results, combine tracking, games and player-play outcomes. Unused game tracking files are not ingested. Raw athlete trajectories, linked individual outputs and private fitted rows remain outside the public repository.

The public source manifest records the owned archive identities, source hashes, official adapter revision and ingestion boundary. The join uses the official nfl_id namespace exactly; recruiting identifiers are not guessed to match numerically. Event and attempt identifiers remain distinct. Nullified plays and unknown/missing outcomes are excluded explicitly.

The pre-fit pilot selects ten eligible athletes in the 2023 combine cohort and ten in 2024. The former train; the latter evaluate. No athlete crosses folds through another attempt. The frozen external outcome is mean non-nullified player_get_off from games in or after the draft season. All compared models use the same twenty athletes. This is a small selected sample, not a cohort-wide population estimate.

For each athlete, the bilinear response uses raw measured acceleration, fixed speed normalization $s/10$, and a declared turning increment normalized by 180 degrees. The target condition is $(1/2, 1/2)$. Exact reported-decimal row checks verify target identifiability; numerical least squares estimates the coefficients. Residual size is empirical model mismatch, not a certified measurement-error radius. No claim identifies the fitted score with intrinsic athletic ability.

Model	Held-out MSE
Conventional combine measurements	0.0457271
Conventional plus identified capability estimate	0.0745024
Conventional plus residual-adjusted capability	0.0690224

Conventional fields are forty, vertical, broad jump and combine weight. Standardization uses training data only; ridge penalty one is fixed rather than tuned on evaluation players. The added capability worsens this frozen task. A formally identifiable statistic can add no predictive value, and the negative result is retained.

The receipt also reports training-residual interval radii, held-out coverage and descriptive player bootstrap intervals. Those are empirical diagnostics, not distribution-free guarantees. The earlier historical 93.2-percent rank statistic is not reused on this new population. Likewise the historical predictive deterioration is context, not a substitute for the newly computed values.

30.3 Replay scope

Exact algebra and independent direct-substitution/repair tests run without athlete data. Public CI verifies restricted-source hash declarations and frozen aggregate integrity; it does not reconstruct restricted raw rows. Authorized owners can rerun the optional empirical_pilots.py with their private input directory. The original execution consumed real measurements and independently recorded game outcomes; publishing only aggregates does not relabel them as synthetic.

Chapter 31

NX30 — Order-sensitive geometry with a finite uncertainty population

The ordered_movement module implements rational level-two signatures, exact phase populations and finite candidate separation. It does not claim that level two reconstructs every trajectory or improves scouting prediction.

31.1 Geometric information that totals discard

For a piecewise-linear path,

$$S_i^{(1)} = \int dX_i, \qquad S_{ij}^{(2)} = \int_{s<t} dX_i(s)\, dX_j(t).$$

Level one is displacement. Antisymmetry $S_{ij}^{(2)} - S_{ji}^{(2)}$ is doubled signed area after straight-line closure. Units are products of channel units. Paths retain sampling times, interpolation convention, channel units and coordinate frame; signatures preserve frame identity. Composition across incompatible units or frames is rejected.

For signatures $(a, A), (b, B)$, Chen composition is

$$(a, A) * (b, B) = (a + b, A + B + a \otimes b).$$

A straight increment v contributes $(v, \frac{1}{2} v \otimes v)$. Streaming therefore updates

$$A \leftarrow A + \tfrac{1}{2} v \otimes v + a \otimes v, \qquad a \leftarrow a + v,$$

using the old displacement a. Updating displacement first introduces a wrong extra tensor term.

The independent integration oracle sums the triangular within-segment integral and every ordered pair of distinct segment increments:

$$S^{(2)} = \frac{1}{2} \sum_r v_r \otimes v_r + \sum_{r<s} v_r \otimes v_s.$$

It does not reuse the streaming recurrence. Associativity follows from tensor expansion and is independently tested over three segments.

Path	First level	Second level	Doubled area
right, then up	$(1,1)$	$\begin{pmatrix} 1/2 & 1 \\ 0 & 1/2 \end{pmatrix}$	1
up, then right	$(1,1)$	$\begin{pmatrix} 1/2 & 0 \\ 1 & 1/2 \end{pmatrix}$	-1

Translation and monotone reparametrization preserve the geometric signature. Timing must be added as an explicit channel when the target uses timing. Reflection and path reversal fixtures retain sign conventions. None of these algebraic invariances makes an incorrectly calibrated camera trajectory physically accurate.

31.2 Segmentation is a population, not one preferred answer

For at most eight candidate breakpoints and three phases, segmentation_population enumerates all admissible ordered cuts and retains their original identities, multiplicities, phase signatures and concatenated feature coordinates. The pilot budget is at most 1,024 variants. Population compression groups only complete signatures with identical units and coordinate frame, and retains every original identifier.

Full signatures are closed under future Chen composition. A smaller current summary may fail that closure: horizontal and vertical single-segment paths both have zero current area, but after appending the same vertical segment their future areas differ. They cannot be safely merged merely because one current feature agrees.

Finite feature uncertainty reports exact coordinate ranges and distinct attainable vectors. Weights are declared population weights, not biological probabilities. No independent feature box is substituted for correlated attainable variants. For two finite candidate sets with minimum squared distance δ^2, common Euclidean error radius ϵ guarantees separation only when $\delta^2 > 4\epsilon^2$. Shared attainable values and touching error balls are explicit refusal cases.

31.3 Real trajectories and a preserved negative result

NX29's recovered official sources support the same frozen twenty-athlete, player-held-out comparison. One earliest valid attempt per athlete is extracted, so twenty attempts remain below the fifty-attempt cap. Each path uses at most 33 uniformly indexed samples, yielding at most 32 geometric segments, and carries official field coordinates in yards with elapsed time in seconds. Full attempts are available for offline scouting; this extraction is not advertised as causal real-time forecasting.

The same geometric channels support the order-insensitive baseline and ordered features. The baseline uses displacement, duration and path length; ordered features add antisymmetric x-y, x-time and y-time areas. A prespecified phase-weighted feature population over three phases supplies segmentation ranges. Across the twenty athletes, 560 admissible variants are retained. Large phase-feature ranges are a reliability finding, not a claim of biological variation.

Matched athlete model	Held-out MSE
Conventional measurements	0.0457271
Same plus order-insensitive trajectory totals	3.05464
Same channels plus ordered areas	6.56017

Matched athlete model	Held-out MSE
Phase-population/residual-adjusted ordered features	1.60353

The primary ordering comparison is ordered versus order-insensitive on the same channels, sampling and normalization. Both perform poorly on this small selected held-out sample. The uncertainty-aware construction is less bad here, but it is not a certified physical uncertainty correction. No feature family is tuned on held-out outcomes, and all attempted families remain in the receipt.

The output includes missingness, identity linkage, training-residual coverage and descriptive bootstrap intervals. With ten evaluation athletes these are exploratory measurements, not general performance estimates. Raw official trajectories and individual predictions are restricted and are not included in public artifacts.

Public replay completely checks the exact geometry, finite uncertainty population, source hashes and aggregate integrity. Authorized owners can rerun the private-data empirical pipeline; ordinary public CI explicitly does not reprocess restricted athlete rows.

Chapter 32

NX31: target-directed rational period transport

This implementation adapts the archived PerfectPower exact differential module and certified Taylor transport engines. Their Apache-2.0 license remains in `handoff/nx36/companion_sources/perfectpower/LICENSE`. The adapter is deliberately separate from finite rational graph observation closure. A supplied connection, a geometrically derived connection and a connection in a marked integral period basis support different interpretations; the software does not conflate them.

32.1 Exact observable differential module

Let a column state satisfy $X' = A(t)X$ over $\mathbb{Q}(t)$. For a row $c(t)$, define $D_A c = c' + cA$. The product rule proves, by induction,

$$\frac{d^j}{dt^j}(cX) = (D_A^j c)X.$$

If $c_r = \sum_{i<r} a_i c_i$ is the first field-linear dependence of these rows, then every solution has $y^{(r)} = \sum_{i<r} a_i y^{(i)}$. Conversely a universal scalar differential relation supplies exactly this row dependence, because arbitrary initial states are permitted at an ordinary point. Hence the first dependence has minimum universal order for that connection and target. A specially chosen solution can have a smaller scalar equation.

The implementation calls the pinned `DifferentialModule.observable` scalar engine; it does not introduce another symbolic scalar-equation implementation. For multiple outputs it forms their field row span and repeatedly applies D_A, retaining independent original rows. Rank increases at most the ambient dimension. A returned basis B, derivative-coordinate matrix T, and output coordinates L are checked by the exact identities

$$B' + BA = TB, \qquad C = LB.$$

The verifier additionally checks basis independence and equality with the span generated by the target derivatives through ambient order minus one. This is essential: an unnecessarily large invariant space can satisfy both displayed identities while failing the minimality claim. For each target a first cyclic dependence occurs by ambient order; differentiating that dependence keeps every later derivative in the same span, including derivatives of its coefficients. Thus the finite generation check is sufficient and rejects forged extra modes.

Numerators and denominators are exact polynomial coefficient arrays, in increasing degree. Input dimensions are at most four and coefficient degrees at most six. The source symbolic engine supplies its own work, degree and coefficient-bit budgets. Public numerical entries reject floats and booleans. Input denominator polynomials and a nonzero basis pivot minor are retained as exclusions; clearing denominators does not license evaluation at their zeros. A conservative rational interval test may refuse an ordinary segment when it cannot prove regularity.

32.2 Initial information and numerical transport

The three-state example $A_{12} = 1, A_{33} = 2$, all other entries zero, has target $c = (1, 0, 0)$. Its closure has rank two, its scalar equation is $y'' = 0$, and $y(t) = a + bt$, independently of the exponentially growing third coordinate. Both a and b must be enclosed initially. Keeping only the initial value of the target is insufficient. Zero targets close at rank zero; supplying all coordinate outputs closes at full rank.

The pinned transport engine encloses the fundamental matrix on pole-free rational disks. If $\|A\| \le M$ on a radius-R disk, the analytic fundamental matrix is bounded by e^{MR}. Cauchy's coefficient estimate gives a Taylor remainder bounded by

$$ne^{MR}\frac{\rho^{N+1}}{1-\rho}, \qquad \rho = |h|/R < 1.$$

The engine uses rational upper bounds for the exponential and separately adds outward center-rounding errors. Errors compose by submultiplicativity. For S with center $\widehat{S}$, row error δ, initial center x, and component radii e_i, the adapter bounds initial uncertainty by $\sum_i |\widehat{S}_{ji}|e_i + \delta \max_i e_i$, and arithmetic error by $\delta\|x\|_\infty$. It reports these contributions separately after applying the target row. A path crossing an excluded pole is refused.

32.3 Source-grounded Legendre application

For the normalized complete elliptic period $F(t) = 2K(t)/\pi$,

$$X = (F, F')^T, \qquad A(t) = \begin{pmatrix} 0 & 1 \\ 1/[4t(1-t)] & (2t-1)/[t(1-t)] \end{pmatrix}.$$

The application transports from $t = 1/2$ to $9/16$, in a regular real chart. The target has observable rank two, equal to the full rank: this family does not provide an automatic dimension or speed improvement. The reduced and full systems carry the same two initial derivative coordinates. The exact nilpotent example above supplies a genuine rank reduction, separately from this geometric validation.

Two independent direct evaluations validate the normalization. First,

$$F(t) = \sum_{k\ge 0} \binom{2k}{k}^2 t^k/16^k,$$

with coefficient bound one, gives positive rational remainder bounds for both F and F'. Second, the substitution $v = \tan(\theta/2)$ gives

$$F(t) = \frac{4}{\pi}\int_0^1 [1 + (2-4t)v^2 + v^4]^{-1/2}\, dv.$$

Interval quadrature bounds the quartic separately on each rational cell, encloses square roots by rational bisection, and bounds π through Machin's alternating arctangent identity. It uses neither the connection nor the hypergeometric coefficients. The narrow transported interval contains the independent hypergeometric enclosure and lies inside the coarser Euler-integral enclosure. Full setup, source seeds, arithmetic errors and output intervals are retained in `results/nx36/nx31_acceptance.json`.

Replay with `PYTHONPATH=src python scripts/nx_analytic_decisions.py --check`. These are exact Python identities and rational analytic enclosures, not a Lean formalization of period integration, and they do not supply an integral flux basis for another geometric family.

Chapter 33

NX32: monodromy as a costed experiment language

The adapter records a basepoint, basis, loop descriptions, positive loop costs, observation costs and source revision. States are columns. In a chronological word $(i_1, \dots, i_m)$, the first loop is executed first, so the reading is $cT_{i_m} \cdots T_{i_1} v$. Every returned witness is replayed in this convention. Every loop matrix must be invertible, including standalone algebraic examples: continuation is an isomorphism of the local-system fiber. General singular actions remain supported by `OperatorSystem`, rather than this adapter.

The finite operator closure from `observability.py` proves all-word visibility. If the final row space W contains the observation rows and is preserved by right multiplication by every available T_i, induction places every future experiment row in W. Its annihilator is therefore permanently invisible in this declared loop language. A finite word census alone would not prove this statement. Cheapest exact witnesses use the existing cost-filtered closure.

Under $v' = Sv$, the matrices and outputs transform as $T_i' = ST_iS^{-1}$, $c' = cS^{-1}$; substitution shows every reading is unchanged. The tests independently enumerate every word through length six under three invertible rational gauges. Inverse loops are added only with explicit positive costs, and exact matrix inversion rejects singular gauges.

For the algebraic toy matrices $T_0 = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, $T_1 = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, the two length-two words have readings 1 and 2 on $v = (1, 0)$, $c = (1, 0)$. For difference $(0, 1/100)$, a cost-one experiment is nonzero but does not tolerate radius $1/100$. The cheapest robust word costs three; equality at twice the error radius is rejected. Robust enumeration retains the complete finite cost-bounded word family, because linear dependence is not safe amplitude pruning.

33.1 Actual marked Legendre loops

The archived PerfectPower `period_matrix_balls.py` supplies the marked Legendre family, its seed normalization and a source-bound integral-cycle premise. Starting at $t = 1/2$, the retained counterclockwise polygons are

$$(1/2, i/2, -1/2, -i/2, 1/2), \qquad (1/2, 1 - i/2, 3/2, 1 + i/2, 1/2).$$

Each loop was independently continued using 215 certified ordinary disks, Taylor order 16 and outward 96-bit center rounding. The resulting marked-cycle matrices are

$$T_0 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad T_1 = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}.$$

Their determinant-one and unipotence relations are exact, and they do not commute. No complete presentation of the fundamental or monodromy group is claimed. Exact recognition is conditional on the independently stated integral cycle marking: a row-norm enclosure of radius below one half identifies at most one integral matrix, but numerical proximity itself does not prove integrality. The two full continuation transcripts are retained alongside their hashes.

The fast receipt checks the source-bound transcripts and exact loop-language claims. `python scripts/nx_analytic_decisions.py --check --slow` recomputes both certified continuations, including their rational analytic errors. These loop replays take much longer than finite row closure and are intentionally separate from the unit-test loop. Their scope is the declared Legendre integral basis; the standalone algebraic toy is not relabeled as geometric monodromy.

Chapter 34

NX33: moving observable quotients

For $x' = A(t)x$, a constant-rank row matrix $B(t)$ defines $z = Bx$. Exact closure exists precisely when

$$B' + BA = TB.$$

Necessity follows by differentiating every possible source solution. Sufficiency is direct substitution. The kernel criterion is equivalent: a row factors through a surjective linear map B if and only if it annihilates its kernel. On a constant-rank chart, solving a nonvanishing pivot minor constructs regular coefficients for T. Surjectivity also makes T unique. The implementation checks this identity over $\mathbb{Q}(t)$, retaining all denominator and pivot exclusions. A sampled identity is never accepted as a neighbourhood proof.

Target preservation is separate: a requested row c must factor as $c = LB$. A dynamically closed quotient can still discard the requested target. Such an input is rejected. Zero and full-rank quotients are supported explicitly.

For $B = (1, t)$, $A = \left(\begin{smallmatrix} 0 & -1 \\ 0 & 0 \end{smallmatrix}\right)$, the derivative term cancels BA, and $z' = 0$. The explicit solution $x_1 = a - bt, x_2 = b$ verifies $z = a$. With $A = \operatorname{diag}(0, 1)$, closure fails: $(-t, 1)^T$ is hidden but is exposed at rate $1 + t$. A rational sample with nonzero exposure is a failure witness; vanishing at the isolated point $t = -1$ does not restore closure on a chart.

For a reduced gauge $\tilde{z} = Gz$, differentiation gives

$$\widetilde{B} = GB, \quad \widetilde{T} = G'G^{-1} + GTG^{-1}, \quad \widetilde{L} = LG^{-1}.$$

All three identities are checked exactly on overlaps. The gauge $G = 1 + t$ turns the constant scalar example into $\widetilde{T} = 1/(1 + t)$. Conjugation alone would give the wrong answer. A path crossing the gauge zero is refused even when the source system itself is regular there. Two subdivided ordinary paths give compatible target enclosures, checked against explicit solutions.

The integrated period example uses the source-grounded Legendre connection from NX31 and the full-rank moving frame $B = \operatorname{diag}(1, t)$. Its reduced connection and target row are computed exactly; both a direct segment and a two-segment path contain the independent normalized hypergeometric period enclosure. The source excludes $t = 0, 1$; the gauge also records its zero at zero. A separate diagonal family $x' = \operatorname{diag}(0, 1)x$, target $(\lambda, 1)x$, has observable rank one at the constant external

parameter $\lambda = 0$ and rank two for $\lambda \neq 0$. The determinant λ proves these strata; the retained samples are checks, not the proof of the stratification. This is a one-parameter differential adapter with an explicit constant external specialization, not a multidimensional flat-connection engine.

All constructions and independent target predictions are retained in `results/nx36/nx33_acceptance.json`. A local gauge identity makes no assertion about loop holonomy; actual loop evidence belongs to NX32.

Chapter 35

NX34: exact local zero bounds and a larger retained disc

This track proves a fourfold enlargement of the retained spectral-pair disc: $|s-1/2| \leq 1/25000$, compared with the earlier radius $1/100000$. It uses the same regularized Euler-product theorem as SNG-F35, with a graph-dependent bound for local-factor motion. It does not assert global meromorphic continuation. The exponent $\delta = 3/32$ is nonintegral, so the comparison uses an explicitly cut fractional branch, not an ordinary pole of that order.

35.1 Polynomial and analytic obligations

For an order-k graph and fixed $0 \leq q \leq 1$, $a_n(q) \leq \binom{n+k-1}{k-1}$. Consequently the exact rational tail on $|z| \leq r < 1$ is bounded by

$$T_N(r) = (1-r)^{-k} - \sum_{n=0}^{N} \binom{n+k-1}{k-1} r^n.$$

A polynomial lower bound larger than this tail throughout a domain proves zero-freeness directly. A boundary Rouche comparison alone supplies a same-zero- count conclusion and still requires a separate polynomial root count. The implementation keeps `zero_count` and `zero_free` distinct. Its exact supported root-count route checks a complete rational-complex factorization and counts factors in a rational disc, rejecting roots on the boundary. A second route subdivides a rational rectangle until Taylor lower bounds on enclosing L1 discs strictly exceed the analytic tail. Budget exhaustion is unresolved, not a zero.

The geometric-series example has tail $1/8$ at cutoff three and radius one half; the roots of $1 + z + z^2 + z^3$ are independently factored as $-1, i, -i$. A polynomial with a genuine interior zero receives count one and no zero-free claim. The rectangle example for $1 + z^2$ needs subdivision: its initial coarse lower bound fails, while the subdivided bounds prove nonvanishing.

35.2 Connected graph Gaussian span improvement

Write $B = (1 - z^k)L$. A minimum-zero profile with span h has volume between h and $(k-1)h$. Along a shortest path from its minimum to its maximum, Cauchy–Schwarz

and integrality give energy at least $\lceil h^2/D \rceil$, where D is graph diameter. The number of box profiles with minimum zero and maximum h is

$$N_k(h) = (h+1)^k - 2h^k + (h-1)^k.$$

For $0 < x < 1$, positive coefficients therefore give

$$B(x) \le 1 + \sum_{h \ge 1} N_k(h) x^h q^{\lceil h^2/D \rceil}, \qquad B'(x) \le \sum_{h \ge 1} (k-1) h N_k(h) x^{h-1} q^{\lceil h^2/D \rceil}.$$

The second-difference integral bounds $N_k(h) \le k(k-1)(h+1)^{k-2}$. Beyond the retained cutoff, successive terms of this upper majorant have ratio at most

$$\left(\frac{h+2}{h+1}\right)^{k-2} x q^{\lfloor (2h+1)/D \rfloor},$$

and the derivative majorant has one additional factor $(h+1)/h$. These ratios decrease. Once the derivative ratio is below one, exact geometric tails bound both infinite sums. No uncomputed infinite remainder is discarded. Differentiating $L = B/(1-x^k)$ then gives

$$L'(x) \le \frac{B'(x)}{1-x^k} + \frac{k x^{k-1} B(x)}{(1-x^k)^2}.$$

For the s-disc with center $1/r$, radius η, put $\sigma = 1/r - \eta$, and let x be a rational upper bound on $2^{-\sigma}$. For every prime, $\log p \, p^{-\sigma} \le 1/(e\sigma)$. Integration along the segment from the center therefore bounds each local-factor change by

$$\eta L'(x)/(e\sigma).$$

The center local factor is real and at least one. A strict bound below one puts the entire local factor in the right half-plane for every prime, hence fixes its principal logarithm. All inequalities use rational upper bounds, including $e > 27/10$ and a bisection enclosure of $2^{-1/r}$. The retained spectral graphs have diameter three. At $q = 1/2$, $r = 2$, $\eta = 1/25000$, the uniform lower bound for the local real part exceeds 0.0873.

The remaining SNG-F35 hypotheses are checked explicitly: $(r+1)\sigma > 1$, $r\eta \le 1/4$, and the original small-radius regularized logarithmic tail with its prime cutoff and Schwarz constant. Thus the larger disc inherits the same proved local branch and uniform Euler-tail conclusion.

Small-prime image records enclose p^{-s} using rational root bisection, the positive atanh series for $\log p$, and $e^a - 1 \le a/(1-a)$ for $0 \le a < 1$. These are outward bounds, not sampled complex plots. The saved limiting inequality is a limitation of this method, not evidence that the actual analytic function becomes singular outside the certificate. Replay is in `results/nx36/nx34_acceptance.json`.

Chapter 36

NX35: proof-directed finite computational refinement

For independently reset scalar observations and target-distinct candidate pair a, b, a robust separating test has positive margin

$$m_{e,ab} = |r_{e,a} - r_{e,b}| - \rho_{e,a} - \rho_{e,b} > 0.$$

This is specifically independent scalar interval geometry. It must not be used as a substitute for joint correlated uncertainty-set separation. Given certified margin intervals $[l_{e,ab}, u_{e,ab}]$, define certain and possible masks by $l > 0$ and $u > 0$, respectively. Equality at zero never counts as coverage. Every true test mask lies between these masks. Every cover feasible in the certain world is truly feasible, and every true cover is feasible in the possible world. Hence

$$\mathrm{OPT}(S^{+}) \le \mathrm{OPT}(S) \le \mathrm{cost}(E^{-}).$$

The implementation solves each finite mask cover by exact prefix dynamic programming. A state stores the least cost for the covered-pair union after processing an experiment prefix. Adding or omitting the next experiment gives the complete recurrence; positive costs and a finite pool ensure termination. Its complexity can be exponential in the required pair count. A declared state budget refusal does not establish infeasibility or optimality.

The deterministic planner examines an unresolved cell on a cheapest optimistic competitor, requests a tighter certified enclosure, intersects it with the old enclosure, and updates both cover bounds. Source/candidate/domain bindings must be identical. Disjoint purported enclosures or reversed fidelity levels are rejected. A stalled oracle is recorded and blocked; exhausted work returns a valid incumbent and unresolved gap. When a concrete certainly feasible panel has cost equal to the optimistic lower bound, optimality follows without evaluating every cell at maximum fidelity.

The three-test exact example initially has lower bound one and upper bound two. Refining the single uncertain zero margin of the first test raises the lower bound to two. The expensive all-pair test need not be refined. Independent enumeration of every tiny panel checks the two inequalities and final optimum.

36.1 Typed NX09 integration

`BoundedReading` separates computational radius from physical uncertainty. If the computed center difference is d, combined computational error is e, and combined physi-

cal radius is ρ, a valid margin enclosure is

$$[\max(0, |d| - e) - \rho,\ |d| + e - \rho].$$

The adapter checks equal units, horizon and reset/input domain. Refining arithmetic or a reduced model can shrink e, but cannot erase an assumed physical or source radius. The integration calls the actual source-pinned DynaComp evaluator separately for every model/sensor/word/fidelity combination; it never transplants a nominal certificate to another fault.

The retained synthetic 4-by-4 example uses the original nominal, conductance-half and heater-half hypotheses, preserving their IDs. Its finite pool has two reset protocols: left-center observation after $(2, 0, 2)$, cost one, and center observation after $(2, 2)$, cost two. Physical error radius is the declared assumption $1/100000$. Six initial rank-one model evaluations give a feasible cost-two plan and an optimistic cost-one competitor. Only the nominal and conductance-half readings of that competitor need enrichment. Two higher-fidelity evaluations certify its true positive gap and the cost-one optimum, compared with six higher-fidelity evaluations for the complete baseline.

Validation performs the remaining four fine evaluations separately and checks all coarse/fine envelopes against full exact sparse thermal predictions. They are reported as validation work, not hidden inside the planner's two-call claim. A second retained case enlarges the physical/source radii so that even exact arithmetic cannot separate the candidates. Its optimistic cover is infeasible: more digits cannot repair that model limitation.

The complete margins, refinement transcript, source evaluator records, calls, incumbent and lower bound are in `results/nx36/nx35_acceptance.json`. The claim is a finite-pool accounting optimum under the stated source and uncertainty premises, not hardware-global experimental optimality.

Chapter 37

NX36: independent end-to-end finite decision certificates

The `nx.experiment_decision/1` format binds the original candidate IDs, target map, complete finite catalogue, positive exact costs, scalar closed observation intervals, source premises, model revision, reset convention, units, parameter and chronological operator order. Unknown mandatory semantics and unsupported geometries are rejected. Exact rationals are reduced integer numerator/positive denominator pairs; booleans and floats are invalid exact integers.

The public checker requires an independently supplied expected problem. This prevents an old valid certificate from being silently rebound to changed endpoints, costs, labels or reset conventions. An integrity digest binds those bytes; it neither authenticates calibration nor proves the source equations. Source-bound model enclosures remain visible premises of the accepted claim.

37.1 Separation and optimality are distinct obligations

For every target-distinct pair, the certificate names one selected test and its strict positive interval gap. Closed intervals meeting at one endpoint are not separated. Checking the endpoints is enough to validate each witness; an optimizer is not involved. A separate exact helper checks symmetric-zonotope contrasts:

$$|w^T(\mu_a - \mu_b)| - \sum_{g \in G_a} |w^T g| - \sum_{g \in G_b} |w^T g| > 0.$$

These contrasts are feasible separation witnesses, not a claim that the helper solves every zonotope intersection problem or that scalar masks suffice for correlated panels. The complete optimality schema currently certifies scalar interval geometry.

For catalogue optimality, the certificate carries a prefix pair-mask dynamic table. At prefix zero only mask zero has cost zero. At the next prefix every state can omit the test or add its coverage mask and cost. The checker validates the complete local recurrence, including absence of extra/missing states and minimum costs, without importing any proposal optimizer or protocol compiler. Induction on prefixes proves every table entry is the minimum cost of an original subset producing that union. At the final full-pair mask, equality of its minimum cost with the explicitly feasible selected panel proves catalogue optimality. The proof table may itself grow exponentially; verification being cheaper than discovery is a measured goal, not an unconditional theorem.

A positive common cost denominator is retained and checked: each cost times that denominator is a positive integer. Natural-cost formal statements thereby apply to a consistently scaled rational cost model. Scaling does not change the comparison order. The Python checker validates the arithmetic scaling; the Lean result does not claim to verify that parser or rational implementation.

37.2 Original-object and physical-source reconstruction

The retained thermal certificate includes all three original hypothesis IDs, their full sparse model matrices, heater and sensor rows, and the two original chronological reset words. Its independent source replay performs plain sparse scalar stepping directly from the serialized definitions. It imports neither `AffineCandidate`, the reduction engine, the protocol population compiler nor a panel search routine. Each output interval must equal the independently reconstructed exact reading plus/minus the stated physical radius. The selected left-pulse experiment costs one and strictly identifies all three supplied target classes. The mask proof excludes every cheaper panel in the two-member catalogue.

This is explicitly a finite-catalogue optimum. The source binding does not call the catalogue complete over all heater words; that separate claim would need the protocol-population completeness evidence. Multiplicities and original candidate identities remain present even when outputs happen to agree.

Run `PYTHONPATH=src python scripts/check_nx_decision.py` for the independent source and decision replay. Mutation tests change endpoints, costs, labels, source revisions, reset domains, proof-table entries and claimed gaps. Every change invalidating the original claim is rejected. Separate plain-loop tests enumerate all tiny subsets and check direct interval intersections.

37.3 Actual compiled formal coverage

`formal/Singularity/NXDecision.lean` proves three scoped semantic results in core Lean 4.12.0: strict ordered gaps prevent a common reading; accepted pair-separation witnesses imply target identification; and a feasible admissible panel plus exclusion of every cheaper admissible panel implies optimality. `formal/NXDecisionAudit.lean` prints their actual axiom dependencies. All three compiled and reported no axiom dependencies. There are no admitted results or custom axiom declarations in this module.

The source-bound audit is replayed by `python scripts/check_nx_decision_formal.py`; `LEAN` or `--lean` selects the actual compiler. The retained audit binds source and audit hashes and records the compiler revision. The theorem treats uncertainty sets, target maps and costs semantically. It does not verify Python execution, the JSON parser, external analytic transport, DynaComp source correctness or real-world physical calibration. Those are separate obligations, kept explicit in the receipt.

Chapter 38

Integration review

The six implementation groups exchanged independent mathematical and code reviews before final acceptance. The integration test run preceded these reviews; a final run and deterministic receipt replay follow the fixes.

Finding	Consequence	Correction and regression
Zero leading coefficient in the exact factored-polynomial checker	The zero polynomial could be reported zero-free	Require a nonzero leading coefficient; reject zero and noncanonical degrees
Differential-closure receipt checked containment and closedness alone	An unrelated invariant mode could inflate a purported minimal observable space	Check independence and equality with the submodule generated by target derivatives; reject forged extra and dependent rows
Forecast decoder on an affine constrained state omitted its offset	Correct perturbation weights could still give a wrong absolute target	Retain the exact affine offset in the decoder contract
Ordered signatures discarded the coordinate frame	Physically incompatible paths could share a key or compose	Preserve frame metadata and reject incompatible frames
Remote-query costs used sequence index instead of physical time	Repeated-block protocols could be underpriced	Bind stride, offset and physical step in query/cost calculations
Evaluator menus did not require the same sensor across candidates	A selector could compare different physical measurements	Validate sensor identities across every candidate menu
Installation tie limit could be zero	An optimal status could have no retained witness	Validate a positive tie budget
Protocol catalogue allowed duplicate candidate identifiers	Original population provenance could be ambiguous	Require distinct original identities
Algebraic parameter isolation could miss a root on a domain boundary	A purported parameter cell could omit an exception	Refine exact isolations against domain endpoints and guard poles
Integer quotient row scales lacked original rational-map binding	A valid Smith identity could describe the wrong serialized map	Bind the original rational matrix, integer scaling and Smith replay
Residue seed budget was checked after initialization	A large initial population could bypass the resource ceiling	Preflight the seed population before materialization
An outer uncertainty enclosure used an impossibility status	Overlapping outer sets need not imply physical overlap	Report enclosure-limited certification status; bind the exact-set flag
A valid evaluator receipt could be paired with an altered reading radius	The optimizer could trust a bound its receipt did not certify	Replay source-bound evaluator options at optimizer entry
Policy/refinement replayers did not bind every status and semantic header	A valid tree could carry a false claim or uncertainty interpretation	Check schemas, statuses, semantics, fidelity direction and impossibility premises

Finding	Consequence	Correction and regression
Duplicate history action labels had different costs	Producer and verifier could resolve the same label differently	Require unique action identifiers
A supplied geometric loop matrix could be singular	An arbitrary transition could be mislabeled monodromy	Reject noninvertible loop operators

The remaining targeted checks include direct flux evolution rather than the sparse transition method, direct integer fibres rather than the Smith solver, literal interval/polytope oracles rather than the production optimizer, explicit polynomial solutions rather than differential closure, and exact public-data normal equations rather than the empirical floating regression.

The review also checked the analytic span derivative majorant, fixed-block all-horizon tail argument, finite decision checker independence, and the scope of original-mask reconstruction. A narrow formal theorem does not verify the Python parser, physical calibration or external interval engine.

Machine timings remain descriptive. The public archive excludes restricted athlete rows; its empirical replay verifies frozen aggregates and provenance while the structural movement results have independent exact oracles.

Chapter 39

Recovery audit of Claude’s interrupted NX36 session

The published working branch ended at `926dce28c4018e55b37e89e48325e802d97376ef`. Main remained at `2ae86788537b8b69c628c9f265d1d7bfc98f1d0b`, so the latest Claude implementation was not visible on main. The branch contains eleven new commits after main, including the original handoff integration.

39.1 What actually exists

Track	Recorded state	Implemented content
NX01	A/B/C complete	Candidate-specific boxes, bounded exact panel optimization, independent cover oracles and the source-derived thermal comparison
NX08	A/B/C complete	Target identifiability, hidden-target witnesses, exact amplification and decoder certificates, sensor-layout selection
NX13	A/B/C complete	Typed rectangular operator closure, legal graph grammar, portable certificates and cheapest legal witnesses
NX16	A/B/C complete	Weighted connected-support compilation, union-support cache invalidation and independently replayable local edits
NX10	A complete	Exact quadratic-metric response tails with nonnormal examples and tamper rejection
NX10	B/C proposed	The module contains substantial search drafts, but their presence alone was not accepted as evidence
Remaining tracks	Proposed	Ninety-three other units, giving ninety-five unfinished units in total

The six new test files contain 83 tests. GitHub’s Python 3.12 job and both formal

jobs were observed passing on the interrupted branch while Python 3.10 was still running; the final recovery records supersede that intermediate observation. The existing formal jobs concern the previously stated Lean scope, not newly formalized versions of these NX results.

The roadmap metadata verifier passed with 36 tracks, 108 units, four complete tracks, one in progress, and thirteen complete units. Metadata verification does not execute the recorded checks or validate physical calibration. The recovery separately runs tests, exact receipts, independent examples and the acceptance-specific evidence.

39.2 Exact source recovery

The command runner's automatic approval review repeatedly failed with a model capacity error during network fetches. No permission settings or sandbox rules were changed. The already available GitHub connector supplied the branch's read-only tree, blobs and signed commit metadata.

All 31 missing blobs were restored and checked against their Git object hashes. The eleven signed commits were reproduced byte for byte from their original payloads and signatures and checked against their published commit hashes. The complete recovered tree is `49cf9378b1d25477425dd83bd91bfd2a6dd5811a`. Local fast-forward integration preserved Claude's commits and all earlier frontier work. No unrelated checkout or user edit was reset.

39.3 Completion scope

The user explicitly requested all unfinished work, superseding the routine three-unit session limit for this recovery. Six disjoint workstreams cover the remaining tracks. Each must provide source, substantive proof, independent tests, a deterministic retained replay, and unit-specific evidence. A missing mandatory empirical dataset or a failed proof remains an explicit acceptance gap; neither is replaced by an invented measurement or a success flag.

The original P and OE ledgers and FR01-FR05 remain intact. NX acceptance is recorded only after its own requirements and dependencies have been checked.